AI Supply Chain Research

From Demand Shock to Bottleneck Relay —Ten Sectors, One Story of the AI Compute Build-Out

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AI Compute Industry-Chain · distilled from SemiAnalysis (2020–2026)

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Part I —The Panorama

Cross-Sector Synthesis

Read together, the ten sectors of this book are cross-sections of a single machine. One demand shock —agentic AI began to really work, Anthropic's ARR ran from $9B toward $44B, coding assistants passed $30B —propagated upstream until every layer broke at once: H100 rentals rose ~40% into a sold-out 2026, AI claims 86% of TSMC's N3 output by 2027, DRAM entered a 'once-in-four-decades shortage,' and a terawatt of load requests queued against a 745GW US grid peak. Because classical scaling broke —N3 is the weakest cost-per-transistor step in 50+ years and SRAM stopped shrinking —the industry now buys performance with packaging, HBM, optics and 800VDC power instead of lithography, and the unit of competition climbed from chip to rack to gigawatt campus. The six through-lines below trace how the shortage travels, where the margin pools, why software is the durable moat, and how the US-China conflict bifurcates.

1. One demand shock, one synchronized shortage across the whole chain

The proximate cause is a repricing of tokens: agentic AI works, so coding is now >70% of OpenAI's and Anthropic's ARR, Claude Code authors ~4% of GitHub public commits, and Anthropic's inference gross margin swung from 38% to above 70%. That demand hit a chain with no slack anywhere. H100 one-year rentals rose ~40% ($1.70→$2.35/hr) with capacity booked through September 2026; every merchant and custom accelerator converges on TSMC N3 in 2026, pushing effective utilization past 100% and AI's share of N3 output to 86% by 2027; DRAM —whose density now doubles per decade, not every 18 months —entered a 'once-in-four-decades shortage' with prices set to double again; and ~1 terawatt of load requests queues against a 745GW US grid peak. Nothing responds quickly: every layer's supply lead time is measured in years, and gas-turbine makers, scarred by past busts, refuse to expand.

2. Moore's Law broke —packaging, memory and process tricks are the repair crew

The economic engine beneath everything —cheaper transistors every node —has stalled: N3 delivered ~15% cost-per-transistor improvement, the weakest in 50+ years, and the SRAM bitcell is identical across N5, N3E and Intel 18A (0.021µm^2). IO is worse —data rates double every four years against transistors' two —so designs hit a pad-limited wall. Value migrates to whatever repairs the gap. TSMC's CoWoS revenue reached $9.6B, 2.5× Apple's InFO; hybrid bonding, 'more transformative than EUV,' pulls packaging into the fab because its cleanliness physics (a 1µm particle makes a 10mm bond void; ISO-3 cleanrooms) structurally lock OSATs out. In equipment, lithography is no longer destiny: High-NA single-patterning costs more than low-NA double-patterning through 1nm, so etch/deposition-led vertical scaling (3D NAND: +30%/yr density) and tricks like Sculpta and DSA carry the roadmap. Memory's repair is HBM —already 60%+ of a Blackwell GPU's manufacturing cost.

3. The unit of competition climbed from chip to rack to gigawatt campus

'Systems beat dies' is now the organizing law of the whole chain. Nvidia's 72-GPU NVLink domain is why the 8-GPU MI355X 'cannot compete head on' in frontier MoE inference; Huawei's CloudMatrix 384 answers with 16 racks, 6,912 LPO transceivers and zero copper —twice GB200 NVL72's compute at 4.1× the power, a rational trade where silicon, not electricity, is scarce. Rack physics then rewrites everything below: copper ends at ~2m reach and a 448G SerDes ceiling, pulling optics into the package (TSMC's COUPE won over even CPO-pioneer Broadcom); 600kW racks break 54V distribution, forcing 800VDC (~15× less current) and liquid cooling; the failure domain grows from node to whole rack. Finally the datacenter itself became the product: at $10-13M revenue per MW-year, xAI's 122-day, gas-turbine-powered Colossus proved that speed-to-power —not just silicon —is the moat.

4. The value-capture war: deliberate under-pricing below, violent rotation to the labs above

Who keeps the margin is a policy choice, not an accident. Nvidia (~75% gross margin, ~4× markup) defends share with equity checks instead of price cuts, and together with TSMC under-prices into scarcity like a 'central bank of AI,' keeping downstream labs profitable so total demand compounds. Custom silicon disciplines from below: OpenAI extracted ~30% off its entire Nvidia fleet before deploying a single TPU; Anthropic committed to ≥1M TPUs (~$10B of Broadcom racks plus ~$42B of GCP RPO); Broadcom became the silent #2 AI-chip company. HBM transfers wealth upstream to SK Hynix at 3×+ per-GB premiums. Then in December 2025 the pool snapped downstream: Anthropic's inference margin crossed 70%, Bedrock's seller-of-record structure lifted AWS EBIT +213bp while peers' margins fell, and Microsoft's capacity pause gifted Oracle >$420B of OpenAI contracts (~$150B of gross profit). Every layer's price is now strategy.

5. Software is the durable moat —CUDA below the API, data and harnesses above it

Hardware gaps close in a product cycle; software gaps compound. Below the API line: MI300X's paper superiority evaporated in out-of-box PyTorch; H100 training MFU rose 34%→54% in twelve months from CUDA kernels alone; one software stack (wideEP + disaggregated serving + MTP) 14×'d DeepSeek throughput on the same B300 silicon —while AMD needed 26 days of hand-written kernels to lift MI355X's Day-0 numbers >100×, the ecosystem-maturity gap made visible. Above the line, RL made data the moat: cloned-UI environments run ~$20k each and OpenAI bought hundreds; Surge passed ~$1B ARR selling expert rubrics. Token economics became harness engineering —Opus's true blended agentic price is ~$0.99/MTok against a $5/$25 sticker, because 300:1 input ratios and 90%+ cache hits set cost-per-task. Even cluster quality is software-rated: ClusterMAX steered ~$400B of contracts, and CoreWeave's SUNK scheduler keeps it the sole Platinum cloud.

6. China's bifurcation: choked at the HBM node, unopposed in robotics

Export controls work and leak at once —and the binding constraint is memory, not logic. SMIC can print millions of Ascend dies (N+3 reaches TSMC N6-class density without EUV; its 32.5nm metal pitch is tighter than shipping Intel 18A), but China's ~13M foreign HBM stacks —11.4M from Samsung, 7M shipped in the one-month pre-enforcement gap —are running out, and CXMT's ~2M stacks in 2026 support only 250-300k 910Cs. So Huawei builds fewer Ascends next year, even as shell fabs and 2.9M smuggled TSMC dies prove the entity list structurally defeatable. China compensates systemically: power-rich, it trades watts for wafers (CloudMatrix 384), while CXMT rationally harvests the DRAM cycle (~$8.6B revenue) instead of low-yield HBM. And on the one front with no US answer, Unitree's $27.3K humanoid carries 67% gross margin on a ~$9K BoM, and Chinese firms hold ~50% of the world's largest robot market.

Chain-Wide Investment Map

The Bottleneck Cascade

Demand no longer gates this cycle —physics and lead times do. Token demand keeps exploding (Anthropic ARR $9B→$44B+, coding >70% of frontier-lab revenue, H100 rentals +40% since Oct-25), so every accelerator roadmap —Nvidia, TPU, Trainium, MI450X —converges on TSMC N3-class silicon: AI takes ~60% of N3 output in 2026, ~86% in 2027, at >100% utilization. Each die then queues for memory and assembly: HBM is 50-60%+ of GPU cost in a once-in-four-decades shortage (prices ~6x, DRAM set to double again in 2026), while CoWoS interposers, TSV and TC bonders gate how fast HBM attaches to logic. Upstream, ASML's EUV output isn't scaling, capping how fast anyone adds fabs. Downstream, watts are the final gate: +21GW of 2026 US datacenter demand versus ~15GW/yr of net grid additions, a ~1TW interconnection queue, 3-4-year transformers, 24-30-month turbines. Chips gate power; power gates chips —the cascade is circular, and every link is sold out.

Winners

TSMC (**TSM**)	The toll-taker on every branch of the war —>90% of advanced-node capacity with AI eating 60%→86% of N3, the CoWoS/SoIC packaging gate ($9.6B in 2025), the HBM4 base-die foundry and COUPE for co-packaged optics —it wins whoever wins.
Nvidia (**NVDA**)	Still king: ~75% gross margin, the NVL72 rack-scale moat that RL-era inference rewards ($0.156/Mtok on GB300), CUDA's Day-0 software flywheel, networking bundled in, and $20B to absorb Groq —three simultaneous challengers only nibble at the flanks.
Broadcom (**AVGO**)	The silent #2 AI chip company: co-designer of Google's TPU, Meta's MTIA and now OpenAI's XPU, merchant switch leader (Tomahawk 5/6) and first to ship production CPO —the anti-Nvidia trade and the networking trade in one ticker.
SK Hynix	The HBM leader on MR-MUF packaging and reliability —60.4% gross margin in FY25, first in line for every content increase on Nvidia's roadmap toward 1TB of HBM4E per GPU (Rubin Ultra).
Micron (**MU**)	The Western pure-play that leapfrogged into HBM with ~30% lower power via superior TSV/power-delivery, a cheaper internal HBM4 base die, and full exposure to the broad DRAM shortage.
ASML & the WFE oligopoly (**AMAT, Lam, KLA, TEL, Kokusai**)	Quiet monopolies at every step —ASML's EUV, TEL's 100% EUV-track lock, KLA's 1,000-step inspection toll, Kokusai's ~70% batch-ALD —with value migrating from litho to deposition/etch as GAA, backside power and 3D stacking arrive, and China rush-orders (46-48% of peak-quarter revenue) as a bonus.

Packaging & bonding toolchain (EVG, Besi, ASMPT/K&S, Disco)	Hybrid bonding and TCB convert packaging into fab-grade capex: EVG owns W2W (sub-50nm alignment), Besi is the de-facto D2W bonder at TSMC, ASMPT/K&S order books rise with every HBM stack height, and Disco 'more than tripled'.
Anthropic & OpenAI (frontier labs)	The swing profit pool on top: Anthropic is the purest value capture (ARR \$9B→\$44B, inference GM 38%→70%+, Claude Code ~4% of GitHub commits), OpenAI the distribution king (#5 website, 700M+ free users awaiting agentic-checkout monetization).
Amazon / AWS (AMZN)	The best-positioned hyperscaler —the only CSP whose AI mix is token-as-a-service (Bedrock ~\$5.5B run-rate, 80-90% Anthropic) plus vertical silicon (Trainium >50% of Bedrock tokens): AI hit 10% of AWS with margins up 213bp.
Power & electrical complex (GE Vernova, Bloom, Delta, Vertiv, Tesla Megapack)	The last physical gate, monetized: turbine 'Big Three' booked into 2028-29, Bloom's fuel cells deployable in weeks, Delta leading the 800VDC transition, Vertiv/Boyd shorting the liquid-cooling ramp, Megapack as the default gigawatt buffer.

Risks

- Demand air-pocket: 1GW of AI capacity must earn ~\$12-13B a year —if token demand or agentic adoption stalls (benchmarks already mislead; automation may prove mere augmentation), the N3 shortage flips to idle capacity (2023: N7 utilization <60%) and hyper-cyclical memory gets crushed first.
- Circular financing & depreciation: Nvidia/hyperscaler equity flows into labs that buy their compute, 4-5-year GPUs sit against 15-year leases and \$420B-scale backlogs —if useful life proves shorter (the Burry thesis), hyperscaler earnings are overstated.
- Taiwan concentration: ~3% of US GDP and 8 of the world's 10 largest firms depend on one square mile for N3 wafers, CoWoS and HBM attach —the tail risk that dwarfs all others.
- ASIC and software erosion of the Nvidia toll: TPU externalization (Meta/xAI talks, ≥1M-TPU Anthropic deal), Trainium3 and OpenAI's XPU eat the merchant TAM from the top, an open-sourced XLA/NKI would flank CUDA, and ~1200x inference cost-down keeps deflating token prices.
- Power as a hard ceiling plus regulatory clampdown: a ~1TW queue against ~15GW/yr of additions, 3-4-year transformers, NERC's Level-3 alert and ERCOT ride-through mandates, an 800VDC code gap until ~2029 —and turbine orders may already be at a 'peak'.
- China cuts both ways: a B30A approval hands China ~10x the FLOPs, rare-earth retaliation and smuggling (2.9M dies in Huawei's 'die bank') blunt the controls, while subsidized legacy-chip floods and SiCarrier-class toolmakers erode Western semicap long-term.

Watchlist

- GPU rental index & InferenceMAX prints —H100 1-yr rentals \$1.70→\$2.35 (+40%); a rollover is the earliest air-pocket signal for the entire chain.

- H2-2026 rack-scale showdown —AMD MI450X (IF64/128) vs Nvidia VR200 NVL144, the first true 72+-GPU apples-to-apples, plus TPUv7 Ironwood and Trainium3 landing on independent inference leaderboards.
- Memory prints —DRAM contract prices modeled to double in 2026, the HBM4 ramp with Samsung's re-entry, Rubin Ultra's 1TB HBM4E per GPU, and the CXMT STAR IPO as China-memory's disclosure event.
- Foundry milestones —TSMC N2/A16 (GAA + backside power) ramp, Intel 18A HVM in late 2026 and its first external anchor customer, and whether smartphone cuts free N3 for AI (~0.7M more Rubins).
- Power signposts —800VDC-native Kyber/Rubin Ultra shipping late-2026/27, the first SST UL certification (DG Matrix, target end-Q2 2026), ERCOT Batch Zero (Jul 2026), and turbine order books versus a surging secondary market.
- CPO adoption curve —Nvidia's 10-15k 'pipe-cleaner' scale-out CPO switches in 2026, Meta/Broadcom field-reliability data (2.6M-hr MTBF), and AWS Trainium4 with Celestial/Marvell scale-up CPO in late 2027.
- Frontier cadence & agentic monetization —GPT-5.5 'Spud' and Anthropic 'Capybara' pre-trains, OpenAI Instant Checkout switching on 700M+ free users, and Claude Code's GitHub-commit share (4% → projected 20%+).
- Hyperscaler prints —AWS AI margins (+213bp), Oracle's >$420B backlog converting to revenue, Microsoft's capacity freeze (>2GW of LOIs dropped), and Anthropic ARR versus its >$100B year-end model.
- China policy —the foreign-HBM stockpile running out (forcing the real CXMT test), the CXMT entity-listing decision, Nvidia B30A license terms, and the ~$7B Unitree IPO as the robotics bellwether.

Part II —The Ten Sectors

Chapter 1

AI Accelerators (GPU / ASIC)

Nvidia is still King of the Jungle, but for the first time three credible challengers —Google TPU, AWS Trainium, and AMD MI450X —have arrived at once, and the real battlefield has moved from chip specs to rack-scale systems, software, and performance-per-TCO.

Overview

SemiAnalysis's coverage frames the accelerator sector as a systems war, not a chip war. Its recurring thesis —'systems matter more than microarchitecture' —explains why Nvidia's GB200/GB300 NVL72 rack (72-GPU NVLink scale-up domain) and its 4-million-developer CUDA moat let it keep ~75% gross margins even as Google's TPUv7 Ironwood, AWS Trainium3, and AMD's MI450X close the raw-silicon gap. The firm's edge is empirical: it runs its own benchmarks (InferenceX/InferenceMAX across ~1,000 GPUs, MI300X-vs-H100 training), rates every GPU cloud (ClusterMAX), models the full bill-of-materials and TCO, and tracks GPU rental prices and per-accelerator HBM supply. Three structural facts dominate. First, the CUDA moat is real but is being eroded on the ASIC flank: Gemini 3 was trained entirely on TPUs, Claude Opus 4.5 was trained on multiple hardware types including TPUs (the majority of Anthropic's training and inference infrastructure sits on Google TPUs and Amazon Trainium), Anthropic committed to ~1M TPUs (~$10B of Broadcom racks plus ~$42B of GCP RPO), and OpenAI cut ~30% off its Nvidia fleet cost merely by threatening to buy TPUs. Second, inference has fractured into prefill vs decode phases, spawning specialized silicon (Rubin CPX, the Groq LPU that Nvidia bought for $20B) and disaggregated serving. Third, the 2026 market is supply-constrained: H100 1-year rentals rose ~40% off their October-2025 low, and all rental capacity through ~September 2026 is booked. The winners are those who own the system —the rack, the scale-up fabric, the compiler, and the developer ecosystem —not just the die.

Core Knowledge

Scale-up world size (NVLink / ICI / NeuronLink / UALink)

The number of accelerators wired into one all-to-all, high-bandwidth coherent domain. It is the single most decisive competitive metric today. Nvidia's GB200 NVL72 = 72 GPUs; TPUv7 Ironwood scales via ICI to a 9,216-chip 3D-torus pod; AMD's MI355X is still stuck at 8, which is why SemiAnalysis says it cannot compete head-to-head with GB200 NVL72 on frontier MoE reasoning inference.

Prefill vs Decode (disaggregated serving)

LLM inference has two phases with opposite hardware needs. Prefill processes the whole prompt in parallel —compute-bound, hungry for FLOPS. Decode emits one token at a time, reloading the KV cache from HBM each step —memory-bandwidth-bound. Running them on the same GPU means prefill batches constantly disrupt decode. 'Disagg' splits them onto separate GPU pools, each tuned independently —the technique used in production at OpenAI, Anthropic, xAI, DeepSeek.

Performance per TCO

SemiAnalysis's core yardstick: not FLOPS or price/chip, but realized throughput divided by all-in total cost of ownership (capex + power + reliability downtime + engineering time). It is why on-paper specs mislead: the GB200 NVL72 has ~1.6x the TCO of an H100 cluster, so it must be $\geq$1.6x faster just to break even on perf-per-TCO. 'The hardware you cannot use has infinite TCO.'

CUDA moat

Nvidia's durable advantage isn't the chip —it's the ~4 million external developers who write kernels, file bugs, and ship day-one CUDA implementations (FlashAttention, Mamba, vLLM all launched CUDA-first). SemiAnalysis's sharp framing: the moat isn't dug by Nvidia's engineers but by the millions of outside developers. AMD's RCCL/ROCm libraries are largely forks of Nvidia's, so every NCCL refactor forces AMD to burn engineering hours just to keep pace.

Systolic array / Tensor Core / MXU

The matrix-multiply engine at the heart of every AI chip. Nvidia's Tensor Core (introduced in Volta 2017) does a fused matrix-multiply-accumulate to amortize instruction overhead —a plain FP op costs ~1.5pJ but issuing the instruction costs ~30pJ, a 20x overhead. Google's TPU uses a systolic array; for Trillium (TPUv6) Google quadrupled the array from 128x128 to 256x256 tiles, doubling FLOPS on the same N5 node.

Number formats: FP8, FP4, MXFP, NVFP4

Lower-precision formats trade numerical range/mantissa for throughput and memory. Each generation adds a narrower format: Blackwell added FP4, which roughly doubles throughput vs FP8. Microscaling formats (MXFP8/MXFP4, and Nvidia's NVFP4) attach a shared scale to blocks of values to preserve accuracy. InferenceX shows GB300 delivering up to 100x on FP8-vs-FP4 relative to an H100 baseline —most of Blackwell's inference gain is the format, not just the silicon.

Optical Circuit Switch (OCS) scale-up

Google's alternative to electrical packet switches. An OCS is a reconfigurable mirror-based 'patch panel' that routes whole optical fibers with near-zero latency and no optical-electrical-optical conversion, making it lower-power. It lets Google carve arbitrary 3D-torus slices out of a 9,216-TPU pod, route around faults, and give complete cube 'fungibility.' Building the max world size needs 48 144x144 OCSs. This is the backbone of Nvidia's only real scale-up rival.

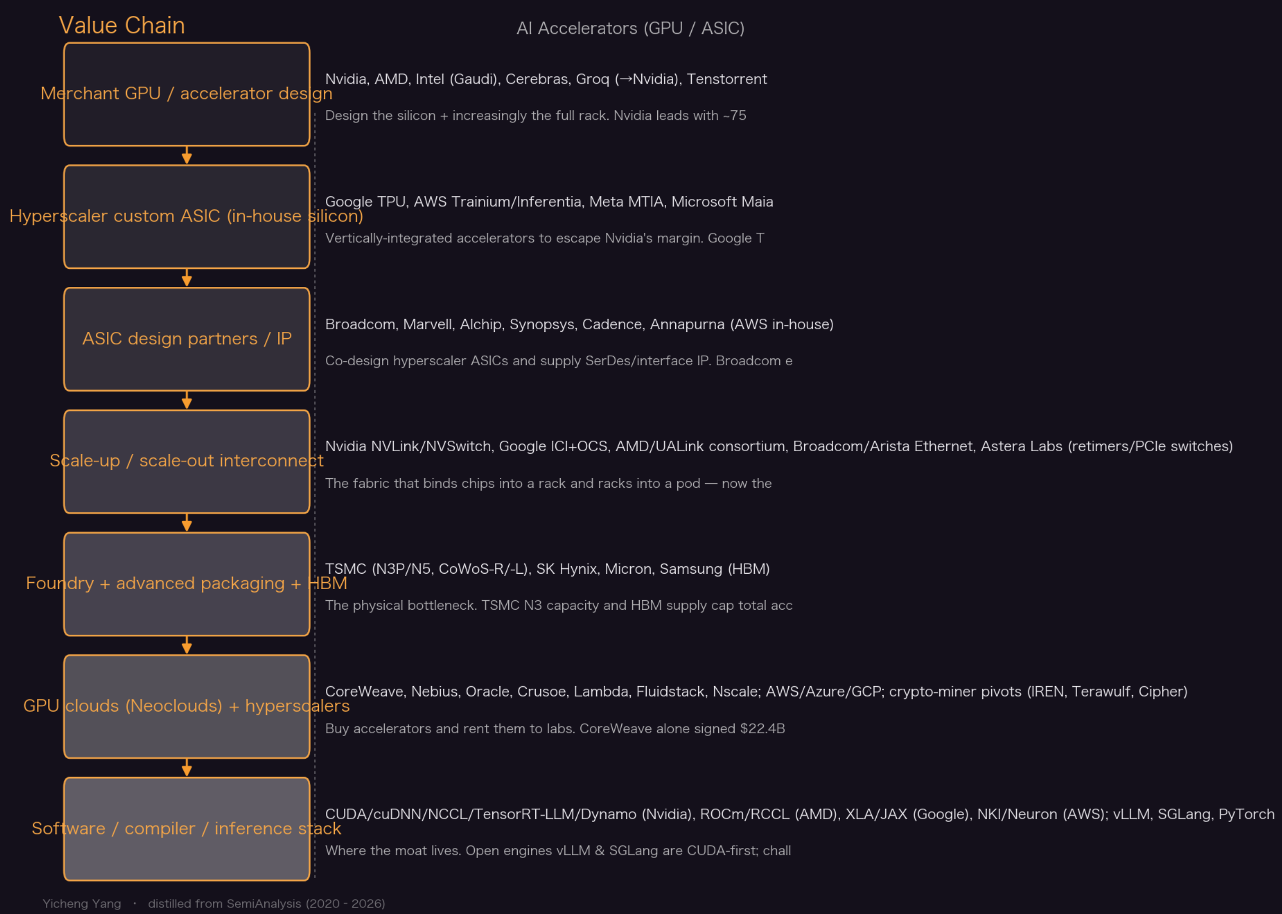
Neocloud / ClusterMAX

Neoclouds are GPU-focused clouds (CoreWeave, Nebius, Crusoe, Lambda) that rent GPUs to labs. ClusterMAX is SemiAnalysis's independent rating system (Platinum$\rightarrow$Underperforming) grading 84 providers on 10 criteria (security, orchestration, storage, networking, reliability, monitoring, pricing $\cdots$). CoreWeave is the sole Platinum. The rating has teeth: top-rated neoclouds have booked ~$400B in RPO since v1.0.

HBM as % of BOM / the memory wall

High-Bandwidth Memory has gone from 80GB/3.4TB/s (H100) to 288GB/8TB/s (GB300) in under three years and is now the single largest component of the GB300 package BOM. Because prefill wastes HBM's expensive bandwidth, Nvidia built Rubin CPX with cheap GDDR7 (>50% lower cost/GB) instead —a direct attack on the memory-cost wall.

Value Chain



Key Theses

1. **Systems, not chips, decide the war —and Nvidia's GB200/GB300 NVL72 rack opened a canyon-sized gap in scale-up.**

   SemiAnalysis's founding thesis is that a strong system beats a strong die: even when TPU silicon lagged Nvidia on paper, Google's system engineering matched Nvidia on perf and cost. The GB200 NVL72's 72-GPU NVLink domain is why AMD's 8-GPU MI355X 'cannot compete head on' on frontier MoE reasoning; competitors only reach a 72-class rack a year later (MI450X, Trainium3).

   > *"AMD and custom silicon competitors may have made a small step forward in emulating Nvidia's 72-GPU rack scale design, but Nvidia has just made another Giant Leap, again leaving competitors very distant objects in the rear-view mirror."*
   > — Another Giant Leap: The Rubin CPX Specialized Accelerator & Rack (2025-09-10)

2. **The CUDA moat is real and still widening on training —AMD's software, not its hardware, is the problem.**

   After a five-month benchmarking quest, SemiAnalysis found the MI300X's on-paper spec advantage evaporated: out-of-box AMD PyTorch training was 'impossible' without workarounds, and public-release performance trailed H100/H200 despite lower TCO. Because RCCL is a fork of NCCL, every Nvidia refactor forces AMD to burn thousands of engineer-hours just to sync. The takeaway: the moat isn't the chip —it's the ~4M external developers.

> *"As fast as AMD tries to fill in the CUDA moat, NVIDIA engineers are working overtime to deepen said moat with new features, libraries, and performance updates."*
> — MI300X vs H100 vs H200 Benchmark Part 1: Training - CUDA Moat Still Alive (2024-12-22)

3. **Google's TPU is the most dangerous merchant challenger —and you get its benefit even before turning one on.**

   Gemini 3 was trained entirely on TPUs; Claude Opus 4.5 was trained on multiple hardware types including TPUs —the majority of Anthropic's training/inference infrastructure runs on Google TPUs and Amazon Trainium. Anthropic committed to ≥1M TPUs (400k Ironwoods = ~$10B of Broadcom racks + 600k via ~$42B of GCP RPO). Meta, xAI, SSI and OpenAI are in the pipeline. Crucially, the mere competitive threat lets labs extract price cuts: 'OpenAI hasn't even deployed TPUs yet and they've already saved ~30% on their entire lab-wide NVIDIA fleet.'

   > *"The more (TPU) you buy, the more (NVIDIA GPU capex) you save! OpenAI hasn't even deployed TPU yet and already increased perf per TCO by getting ~30% off their compute fleet due to competitive threats."*
   > — TPUv7: Google Takes a Swing at the King (2025-11-28)

4. **'Systems matter more than microarchitecture' —but TPU silicon has now nearly closed the gap too.**

   Historically Google under-specced TPU FLOPS/memory versus Nvidia (prioritizing RAS/uptime, and because RecSys had low arithmetic intensity). That changed post-LLM: Trillium (TPUv6, same N5 node) doubled FLOPS by quadrupling the systolic array to 256x256; TPUv7 Ironwood nearly matches GB200 on FLOPS, bandwidth and 8-Hi HBM3E capacity —arriving only ~a year after Blackwell.

5. **AWS Trainium3 is a credible third front, built on an 'Amazon Basics' perf-per-TCO obsession.**

   Trainium3 moves to N3P, doubles MXFP8 FLOPS, upgrades to 12-Hi HBM3E (144GB) and —by switching from sub-par Samsung to Hynix/Micron HBM —lifts bandwidth 70%. It adds an all-to-all switched scale-up fabric (Nvidia Oberon-like), making AWS the first outside Nvidia to ship a 72-class switched rack. AWS deliberately drives suppliers hard: 'Annapurna places greater emphasis on driving down the TCO denominator.'

6. **AMD is finally in 'wartime mode'; its real shot at Nvidia is the H2-2026 MI450X rack, not the MI355X.**

   After the December-2024 article, Lisa Su personally engaged; AMD stood up devrel (Anush Elangovan), added MI300 to PyTorch CI/CD, and is launching a dev cloud. But the MI355X's 8-GPU world size relegates it to competing with air-cooled HGX B200/B300, not GB200 NVL72. SemiAnalysis's window: the rack-scale MI450X IF64/IF128 (UALink/Infinity-Fabric-over-Ethernet) could match Nvidia's VR200 NVL144 in H2 2026 —with 'sweetheart pricing' + OpenAI's equity rebate as sweeteners.

   > *"AMD needs to invest significantly more GPUs, they have less than 1/20th of Nvidia's total GPU count."*
   > — AMD 2.0 - New Sense of Urgency | MI450X Chance to Beat Nvidia | Nvidia's New Moat (2025-04-23)

7. **Inference has split into prefill and decode —spawning specialized silicon like Rubin CPX and the Groq LPU.**

   Because prefill is compute-bound and decode is bandwidth-bound, running both on one HBM-rich GPU wastes expensive memory. Rubin CPX answers with 20 PFLOPS dense FP4 but only 128GB of cheap GDDR7 (2TB/s) —HBM-free, >50% lower memory cost/GB. SemiAnalysis calls this second only to the GB200 NVL72 in significance, and says it sends every competitor ′back to the drawing board′ to build their own prefill chip.

   > *"Only with hardware specialized to the very different phases of inference, prefill and decode, can disaggregated serving achieve its full potential."*
   > — Another Giant Leap: The Rubin CPX Specialized Accelerator & Rack (2025-09-10)

8. **On real inference, rack-scale Blackwell ′framemogs′ Hopper —and Nvidia dominates energy efficiency.**

   InferenceX v2 (formerly InferenceMAX) benchmarks ~1,000 GPUs across all SKUs from the past 4 years. GB300 NVL72 hits up to 100x FP8-vs-FP4 and 65x FP8-vs-FP8 over an H100 disagg+wideEP baseline; H100→GB200 NVL72 shows up to 55x at 75 tok/s/user. Jensen ′under-promised and over-delivered′ on his GTC-2024 30x claim. AMD is competitive on subsets, but its weakness is composability —combining disagg+wideEP+FP4 breaks down.

   > *"Rack scale Blackwell NVL72 is framemogging hopper and makes hopper looks like it is jestermaxxing."*
   > — InferenceX v2: NVIDIA Blackwell Vs AMD vs Hopper (2026-02-16)

9. **GB200 NVL72 costs ~1.6x an H100 cluster′s TCO —and its rack-scale reliability is the hidden tax.**

   Factoring capex + opex, GB200 NVL72 TCO is ~1.6x an H100 cluster, so it must be ≥1.6x faster to win on perf-per-TCO. The catch: with Blackwell the failure domain jumped from the node to the entire 72-GPU rack —a single faulty component can drain a whole rack. NVLink copper backplane and cross-rack ACC cables have persistent signal-integrity/reliability issues; top providers now offer 99% rack-level SLAs. Software also matters enormously: H100 MFU rose from 34%→54% in 12 months purely from CUDA-stack improvements.

   > *"TCO for the GB200 NVL72 is about 1.6x higher than TCO for the H100. This means that the GB200 NVL72 needs to be at least 1.6x faster than the H100 in order to have an performance per TCO advantage."*
   > — H100 vs GB200 NVL72 Training Benchmarks (2025-08-20)

10. **The 2026 GPU market is acutely supply-constrained: H100 rentals rose ~40% and everything through ~Sept-2026 is booked.**

    Contrary to the consensus that Hopper would collapse in price as Blackwell ramped, H100 1-year rental prices rose ~40% from a $1.70/hr low in Oct 2025 to $2.35/hr by March 2026. On-demand is sold out across every GPU type; AWS p6-b200 spot fetches $14/hr; all capacity online through Aug – Sept 2026 is booked. Drivers: open-weight model adoption, agentic/multi-step workloads (Claude Code), native media generation, and a memory-price spike (DDR5 ~5x YoY) that made OEMs reprice servers and withhold supply.

    > *"Trying to find GPU compute in early 2026 has been like trying to book airplane tickets on the last flight out, high prices, and almost no availability."*
    > — The Great GPU Shortage –Rental Capacity (2026-04-02)

11. **Nvidia's $20B Groq deal is really about building an SRAM 'fast-token' front for disaggregated decode.**

    Nvidia paid Groq $20B to license IP and hire the team (structured to dodge antitrust). Standalone, a Groq LPU is uneconomic at scale —its ~500MB SRAM saturates fast and leaves little for KV cache —but it serves tokens extremely fast, commanding a premium. Nvidia will fuse the LPU into Vera Rubin via Attention-FFN Disaggregation (AFD): latency-sensitive decode on the SRAM-heavy LPU, memory-hungry attention on GPUs. Bonus: the LPU runs on Samsung SF4, sidestepping constrained TSMC N3/HBM —'true incremental revenue and capacity that no one else can access.'

    > *"Nvidia paid Groq $20B to license their IP and hire most the team···if this transaction were structured as a full acquisition and were put to anti-trust review, such a transaction would likely not go through."*
    > — Nvidia –The Inference Kingdom Expands (2026-03-24)

12. **CoreWeave is the only 'Platinum' GPU cloud —and ClusterMAX ratings now steer hundreds of billions in compute contracts.**

    ClusterMAX 2.0 reviewed 84 providers (from 26); CoreWeave alone holds Platinum, the only cloud that can consistently command a pricing premium because its all-in TCO is better even at a higher $/GPU-hr. Its differentiator SUNK (Slurm-on-Kubernetes) is the only viable way to run Slurm and Kubernetes jobs on one cluster. CoreWeave signed $22.4B with OpenAI and a $14.2B/6-year Meta deal; ClusterMAX's top-rated clouds have collectively booked ~$400B in RPO.

    > *"CoreWeave retains top spot as the only member of the Platinum tier···the only cloud to consistently command premium pricing in our interviews with end users."*
    > — ClusterMAX™ 2.0: The Industry Standard GPU Cloud Rating System (2025-11-06)

13. **Nvidia's investments in AI startups are a margin-defense strategy, not a fake 'circular economy.'**

    Skeptics claim Nvidia props up cash-burning startups to book revenue. SemiAnalysis disagrees: Nvidia offers equity investment rather than cutting prices, because a price cut would compress its ~75% gross margin and trigger investor panic. It's how Nvidia keeps foundation labs on GPUs while labs use the TPU/AMD threat to negotiate. The competitive pressure is real —hence Nvidia's own defensive PR and finance-team responses reproduced by SemiAnalysis.

    > *"Nvidia aims to protect its dominant position at the foundation labs by offering equity investment rather than cutting prices, which would lower Gross margins and cause widespread investor panic."*
    > — TPUv7: Google Takes a Swing at the King (2025-11-28)

Article Deep-Dives

ClusterMAX™ 2.0: The Industry Standard GPU Cloud Rating System (2025-11-06)

The definitive 46,000-word GPU-cloud rating: 84 providers hands-on tested against 10 criteria, 209 tracked, 140+ end users interviewed. CoreWeave is the sole Platinum (its SUNK Slurm-on-Kubernetes and premium pricing power stand out); Nebius/Oracle/Azure lead Gold; Google/AWS lead Silver. Key trends: the shift to Slurm-on-Kubernetes, GB200 NVL72 rack-level failure domains forcing 99% rack-level SLAs, crypto-miners (IREN, Terawulf) pivoting to AI hosting, and —pointedly —that any given firm's AMD cloud offering is materially worse than its Nvidia one. Rating carries commercial weight: top clouds have booked ~$400B in RPO.

TPUv7: Google Takes a Swing at the King (2025-11-28)

The 10K-word case that Google is Nvidia's most dangerous merchant challenger. Gemini 3 trained entirely on TPUs; Opus 4.5 trained on multiple hardware types including TPUs (most of Anthropic's training/inference infra sits on TPUs and Trainium); Anthropic committed ≥1M TPUs; Meta, xAI, SSI, OpenAI are lining up. Ironwood nearly closes the silicon gap to GB200 (FLOPS/bandwidth/HBM), and Google's real edge is the ICI 3D-torus + Optical Circuit Switch scale-up fabric —the only true NVLink rival —reaching a 9,216-TPU pod with reconfigurable, fungible slices. Google earns superior EBIT margins even after Broadcom's cut. The missing ingredient vs the CUDA moat: Google still won't open-source XLA:TPU, the runtime, and the MegaScaler multi-pod code.

AWS Trainium3 Deep Dive | A Potential Challenger Approaching (2025-12-04)

SemiAnalysis's most detailed accelerator teardown, covering silicon (N3P, CoWoS-R dual-die), the new all-to-all switched scale-up fabric (making AWS the first outside Nvidia to ship a 72-class switched rack), a 'don't decide' multi-vendor switch strategy, and software (open-sourcing NKI + a native PyTorch backend to seed a developer ecosystem). Trainium3 doubles MXFP8 FLOPS, moves to 144GB 12-Hi HBM3E with +70% bandwidth (Samsung→Hynix/Micron), and is architected for perf-per-TCO with cableless serviceability and redundant scale-up lanes. Alchip beat Marvell for the backend; Astera Labs gives AWS an equity 'rebate' (~23%) on PCIe retimers. Trainium4 splits into UALink and NVLink-fusion tracks.

InferenceX v2: NVIDIA Blackwell Vs AMD vs Hopper (2026-02-16)

The open-source (Apache 2.0) benchmark run across ~1,000 GPUs —every western Nvidia SKU of the past 4 years and every AMD SKU of the past 3 —measuring the full throughput-vs-interactivity Pareto frontier with production techniques (disagg prefill + wideEP + FP4/FP8). Results: rack-scale Blackwell dominates (GB300 up to 100x FP8→FP4 over H100), and Nvidia leads energy-per-token everywhere. AMD is competitive on subsets and doubled its DeepSeek-R1 FP4 throughput in <2 months, but its Achilles' heel is composability —stacking disagg+wideEP+FP4 breaks. Widely reproduced/validated by Google Cloud, Azure, Oracle, OpenAI.

MI300X vs H100 vs H200 Benchmark Part 1: Training - CUDA Moat Still Alive (2024-12-22)

The five-month training-benchmark quest that reset the AMD narrative and triggered Lisa Su's personal response. Despite the MI300X's superior on-paper memory and lower TCO, out-of-box AMD PyTorch training was broken and needed workarounds; public-release performance trailed H100/H200. Root causes: weak QA culture, forked libraries (RCCL from NCCL), too few internal GPUs for CI/CD, and poor scale-out. SemiAnalysis open-sourced its benchmarks and issued a detailed roadmap to AMD leadership —the article that arguably kick-started AMD's 'wartime mode.'

Key Data

Nvidia gross margin / markup	~75% GM, ~4x markup	The margin umbrella that leaves room to invest in labs rather than cut price; Broadcom takes a chunk on TPU BOM.
GB200 NVL72 scale-up world size	72 GPUs (NVLink)	vs AMD MI355X's 8; the gap that keeps AMD off frontier MoE reasoning inference until MI450X.

TPUv7 Ironwood max pod size	9,216 TPUs (ICI 3D torus)	Built from 144 4x4x4 cubes via 48 144x144 OCSs; ~8,000 is the practical training block due to slice availability.
H100 1-year rental price	$1.70→$2.35/hr/GPU (+~40%, Oct-25→Mar-26)	Defied the consensus of a Hopper price collapse; on-demand sold out across all SKUs.
Cost to train GPT-3 175B (FP8, H100)	72¢→54.2¢ per 1M tokens (2024, software gains)	= $218k→$162k for a 300B-token run, purely from CUDA-stack MFU improvements at ~$1.42/hr/GPU.
H100 MFU improvement (software only)	BF16 34%→54%, FP8 29.5%→39.5% in 12 mo	+57% BF16 throughput from cuDNN/cuBLAS/NCCL kernels alone — the CUDA moat compounding.
GB200 NVL72 vs H100 TCO	~1.6x higher	Must be ≥1.6x faster to win perf-per-TCO; rack-level reliability is the hidden tax.
GB300 NVL72 inference uplift vs H100	up to 100x (FP8→FP4), 65x (FP8→FP8)	InferenceX v2 across ~1,000 GPUs; H100→GB200 shows up to 55x at 75 tok/s/user.
Anthropic TPU commitment	≥1M TPUs (400k Ironwood ≈ $10B racks + ~$42B GCP RPO)	400k sold directly via Broadcom; 600k rented via GCP —most of GCP's $49B Q3 backlog jump.
OpenAI Nvidia-fleet cost cut from TPU threat	~30%	Achieved before deploying a single TPU —the perf-per-TCO threat alone extracts the discount.
Trainium3 spec deltas vs Trn2	2x MXFP8 FLOPS; 144GB 12-Hi HBM3E; +70% bandwidth	On N3P; HBM switched from sub-par Samsung to Hynix/Micron (9.6Gbps pins, highest seen).
Rubin CPX prefill chip	20 PFLOPS dense FP4, 128GB GDDR7, 2TB/s	HBM-free, >50% lower memory cost/GB vs R200's 288GB HBM/20.5TB/s —attacks the memory-cost wall.
Nvidia–Groq deal	$20B IP license + team hire	Structured to dodge antitrust; LPU3 (LP30) has 500MB SRAM, 1.2 PFLOPS FP8, runs on Samsung SF4.
CoreWeave contract wins	$22.4B OpenAI + $14.2B/6yr Meta; $9B Core Scientific	IPO'd on NASDAQ ($CRWV), stock +200% in 6 months; acquired Core Scientific for 1.3GW of DC.

ClusterMAX top-cloud RPO booked	~$400B since v1.0 (Mar 2025)	84 providers reviewed; CoreWeave sole Platinum; AMD cloud offerings rated worse than same firm's Nvidia offering.
Astera Labs (ALAB) equity 'rebate' to AWS	~23% effective discount (strike $20.34)	AWS earns warrants for hitting PCIe-retimer purchase milestones on Trainium3 —a component 'rebate.'
Opus 4.6 fast-mode economics	6x price for ~2.5x (now ~1.75x) interactivity	Revealed preference for fast tokens over smart tokens; ~80% of SemiAnalysis's ~$10M AI spend was on it.
Cerebras–OpenAI deal	750MW compute (by 2028)	WSE-3 wafer-scale chip wins on speed (fast tokens); Cerebras filed to IPO on the strength of it.

Investment Angle

Winners

Nvidia (**NVDA**)	Still 'King of the Jungle': ~75% GM, the 72-GPU NVLink rack moat, the CUDA developer flywheel, and it keeps out-innovating (Rubin CPX, $20B Groq LPU, CPO). Even three simultaneous challengers only nibble at the flanks —as long as Jensen keeps accelerating.
Broadcom (**AVGO**)	Co-designs the TPU (largest BOM item, fat margin) and is central to the merchant-TPU externalization; the biggest beneficiary of the Google/Anthropic ASIC ramp and custom-silicon supercycle.
TSMC	Every leading accelerator (Nvidia, TPU, Trainium, MI450X, LPU4) is on N3-class + CoWoS. N3 capacity is the binding constraint on total industry output —pure toll-taker on the whole war.
CoreWeave (**CRWV**)	Sole ClusterMAX Platinum, premium pricing power via SUNK; $22.4B OpenAI + $14.2B Meta contracts; +200% in 6 months post-IPO. The reference neocloud.
AMD (**AMD**)	The credible #2 —but the bet is on the H2-2026 MI450X rack matching VR200 NVL144, plus OpenAI's equity rebate and sweetheart pricing. Software (ROCm) execution is the swing factor; 'wartime mode' is real but Nvidia is still sprinting.
Astera Labs (**ALAB**) / **interconnect suppliers**	PCIe retimers/switches into AWS Trainium (with equity 'rebate'); scale-up/scale-out connectivity is the new decisive layer —a rising tide across ASIC and GPU builds.

Crypto-miner-to-AI pivots (**IREN, Terawulf, Cipher**)	Own scarce power + PPAs; the 'hyperscaler backstop' financing template (Google's off-balance-sheet IOU) unlocks a NeoCloud growth wave; IREN scored a 200MW GB300 deal with Microsoft.

Bottlenecks

- TSMC N3 wafer capacity —caps total accelerator output; N3P defect density is improving slower than expected, forcing re-spins/queues.
- HBM supply —the largest single BOM item on GB300; constrained across Hynix/Micron/Samsung and repriced parabolically (DDR5 ~5x YoY).
- Power / datacenter capacity —Google's true bottleneck is power (3-yr MSA cycles); all rental capacity through ~Sept-2026 is booked.
- Advanced packaging (CoWoS-R/-L) —the reticle-scale interposers and 20-layer ABF substrates that gate chiplet + HBM integration.
- Software talent —AMD is uncompetitive on AI-software-engineer comp; ROCm/RCCL forever chasing NCCL refactors.

Risks

- ASIC substitution accelerates: if Google open-sources XLA:TPU/runtime and AWS open-sources NKI, the CUDA moat erodes on the software flank faster than expected.
- GB200/GB300 rack reliability: NVLink copper backplane + cross-rack ACC cables have persistent signal-integrity failures; the 72-GPU failure domain raises the hidden TCO tax.
- GPU terminal-value / rental-price reversal: the ~40% H100 rental rally could unwind if inference demand growth slows, hitting neocloud IRRs and 6-yr depreciation assumptions.
- 'Circular economy' financing: Nvidia/hyperscaler equity investments and off-balance-sheet backstops concentrate counterparty risk if a marquee lab's demand disappoints.
- Duration mismatch: 4-5yr GPU life vs 15-yr datacenter leases (~8yr payback) —solved for now by the hyperscaler backstop, but a fragile structure.

Catalysts

- H2-2026 rack-scale showdown: AMD MI450X (IF64/IF128) and Nvidia VR200 NVL144 both hit production —the first apples-to-apples 72+ rack comparison.
- TPUv7 Ironwood + Trainium3 added to InferenceX —the first independent, apples-to-apples ASIC-vs-GPU inference numbers.
- More TPU externalization deals (Meta, xAI, SSI, OpenAI) + Cerebras IPO —re-rating the ASIC/merchant-challenger supply chain.
- Rubin CPX / Vera Rubin ramp —validates prefill-decode specialization and disaggregated serving at scale.
- GPU rental index inflection —SemiAnalysis's public H100 1yr index is the real-time barometer for the whole cycle.

Open Questions

- Will Google open-source XLA:TPU, the runtime and MegaScaler? SemiAnalysis argues it's the single missing ingredient for TPU to threaten the CUDA moat at scale —but Google has resisted for years.
- Can AMD's MI450X rack actually match VR200 NVL144 in H2-2026, and —more importantly — will ROCm software composability be there? Hardware parity without software parity has failed before.
- Is the ~40% H100 rental surge structural (insatiable inference/agentic demand) or a memory-price-driven supply squeeze that reverses once OEM server pricing normalizes?
- Does prefill-decode specialization (Rubin CPX, LPU) become the dominant inference architecture, forcing every competitor to build two chip families —and can anyone match Nvidia's system integration to exploit it?
- How durable is the 'hyperscaler backstop' financing template that papers over the 4-5yr-GPU vs 15yr-lease duration mismatch if a frontier lab's demand or funding falters?

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Chapter 2

Foundry & Advanced Process

One square mile in southern Taiwan makes the transistors the entire AI economy runs on —and the whole industry is now fighting over N3 wafers, backside power, and whether anyone can ever build a second TSMC.

Overview

Leading-edge logic is the most concentrated critical industry on earth. TSMC holds >90% of advanced-node capacity, and by SemiAnalysis's count 8 of the 10 largest companies in the world depend on it —roughly 3% of US GDP riding on one square mile in Tainan. The AI buildout has turned this from a smartphone business into an HPC business: HPC rose from 36% of TSMC revenue in 2020 to 58% in 2025 while smartphone fell from 46% to 29%, and Nvidia is modeled to consume more N3 wafers than Apple by 4Q27. That collision has produced 'the great AI silicon shortage' —every AI accelerator family is converging on N3 in 2026, pushing effective N3 utilization above 100%, with AI taking ~60% of N3 output in 2026 and 86% by 2027. Underneath the economics sits a physics inflection: FinFET is dead, so 2nm-generation nodes must simultaneously adopt gate-all-around (GAA) transistors and backside power delivery (BSPDN) —the first interconnect paradigm shift since copper in 1997. The competitive field has thinned to TSMC (dominant), a process-competitive-but-broke Intel (first to GAA+BSPDN with 18A), a customer-starved Samsung, and Japanese startup Rapidus. China is a separate track: export controls froze SMIC out of EUV, yet its N+3 node reaches TSMC N6-class density (32.5nm metal pitch, tighter than Intel 18A's shipping 36nm) through brutal DUV multi-patterning —density it cannot translate into competitive power or cost. Cost-per-transistor scaling has nearly stopped (N3 delivered the weakest scaling in 50+ years), pushing value toward advanced packaging, chiplets, and DTCO, and inviting moonshots like X-ray lithography startup Substrate that could halve wafer cost if it works.

Core Knowledge

Process node (N5 / N3 / N2 / A16)

A named generation of transistor manufacturing. Names ('3nm', '2nm', A16=1.6nm) are marketing, not physical dimensions; what matters is contacted gate pitch, metal pitch, cell height and resulting density/power/performance (PPA). Each family spawns many variants (N3B, N3E, N3P, N3X, N3S) tuned for cost, mobile, or HPC.

FinFET → GAA (Gate-All-Around)

FinFET (fins the gate wraps on 3 sides) has scaled since 22nm but can no longer improve. GAA stacks horizontal 'nanosheets' so the gate surrounds the channel on all 4 sides, boosting drive current. All 2nm-generation nodes (TSMC N2, Intel 18A, Samsung SF2) adopt GAA 2025+. Successors: forksheet (denser, worse electrostatics) and CFET (stacking N over P for ~1.5x scaling, but ~a decade from HVM).

Backside Power Delivery (BSPDN)

Moving power wiring to the wafer backside, freeing the frontside entirely for signals —the first interconnect innovation since aluminum→copper (1997). Enables cells shorter than 6-track (~15% density) and cuts power ~15-20%. Three flavors: buried power rail (simplest, contamination risk, won't reach HVM), Intel's PowerVia (easier, less scaling), and direct backside contacts (highest risk/reward, ~25% cell shrink, needs <5nm post-bond overlay). TSMC calls its HPC version 'Super Power Rail.'

EUV vs DUV multi-patterning

Extreme UV (13.5nm) prints tiny features in one shot; a single EUV tool costs $225M and can produce >$650M of wafers a year. Deep UV (193nm immersion) must use SADP/SAQP multi-patterning (2 or 4 passes) to hit the same pitch, adding masks, overlay error, complexity and cost. Export controls deny China EUV, forcing SMIC into aggressive DUV multi-patterning.

DTCO & the death of SRAM scaling

Design-Technology Co-Optimization: squeezing density from layout/design tricks (fin depopulation, contact-over-active-gate, single diffusion break) rather than raw lithography. Increasingly the main scaling lever because SRAM bitcells have barely shrunk since N5 —N3E's 6T cell (0.021μm^2) is identical to N5's. Logic keeps scaling; cache does not, raising the importance of architecture and 3D stacking.

Anchor tenant / 'first and best' customer

A leading-edge fab needs one huge, deep-pocketed customer to pre-fund capacity and prove out yield/IP on each new node. Apple was TSMC's anchor for a decade (funded the learning curve, often >50% and near-100% of node launches). AI's cash now gives TSMC a second anchor in Nvidia. Intel Foundry's core problem: no anchor tenant to fill and de-risk its leading-edge fabs.

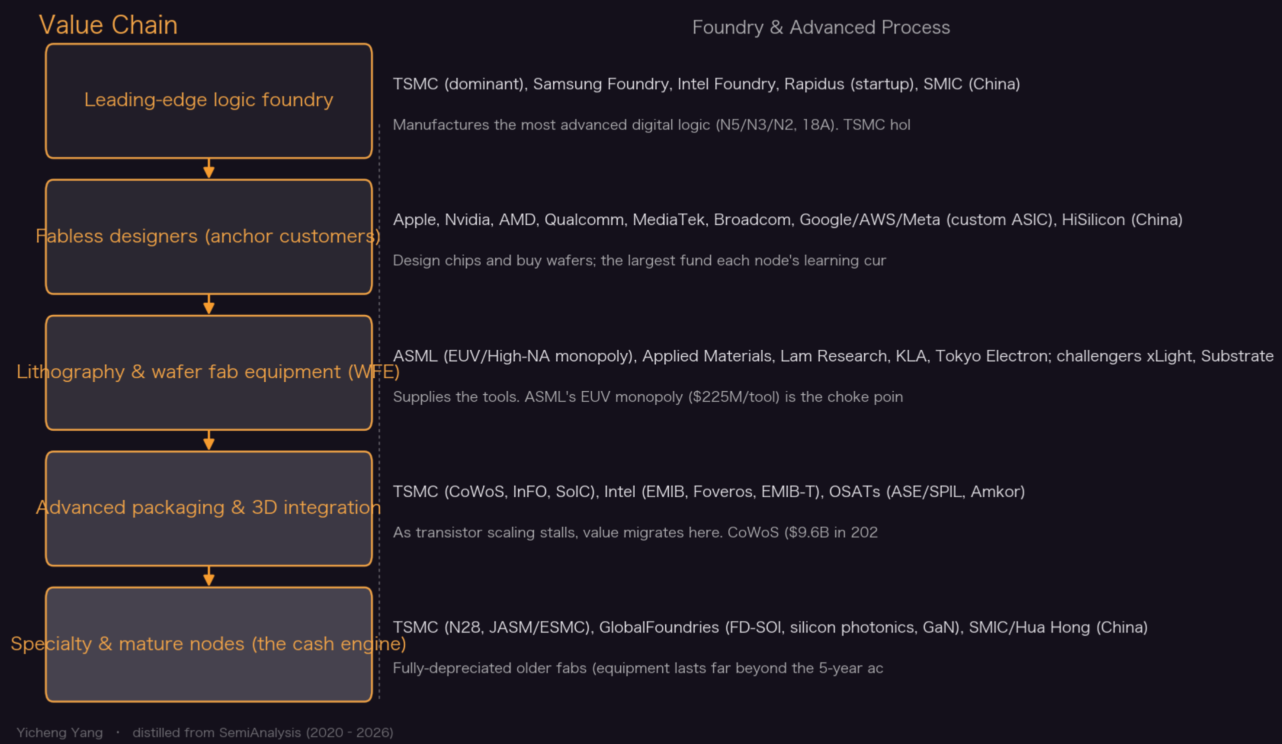
Cluster effect / 'one-hour ecosystem'

TSMC's structural moat: Hsinchu/Taichung/Tainan science parks put semiconductor firms, suppliers and talent within an hour, so anything needed arrives fast and yield problems are solved 24/7. Over 90% of TSMC capacity and 87% of staff are in Taiwan. This cluster cannot be cheaply replicated abroad —the core reason Morris Chang called the Arizona fabs 'a very expensive exercise in futility.'

STCO / 3D stacking (Huawei's answer to no-EUV)

System-Technology Co-Optimization: when you can't shrink transistors, shorten wires and stack active logic vertically. Huawei's '□ (tau) scaling law' and 'LogicFolding' split a logic block across face-to-face bonded dies so the bond behaves like an extra metal layer, recovering frequency and density. Density claimed 'per package footprint' is not comparable to a foundry's per-mm^2 number.

Value Chain



Key Theses

1. **One square mile in Taiwan carries ~3% of US GDP —the most extreme concentration risk in the global economy.**

   Over 90% of TSMC capacity and >90% of global advanced-node output sit in Taiwan. 8 of the world's 10 largest companies depend on TSMC (the 9th is TSMC), and more than a third of their combined $2T revenue rides on TSMC hardware. SemiAnalysis frames onshoring as a security imperative: the 2021 shortage of *mature* chips cost ~1% of US GDP ($240B); zeroing out advanced logic would be 'a 100-foot tsunami.'

   > *"That alone is 3% of U.S. GDP dependent on 1 square mile in southern Taiwan···The world economy would grind to a halt without TSMC."*
   > — TSMC Overseas Fabs –A Success? (2025-12-01)

2. **AI turned TSMC from a smartphone company into an HPC company —and Nvidia will out-consume Apple at N3 by 4Q27.**

   HPC revenue at TSMC rose from 36% (2020) to 58% (2025) while smartphone fell from 46% to 29%. Apple, TSMC's decade-long anchor, has been consistently >50% of node launches since 20nm (near-100% in some cases, e.g. N3) and funded the yield learning curve of every node; now Nvidia's AI cash makes it a second anchor. SemiAnalysis models Nvidia consuming more N3 wafers than Apple by 4Q27, and Apple's share of N2 dropping to 48% —the first time in a decade Apple isn't the dominant customer on a new node —because A16 is architected for HPC (backside power, GAA) and smartphones skip it.

> *"Our model shows Nvidia will consume more N3 wafers than Apple by 4Q27. Apple's share of N2 drops to 48%, the first time in a decade Apple is not the dominant customer on a new node."*
> — Apple-TSMC: The Partnership That Built Modern Semiconductors (2026-01-08)

3. **The 'great AI silicon shortage' is now front-end wafers: every accelerator converges on N3 in 2026, pushing utilization past 100%.**

   After CoWoS and power were the binding constraints, TSMC N3 logic capacity is now the choke point. Nvidia (Blackwell→Rubin), AMD (MI350/MI400), Google TPU v7, AWS Trainium3 and Meta MTIA all move to N3 in 2026. AI takes just under 60% of N3 output in 2026 and 86% by 2027, nearly squeezing out all smartphone/CPU wafers. Effective N3 utilization exceeds 100% in 2H26; TSMC is constrained by cleanroom space, so gains for one customer mean another loses allocation.

   > *"We model AI demand to be 86% of 2027 N3 wafer output nearly entirely squeezing out smartphone and CPU wafers."*
   > — The Great AI Silicon Shortage (2026-03-12)

4. **2nm-generation logic requires two simultaneous paradigm shifts —GAA transistors and backside power —the biggest transition in decades.**

   FinFET cannot scale further and SRAM shrink is dead, so leading-edge logic must adopt GAA and BSPDN in 2-3 years. BSPDN is the first interconnect innovation since aluminum→copper in 1997; it delivers ~15-20% power improvement and enables sub-6T cells. Foundries diverge on aggressiveness: Intel 18A ships PowerVia first (easier, less scaling); TSMC A16 and Samsung SF2Z dive into higher-risk direct backside contacts. This inflection is the rare opening that could force Samsung or Intel out —or let in Rapidus.

   > *"Leading-edge logic must adopt two new paradigms in the next 2-3 years: gate all around (GAA) and backside power delivery."*
   > — Clash of the Foundries: Gate All Around + Backside Power at 2nm (2024-10-01)

5. **SMIC N+3 reaches TSMC N6-class density without EUV —but density it cannot turn into competitive power or cost.**

   SemiAnalysis's STEEL lab teardown of Huawei's Kirin 9030 measured a 32.5nm minimum metal pitch on SMIC N+3, ~10% tighter than the 36nm shipping in Intel 18A Panther Lake, and 113.4 MTr/mm^2 Bohr density (just above N6's 107.7). But it's achieved via SAQP quadruple-patterning DUV and DTCO, so it costs far more and yields less. The chip performs like a 3-year-old Android flagship: Apple's efficiency core delivers 20% more integer performance at 1W vs Huawei's prime core at 4.5W.

   > *"SMIC N+3 reaches TSMC N6-class logic density, but it requires far more aggressive DUV multi-patterning, so it does not match N6 on process maturity or cost."*
   > — Is SMIC N+3's Metal Pitch Smaller than Intel 18A's? (2026-06-14)

6. **Apple built TSMC as much as TSMC built Apple —a $10B 'bet the company' gamble that reshaped computing.**

   In 2013 Morris Chang committed $10B to 20nm on Apple's promise; the A8 (2014) proved it. Apple's TSMC spend grew from ~$2B (2014) to $24B (2025) —12x in 12 years —going from 9% of TSMC revenue to a 25% peak, funding the yield learning curve of every node since 20nm. Intel's Otellini declined the same deal over margins: SemiAnalysis calls it 'the biggest misstep in the history of chip foundries.' Vertical integration paid off —Mac gross margin expanded ~11pp after dropping Intel.

> *"I bet the company, but I didn't think I would lose."*
> — Apple-TSMC: The Partnership That Built Modern Semiconductors (2026-01-08)

7. **Cost-per-transistor scaling has nearly stopped: N3 delivered the weakest scaling in 50+ years.**

   N3's total spend per wafer-start rose 38-55% vs N5 (N3E pricing ~35% above N5), with density up ~56% only in the best FinFlex config —an ~15% cost-per-transistor improvement, the weakest ever for a major node. Most designs get ~30% density, implying flat-to-negative cost scaling. N3B had ~25 EUV layers (nearly double N5); N3E cut that to 19 for cost. SRAM barely scaled (N3B 6T cell only 5% smaller vs the promised 20%). Result: an explosion of chiplets and advanced packaging.

   > *"This results in an ~15% cost per transistor improvement, the weakest ever scaling for a major process technology in 50+ years."*
   > — TSMC's 3nm Conundrum, Does It Even Make Sense? (2022-12-21)

8. **Intel Foundry has the process (18A on track) but not the cash —it needs 150k+ wafers/month and can't self-fund the fabs.**

   18A defect density is on-track and Intel will be first to market with GAA+BSPDN (PowerVia), a genuine bright spot. But Intel needs $25-30B capex per 10k wafers/week and 150k+ wafers/month to be competitive, at a time its core business is barely breakeven. TSMC's mature nodes throw off cash to fund the leading edge; Intel historically deprecated old nodes and has no such 'forever node' cash pool. It has ~$120B of capital to tap but may need more —'we are not sure where they can get this cash from.'

   > *"Building 150,000+ wafers a month of leading edge capacity requires a lot of investment···Today they cannot afford to build these fabs with their own cashflow."*
   > — Is Intel Back? Foundry & Product Resurgence Measured (2024-04-02)

9. **TSMC's overseas fabs are real but structurally costlier —Chang cites the 1996 WaferTech precedent, whose margins ran 20-25% below Taiwan.**

   Arizona, Japan (JASM) and Germany (ESMC) all yield wafers, but the cluster can't be replicated: 90%+ of TSMC capacity and 87% of staff remain in Taiwan; US permitting takes ~2x as long; the supply chain isn't there (a Linde gas outage caused a multi-million-dollar scrap event, cutting one quarter's AZ fab profit from $140M to $1.4M). The 1996 WaferTech venture ran 20-25% lower margin on identical products —a precedent Chang cites. Arizona remains a small single-digit share of leading-edge output and won't materially cut Taiwan dependence in the near term.

   > *"I think it will be a very expensive exercise in futility."*
   > — TSMC Overseas Fabs –A Success? (2025-12-01)

10. **The next lithography war could be won outside ASML: X-ray litho startup Substrate claims 50% cheaper wafers.**

    The industry is ripe for disruption (ASML admits its own hyper-NA roadmap 'may not be economically viable'). Substrate's X-ray litho tool claims single-patterning at 2nm/1nm, high-NA-class resolution, 12nm features, <1.6nm overlay —at a tool cost ~$40M vs High-NA EUV's $400M. SemiAnalysis's model is skeptical of the full 50% claim (its own analysis shows ~25% cost reduction) but even that is 'massive.' Combined with xLight's free-electron-laser source, these are the first credible threats to on-shore the litho monopoly —with China running parallel efforts.

> *"Substrate promises the same great taste but without $400M in calories."*
> — How to Kill 2 Monopolies with 1 Tool (2025-10-29)

11. **Samsung Foundry's decline is cultural, not just technical —it lied about yields and lost its two biggest customers.**

Samsung was technically first to GAA (SF3E, 2022) but only in a low-volume bitcoin/watch chip. Its 4LPE node yielded as low as ~20% (parametric), Exynos 2200 GPU clocks were cut from 1.69GHz to 1.29GHz, and Korean media reported the foundry lied about 5/4/3nm yields to customers and its own chairman. Qualcomm and Nvidia both fled to TSMC; both negotiated to pay per yielded die rather than per wafer. SemiAnalysis: TSMC N3's 30-40% yield problems don't mean Samsung can catch up —'you are sorely mistaken.'

> *"The foundry is even allegedly lying about the yields···lied to customers and the Samsung chairman on 5nm, 4nm, and 3nm yields."*
> — Samsung Electronics Cultural Issues Are Causing Disasters In Samsung Foundry (2022-04-17)

12. **China's chip choke point is shifting from one fab to an ecosystem —SMIC is being forced to license N+2/N+3 to peers.**

Export controls changed China's optimization problem rather than ending it. SMIC is licensing N+2/N+3 to HLMC/Hua Hong at government direction, and the process learning could feed Ascend AI accelerators, Alibaba's T-Head, and Cambricon (a likely ByteDance supplier). Peking University even built a prototype EDA tool for Huawei's LogicFolding. Once manufacturing knowledge spreads, sanctions aimed at SMIC alone lose bite. 'If domestic chips become good enough for phones, inference, networking···they can matter strategically without matching TSMC.'

> *"Sanctions aimed at SMIC alone become less effective once the manufacturing knowledge has spread to other fabs and design houses."*
> — Is SMIC N+3's Metal Pitch Smaller than Intel 18A's? (2026-06-14)

13. **Advanced packaging, not lithography, is now where value and constraint migrate —CoWoS is 2.5x InFO and stacking is the future.**

With cost-per-transistor scaling stalled, TSMC's packaging platforms carry the growth. CoWoS revenue hit $9.6B in 2025, 2.5x Apple's InFO ($3.5B+), fueled by Nvidia/AMD HBM. Apple (InFO-PoP) and Nvidia (CoWoS-L) don't compete for packaging lines today, but converge on 3D SoIC/hybrid bonding by AP6/AP7. WFE litho spend intensity plateaus after 3nm; share shifts to deposition, etch, metrology and packaging tools. SemiAnalysis: 'packaging is now a key avenue for driving compute scaling.'

> *"CoWoS revenue hit $9.6B in 2025, 2.5x InFO, fueled by Nvidia and AMD demand."*
> — Apple-TSMC: The Partnership That Built Modern Semiconductors (2026-01-08)

Article Deep-Dives

Is SMIC N+3's Metal Pitch Smaller than Intel 18A's? (STEEL teardown of Huawei Kirin 9030) (2026-06-14)

The first public report from SemiAnalysis's own Oregon teardown lab (STEEL), physically reverse-engineering Huawei's Kirin 9030 on SMIC N+3 against a MediaTek Helio G99 on TSMC N6. Headline finding: SMIC N+3 hits a 32.5nm minimum metal pitch —~10% tighter than Intel 18A Panther Lake's shipping 36nm M0 —and 113.4 MTr/mm^2 Bohr density, edging TSMC N6. But it's a cherry-picked metric: N+3 gets there via SAQP quadruple-patterning DUV (M0 below single-DUV resolution), COAG, single diffusion break and a 3:2 M1-to-gate ratio, all adding masks, overlay error and cost. Performance lags a 3-year-old flagship (Apple's efficiency core: +20% integer at 1W vs Huawei's prime at 4.5W). SMIC has scaling levers left (theoretical N+4 ~138 MTr/mm^2 = N5-class, N+5 via backside contacts ~164 = 18A-class) but each is harder and more expensive. Huawei's answer to no-EUV is the '□ scaling law' / LogicFolding —3D-stacked active logic targeting ~5GHz and '14A-equivalent' density by 2031, a packaging roadmap not a foundry one.

Apple-TSMC: The Partnership That Built Modern Semiconductors (2026-01-08)

A five-phase history and forward model of the most consequential partnership in chips. Built on SemiAnalysis's proprietary Foundry and Apple Wafer Demand models: Apple's TSMC spend grew 12x ($2B→$24B) in 12 years —consistently >50% of node launches since 20nm (near-100% in some cases, e.g. N3), funding the yield learning curve of every node. TSMC itself grew 9.4x ($13B→$122B revenue), capex 7x, gross margin +13.5pp to 59%+. Phases run from Intel's 2010 rejection (Otellini's 'biggest misstep in foundry history') through mutual lock-in (Samsung 3nm yields 30-40% vs TSMC 80%+; switching cost $2-5B) to today's 'diversified dependence,' where Nvidia becomes TSMC's second anchor and Apple's N2 share drops to 48%. Also maps Apple's 5 transformational acquisitions ($278M P.A. Semi → A4; Intel modem $1B), 8,000+ engineers across 15 design centers, and the packaging bifurcation (Apple InFO vs Nvidia CoWoS, converging at 3D SoIC by AP6/AP7). Apple's real diversification isn't leading-edge —it's PMICs, display drivers, CIS (a new Samsung Austin deal), and a possible Intel 18A-P base-M-series qualification.

Clash of the Foundries: Gate All Around + Backside Power at 2nm (2024-10-01)

The definitive technical primer on the 2nm-generation inflection. FinFET is exhausted and SRAM shrink is dead, so all four foundries must adopt GAA transistors and backside power delivery (BSPDN) in 2-3 years —the first interconnect paradigm shift since aluminum→copper (1997), worth ~15-20% power. Deep-dives the three BSPDN schemes: buried power rail (simplest but contaminates FEOL, won't reach HVM), Intel's PowerVia (easier, self-aligned, less scaling), and direct backside contacts (highest risk, ~25% cell shrink, needs <5nm post-bond overlay and ~50nm-pitch backside patterning). Roadmaps diverge: Intel 18A first to GAA+BSPDN (PowerVia) but at ~3nm-class density; TSMC N2 is GAA-only, A16 adds conservative backside contacts (7-10% density); Samsung SF2Z in 2027; Rapidus a heavily-subsidized long shot (IBM 2nm license, 25k wpm, no BSPDN, no signed volume customers). SRAM 'beating a dead horse' —the ~22% N2 gain is periphery, not bitcell.

The Great AI Silicon Shortage (2026-03-12)

The constraint has moved down the stack: from CoWoS packaging, to datacenter power, and now to front-end wafers themselves. Every AI accelerator family converges on TSMC N3 in 2026 (Nvidia Blackwell→Rubin, AMD MI350/MI400, Google TPU v7, AWS Trainium3, Meta MTIA) plus Vera CPUs, NVLink6 and Tomahawk6 switches. AI takes just under 60% of N3 output in 2026, rising to 86% in 2027, nearly squeezing out smartphone/CPU. Effective N3 utilization exceeds 100% in 2H26; TSMC is gated by cleanroom space, so one customer's gain is another's loss —TSMC acts as 'kingmaker,′ prioritizing high-ASP, multi-year-visibility AI over saturated mobile. Smartphones are the ′release valve′: reallocating 25% of smartphone N3 starts could yield ~0.7M more Rubins or ~1.5M more TPU v7s. Memory is the parallel shortage: HBM consumes ~3x commodity DRAM wafers per bit, and DDR margins have surged to rival HBM, removing suppliers′ incentive to add HBM capacity.

How to Kill 2 Monopolies with 1 Tool (Substrate X-ray lithography) (2025-10-29)

SemiAnalysis surfaces Substrate, a Bay Area startup building an X-ray lithography (XRL) tool that could break both ASML′s litho monopoly and TSMC′s foundry dominance. Claims: single-patterning at 2nm/1nm/beyond, high-NA-class resolution, 12nm features, <1.6nm overlay, 0.25nm CDU —at a tool cost ~$40M vs High-NA EUV′s $400M, promising 50%-cheaper wafers. SemiAnalysis is rigorously skeptical (its own model shows ~25%, not 50%) but concludes external reports say ′the litho tool is legit.′ If real, it opens process design flexibility, eliminates multi-patterning, and threatens a >$200B TSMC-share TAM by 2030. The article catalogs the monumental non-litho challenges (stochastic defects, secondary-electron blur, HAR etch, X-ray damage) and contrasts Substrate (novel exposure tool, own fab) with xLight (free-electron-laser source that plugs into existing EUV). Framed as a US strategic imperative —′seven American mega-caps depend almost completely on TSMC Taiwan for nearly $2T′ —with China running parallel FEL/synchrotron/XRL efforts.

TSMC Overseas Fabs –A Success? (Arizona / Japan / Germany economics) (2025-12-01)

A candid audit of TSMC′s three overseas ventures and why the Taiwan cluster is so hard to replicate. Over 90% of capacity and 87% of staff remain in Taiwan; the Hsinchu/Taichung/Tainan ′one-hour ecosystem′ —TSMC can obtain anything it needs within an hour —is the structural moat. Arizona has ramped —early yield reportedly ~4pts above comparable Taiwan fabs —but on a mature 4nm process already optimized for 2+ years, with an easy high-yield product mix (AMD chiplets, Apple SoCs ~100mm^2 vs Nvidia ~800mm^2 reticle GPUs in Tainan). The supply chain isn′t there: a Linde gas outage caused a scrap event that cut quarterly AZ profit from $140M to $1.4M; US permitting takes ~2x Taiwan′s. The 1996 WaferTech precedent ran 20-25% lower margin (′a nightmare fulfilled,′ per Chang). JASM (Japan, 28/16nm, Sony) and ESMC (Germany, 28/16nm, Bosch) are legacy-focused; a UAE fab was floated then killed over sovereign-control and IP-leak concerns. TSMC′s $165B US commitment may reach 12 Arizona phases by the 2030s —the world′s largest advanced fab —if the cluster can be built.

Key Data

TSMC share of global advanced-node capacity	>90%	Over 90% of TSMC capacity and 87% of staff are in Taiwan; supports >60% of world semiconductor output.
TSMC HPC vs smartphone revenue mix (2020→2025)	HPC 36%→58%; smartphone 46%→29%	The AI platform shift; Nvidia modeled to out-consume Apple at N3 by 4Q27.
AI share of TSMC N3 wafer output (2026→2027)	~60% → 86%	Effective N3 utilization exceeds 100% in 2H26; smartphone/CPU squeezed out.
Apple annual spend at TSMC (2014→2025)	~$2B → $24B (12x)	Apple went from 9% of TSMC revenue to a 25% peak; funded yield learning on every node since 20nm.

SMIC N+3 minimum metal pitch vs Intel 18A	32.5nm vs 36nm (shipping)	~10% tighter M0 than 18A Panther Lake, via SAQP DUV; 113.4 MTr/mm^2 Bohr density (>N6's 107.7).
N3 cost-per-transistor improvement vs N5	~15% (best case)	Weakest scaling for a major node in 50+ years; wafer spend up 38-55%, most designs get ~30% density.
N3B EUV layers vs N3E	~25 → 19	N3B nearly doubled N5's EUV count and yielded poorly; N3E relaxed pitches to cut cost, is the volume node.
SRAM bitcell size, N5 = N3E = Intel 18A (HD)	0.021 μm^2	SRAM scaling is dead: N3E's HD cell identical to N5's; 18A HD cell also 0.0210 μm^2 on par with N5/N3E.
Backside power delivery benefit	~15-20% power; up to ~25% cell shrink (direct backside contacts)	First interconnect paradigm shift since aluminum→copper (1997); Intel 18A PowerVia first to HVM.
EUV tool economics	$225M/tool → >$650M wafers/yr	ASML monopoly; High-NA EUV EXE:5000 ~$400M. Substrate X-ray litho targets ~$40M/tool.
TSMC 2022 capex 'gauntlet'	$40-44B (70-80% on 2/3/5/7nm)	Set a ceiling on Intel/Samsung's $25B+ plans; 10% on masks/advanced packaging (CoWoS, SoIC, InFO).
TSMC Arizona quarterly profit after Linde gas outage	$140M → $1.4M	Impure gas caused a multi-million-dollar scrap event; illustrates immature US supply-chain cluster.
TSMC N7 utilization trough (2023 downturn)	<60% (from 100%)	Foundry cyclicality: N7 below 70% in Q1'23, below 60% in Q2; even best-in-class N5 fell to ~88%.
HBM wafer intensity vs commodity DRAM	~3x → ~4x at HBM4 (2026), even larger at HBM4E	HBM crowds out commodity DRAM per bit; the memory shortage runs parallel to the N3 logic shortage.
Intel leading-edge fab capex requirement	$25-30B per 10k wafers/week; needs 150k+/month	TSMC quoted ~$42B per 10k wafers/week for Arizona 3nm; Intel core business ~breakeven, can't self-fund.
TSMC advanced packaging CoWoS revenue (2025)	$9.6B (2.5x InFO)	CoWoS grew from $0.6B (2018) to $8.4-9.6B (2025), 14x; advanced packaging now >10% of TSMC revenue.

Huawei prime-core frequency roadmap (LogicFolding)	2.75 GHz (2025) → ~5 GHz (2031)	Via 3D active-logic stacking, not planar scaling; claims ′14A-equivalent′ density per package footprint by 2031.

Investment Angle

Winners

TSMC (TSM)	The dominant leading-edge foundry (>90% advanced-node share), sole viable N3/N2 supplier during the AI silicon shortage, kingmaker over allocation with pricing power. Two anchor tenants now (Apple + Nvidia). Cluster moat and cash-generating mature nodes fund the roadmap. Concentration risk in Taiwan is the flip side.
ASML	EUV/High-NA monopoly; every leading-edge wafer needs its tools ($225M each, >$650M wafers/yr). Litho spend intensity plateaus post-3nm, but the installed base and High-NA transition sustain it. Tail risk: a working X-ray litho (Substrate) or FEL source (xLight) could eventually erode the monopoly.
Deposition/etch/metrology WFE (Applied Materials, Lam, KLA, TEL)	As litho intensity plateaus after 3nm, share shifts to deposition, etch, metrology and packaging tools —exactly what GAA, backside power (CMP, fusion bonding, TSV etch) and 3D stacking demand most. Beneficiaries of the paradigm shift regardless of which foundry wins.
Intel Foundry (INTC) — high-risk turnaround	18A defect density on-track and first to market with GAA+BSPDN (PowerVia); US-government backing and onshoring tailwind (any AI-capacity diversification toward Intel earns ′brownie points′). But the capital wall (needs 150k+ wafers/month, can′t self-fund) and lack of an anchor tenant make it a speculative bet, not a sure thing.
GlobalFoundries (GFS) —specialty niche	Exited Moore′s Law but leads in FD-SOI (22FDX/12FDX), RF/5G-6G, GaN-on-Si and silicon photonics —differentiated ′leading edge′ insulated from the leading-edge capex arms race, with defense/US-industry partnerships. A structurally different, less capital-intensive profile.

Bottlenecks

- TSMC N3 front-end wafer capacity —the single biggest constraint on AI accelerator output; utilization >100% in 2H26, gated by cleanroom construction lead time.
- EUV lithography tools —ASML production isn′t scaling measurably; a hard ceiling on how fast any foundry can add leading-edge capacity.
- HBM and advanced packaging (CoWoS/SoIC) —the co-requisite for AI accelerators; HBM eats ~3x DRAM wafers per bit, packaging capacity moves in lockstep with N3.

- Taiwan engineering talent —only so many Taiwanese engineers exist to spin up foreign fabs, structurally capping the speed of onshoring.

Risks

- Taiwan concentration / cross-strait conflict —~3% of US GDP and 8 of the 10 largest firms depend on one square mile; the tail risk that dwarfs all others.
- AI capex air-pocket —the N3 shortage is built on multi-year AI compute commitments; a demand reset would leave TSMC with the industry's most expensive idle capacity (see 2023: N7 utilization <60%).
- Overseas-fab margin drag —Arizona/Japan/Germany are structurally costlier than Taiwan (the 1996 WaferTech precedent ran margins 20-25% lower); politically mandated capacity that dilutes returns.
- China ecosystem diffusion —SMIC licensing N+2/N+3 to Hua Hong/HLMC spreads the choke point; domestic 'good-enough' chips for inference/networking matter strategically even without matching TSMC.
- Lithography disruption —if Substrate's X-ray litho or xLight's FEL source industrializes, ASML's and TSMC's moats erode; low near-term probability, end-of-decade timeline.

Catalysts

- Intel 18A / 18A-P HVM ramp (late 2026) and first external anchor customer —the swing factor for whether Intel Foundry survives as a #2 option.
- TSMC A16 (Super Power Rail) and N2 ramp —first GAA+backside-power volume for the leading pack; validates the 2nm paradigm.
- Smartphone demand cuts freeing N3 for AI —a 25% reallocation could yield ~0.7M more Rubins / ~1.5M more TPU v7s.
- Samsung/Intel design wins (Tesla AI5/AI6, Nvidia supply-chain entry) —foundry diversification under capacity scarcity and US political pressure.
- Substrate XRL tape-outs (as soon as 2028) —milestones that would 'silence the skeptics' on a credible ASML/TSMC disruptor.

Open Questions

- Can any second source (Intel 18A, Samsung SF2, Rapidus) become a real anchor-fundable leading-edge foundry, or does the capital + anchor-tenant + cluster trap keep TSMC a de facto monopoly through the decade?
- How far can SMIC push DUV-only scaling before it hits a wall —is theoretical N+4 (~N5-class) / N+5 (backside contacts, ~18A-class) achievable at any usable yield and cost without EUV?
- Which backside-power scheme wins in volume —Intel's easier PowerVia or the higher-reward direct backside contacts of TSMC/Samsung —and does the post-bond overlay/thinning yield hold at scale?
- Is the N3 shortage a durable structural constraint or an AI-capex bubble that reverses into the industry's most expensive idle capacity, as N7 did in 2023?
- Does a lithography disruptor (Substrate X-ray, xLight FEL) actually industrialize —and if so, does it on-shore the moat to the US or get replicated by China's parallel FEL/synchrotron efforts?

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Chapter 3

Memory & HBM

DRAM stopped scaling a decade ago; AI turned that broken physics into the highest-margin product memory has ever sold. HBM is where the money, the yield pain, and the geopolitics all collide.

Overview

Memory is the quiet chokepoint of the AI buildout. SemiAnalysis's throughline across this bucket: DRAM bit-density scaling collapsed from 2x-every-18-months to 2x-per-decade, so the industry can no longer cushion price with cost-down, leaving a consolidated 3-player oligopoly (SK Hynix, Samsung, Micron) plus rising CXMT that swings violently on supply. HBM is the release valve —vertically stacked DRAM with a 1,024-bit (soon 2,048-bit) bus, glued to the GPU by CoWoS, now ~60%+ of Blackwell's manufacturing cost and the single largest BOM increase generation-over-generation. It is also the hardest DRAM ever made: front-end TSV yield and 8-to-16-high stacking losses run far below normal DRAM, which is exactly why it is margin-accretive. The frontier is HBM4, which splits the base die onto a foundry logic node (TSMC N12 for Hynix, SF4 for Samsung, internal for Micron) and opens the door to custom base dies for OpenAI/Nvidia/AMD/Amazon. Beneath HBM sits the slower war over how to keep DRAM alive at all —4F2 vertical-channel cells, hybrid-bonded cell-on-peri, and eventually 3D DRAM (~2030-2035) —plus a once-in-four-decades shortage that has DRAM pricing set to double again in 2026. NAND runs a parallel story: layer-count racing (332L), 5-bits-per-cell, and China's YMTC/CXMT breaking the cartel from below.

Core Knowledge

HBM (High Bandwidth Memory)

A 3DIC stack of 8-16 DRAM dies on a logic base die, connected by 1,000+ TSV wires and a 1,024-bit bus (2,048-bit in HBM4). It trades a warranted 3x+ price premium over DDR5 for the bandwidth AI accelerators need; every leading GenAI training/inference chip uses it. Must sit at the GPU 'shoreline' via a 2.5D interposer (CoWoS) because a PCB can't route the density.

The Memory Wall

DRAM density doubled every 18 months in its golden age (100x/decade); in the last decade it managed only ~2x. Compute keeps racing ahead while memory bandwidth crawls, so most LLM inference is memory-bandwidth-bound, not compute-bound —the GPU waits on HBM. 'Moore's Law is dead for DRAM' —and it died a decade ago with zero fanfare.

TSV & stacking yield

Through-Silicon Vias carry power/signal up the HBM stack; forming them (etch/plate/grind/bond) is the incremental step that converts DDR wafers to HBM, so HBM capacity is quoted in 'TSV capacity.' Yield compounds badly: at 99% per-layer bond yield an 8-hi stack yields 92%, a 12-hi only 87%. Front-end (power-TSV/PDN) is the bigger yield problem, not the stack.

MR-MUF vs TC-NCF (packaging)

SK Hynix's Mass Reflow-Molded Underfill uses a single over-mold and batch reflow (co-developed with Namics), giving higher productivity and better thermal dissipation than the Non-Conductive Film (TC-NCF) that Micron and Samsung use, where each layer needs a full thermocompression bond. This is a core reason Hynix leads HBM3E while Samsung struggles to yield.

HBM4 custom base die

HBM4's big architectural break: the core DRAM dies stay on a DRAM node, but the base die moves to a foundry logic node (FinFET), enabling higher speed, lower voltage, and customer-specific customization. Hynix uses TSMC N12; Samsung uses its own SF4; Micron uses an internal CMOS base. Opens memory-controller offload, repeater PHYs and per-customer base dies for OpenAI/Nvidia/AMD.

4F2 & Vertical Channel Transistor (VCT)

The next DRAM cell shrink: 4F2 is the theoretical minimum single-bit cell (a 6F2->4F2 move gives ~30% density in theory) and requires a vertical-channel transistor stacked under the capacitor. Paired with hybrid-bonded cell-on-peripheral (COP/PUC), core circuitry hides under the array —Samsung cut core-circuit area from 17.0% to 2.7%. China's CXMT already ships a 4F2 18nm DRAM.

3D DRAM

The endgame when vertical capacitors can't shrink further: lay capacitors horizontally and stack many atop each other (like 3D NAND did). Micron calls timing 'the quintessential question'; IMEC/Micron roadmaps put insertion ~2030-2035. Tellingly, none of the big-3 present serious 3D DRAM papers — it's the race that will reshuffle market share.

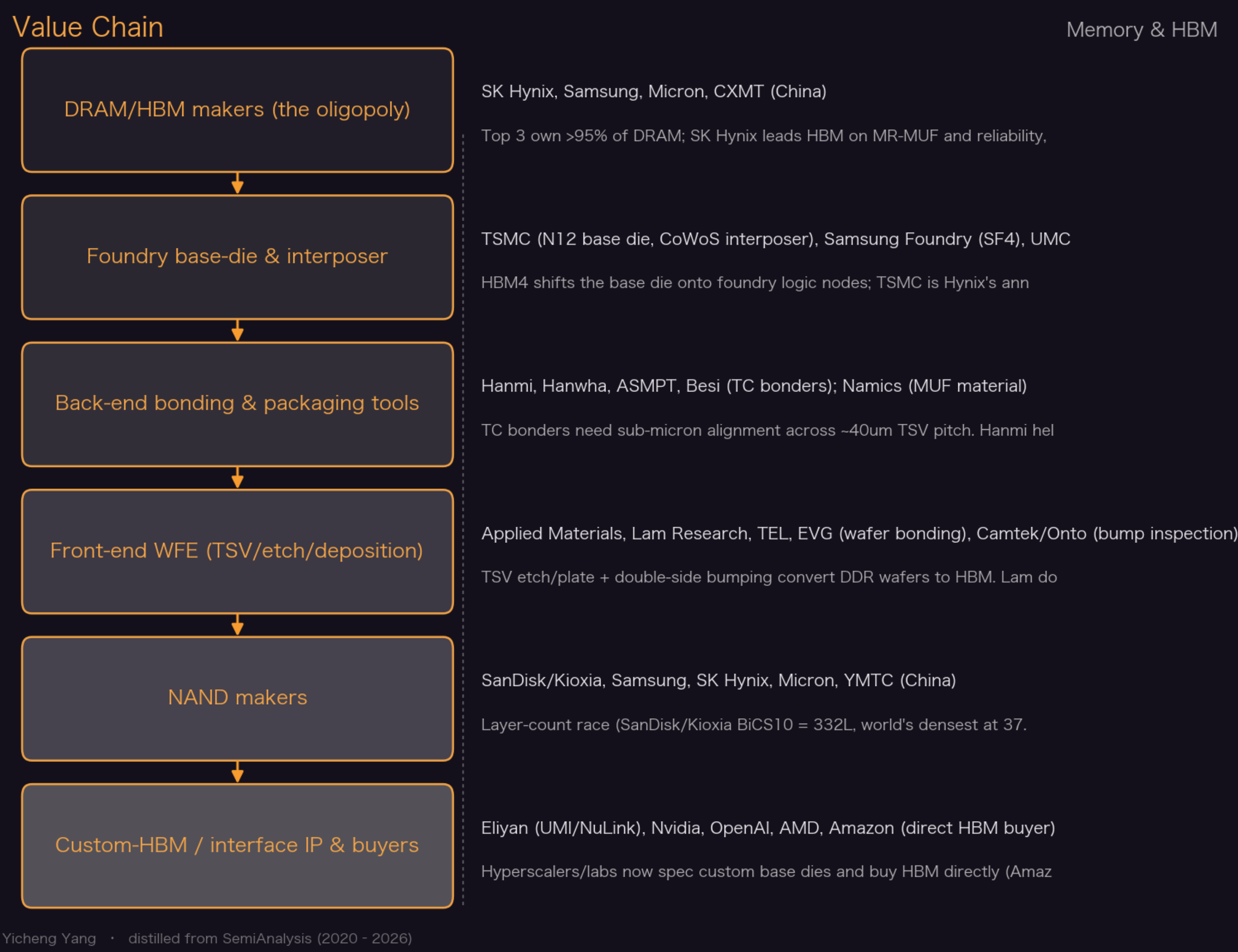
LPDDR as 2nd-tier memory / KVCache offload

HBM is scarce, so systems tier memory: hot KV/weights in HBM, warm data in cheaper LPDDR/DDR, cold in NVMe (Nvidia's Dynamo KVCache Manager). LPDDR uses ~10x less energy/bit than DDR5 DIMMs (Grace ships 480GB usable). Crucially SemiAnalysis argues these don't cannibalize HBM — like L1/L2 cache vs DRAM, they're complements, not substitutes.

Emerging memory (FeRAM/MRAM)

Non-volatile alternatives researched for decades but none has 'emerged' to high volume. Micron's FeRAM matched D1b DRAM density at IEDM 2023 but is too costly (exotic materials). MRAM (magnetic tunnel junction) reached 0.49 Gb/mm2 —denser than D1b DRAM —but ships in MB, not GB. TSMC's N16 STT-MRAM is real, but only as embedded NVM for automotive/edge, not a DRAM replacement.

Value Chain



Key Theses

1. **DRAM's scaling death is the whole story: density went from 2x every 18 months to 2x per decade.**

   Moore's Law for DRAM died a decade ago with zero fanfare. Because DRAM is commoditized and has almost no cost-down to cushion price, it can't behave like logic —the only way to add supply is to build new fabs, so wild price swings and huge capex leave a top-3 oligopoly owning >95% of the market. This is the structural reason memory is so cyclical and why AI-driven scarcity is so violent.

   > *"The world increasingly questions the death of Moore's Law, but the tragedy is that it already died over a decade ago with 0 fanfare or headlines."*
   > — The Memory Wall: Past, Present, and Future of DRAM (2024-09-03)

2. **HBM is now the majority of an AI GPU's manufacturing cost —~60%+ on Blackwell.**

   For the H100, 50%+ of manufacturing cost is HBM; on Blackwell it grows past 60%. Nvidia's roadmap scales HBM from the A100's 80GB HBM2E to 1,024GB HBM4E on Rubin Ultra, and from

Ampere to Blackwell Ultra the single biggest absolute and relative BOM increase is HBM —a direct wealth transfer to memory vendors, primarily SK Hynix. HBM3E costs 3x+ per GB versus DDR5 and customers pay it because no other DRAM makes a competitive accelerator.

> *"For the H100, ~50%+ of the cost of manufacturing is attributed to HBM and with Blackwell, this grows to ~60%+."*
> — The Memory Wall: Past, Present, and Future of DRAM (2024-09-03)

3. **The memory market has entered a 'once-in-four-decades shortage'; DRAM pricing is set to double again in 2026.**

   SemiAnalysis first flagged AI's insatiable reasoning/agentic memory demand in late 2024. The shortage now spans not just HBM but DDR5, DDR4 and even DDR3, with the firm 'even more confident' pricing could exceed expectations by year-end. Even factoring in CXMT and peer wafer adds at high-90s% utilization, they model DRAM undersupplied by a high-single-digit % in 2026, widening to low-to-mid-teens bit undersupply in 2027, potentially through 2028 —because fab construction timelines are simply too long to react.

   > *"we described the memory market as entering a 'once-in-four-decades shortage,' we believe DRAM pricing remains on track to double again this year driven by sustained supply-demand imbalance"*
   > — China's CXMT Is Set to Challenge DRAM Incumbents (2026-06-23)

4. **HBM's yield pain —front-end TSV/PDN, not just stacking —is precisely why it's margin-accretive.**

   All makers run absolute HBM yields well below their conventional DRAM, so it's a game of relative yields and end economics. Yield compounds with layers (99% per-layer -> 92% at 8-hi, 87% at 12-hi). The bigger problem is front-end: delivering power up the stack via power-TSVs. Hynix's all-around power TSVs cut IR drop up to 75% for VPP; Micron's TSV/PDN focus lets it claim 30% lower power. Because pricing more than covers yield loss, HBM is the most valuable, highest-margin product the memory industry has ever had.

   > *"As data from hyperscalers have shown, HBM failures are the number one cause of GPU failures, which happen more frequently than other chips in the data center."*
   > — Scaling the Memory Wall: The Rise and Roadmap of HBM (2025-08-12)

5. **HBM4 is a tectonic shift: the base die moves to a foundry logic node, and Samsung has clawed back into contention.**

   HBM4 splits process nodes —DRAM node for core dies, advanced logic node for the base die. At ISSCC 2026 Samsung showed a 36GB 12-hi HBM4 at 3.3 TB/s with 2,048 IO on 1c DRAM + an SF4 logic base, hitting 13 Gb/s/pin (>2x the JEDEC HBM4 spec of 6.4 Gb/s) while dropping VDDQ 32% from 1.1V to 0.75V. Hynix uses cheaper TSMC N12, Micron an internal CMOS base —both lower-cost than Samsung's SF4. Samsung's 1c yields ran ~50% in 2025, a margin risk, but its technology gap to Hynix has meaningfully closed.

   > *"Samsung's paper claimed that the ABB and the 4x higher TSV count allow their HBM4 to achieve operating speeds up to 13 Gb/s per pin."*
   > — ISSCC 2026: NVIDIA & Broadcom CPO, HBM4 & LPDDR6... (2026-04-15)

6. **The DRAM interface is grotesquely wasteful —95-99% of read/write energy is spent in the interface, not the cells.**

   DRAM is a ′dumb′ state machine with no on-chip control logic; commands funnel through one ancient half-duplex interface. DDR5 DIMMs burn >99% of read/write energy in the host controller/interface; even HBM is ~95% interface, 5% cell. Worse, an HBM die′s 256-bit bus taps only ~1/16 of the banks′ true ~4 TB/s potential. This is the opening for compute-in-memory and modernized base-die PHYs (UCIe: ~11 Tbps/mm, ~12x HBM3E; energy/bit from 2pJ to 0.25pJ). ′Someone is going to win big in the IP wars.′

   > *"DDR5 DIMMs, the most common on servers, expend more than 99% of read or write energy in the host controller and interface."*
   > — The Memory Wall: Past, Present, and Future of DRAM (2024-09-03)

7. **CXMT′s explosive earnings are a cycle story, not a competitiveness story —its cost-per-bit is still 30%+ worse.**

   CXMT FY25 revenue rose 156% to ~$8.6B, first-ever net profit ~$1B; 1Q26 revenue $7.3B (~700% YoY) at ~70% op-margin. But its 1Q26 bit shipments grew only 11% while ASP rose ~57% —the earnings are ASP, not share. On DDR5, CXMT′s cost-per-bit is still >30% higher than the big-3; ~38% gross margin is pricing, not product. The old ′Chinese memory will flood and crush pricing′ fear is, this cycle, inaccurate —CXMT ASP sits only 5-10% below the leaders.

   > *"such significant upside to CXMT′s earnings is clearly driven more by the cycle itself than company′s technology or market positioning."*
   > — China′s CXMT Is Set to Challenge DRAM Incumbents (2026-06-23)

8. **CXMT is deliberately choosing commodity DRAM over HBM —because commodity DRAM currently earns more.**

   ~99% of CXMT′s 2025 revenue is DDR/LPDDR; only ~5 of ~265 kwspm went to HBM in 2025. Its HBM3 8-hi is modeled at ~35% front-end and ~70% back-end yield = ~25% overall, so HBM would be low-margin and consume scarce DRAM wafers that yield 3x more bits as commodity. So prioritizing commodity is rational —but Beijing′s self-sufficiency push conflicts with that, forcing CXMT′s HBM wafer capacity up to a modeled 55/100 kwspm in 2027/2028 (1%->12% of global HBM wafers). The IPO prospectus discloses NO dedicated HBM project.

   > *"The commodity DRAM currently offers materially higher margins than CXMT′s HBM products, while also delivering more than 3x the bits per wafer on a like-for-like basis."*
   > — China′s CXMT Is Set to Challenge DRAM Incumbents (2026-06-23)

9. **Hybrid bonding for HBM keeps slipping; JEDEC chose to just make the stack taller (720um->775um) instead.**

   Hybrid bonding is bump-less, freeing room for more layers, but D2W HB yield for even 2 layers is hard/expensive —imagine 16. HBM3/3E hit the 12-hi limit inside the 720um JEDEC cube; the two ways up are bump-less or taller. In a blow to HB, JEDEC relaxed height to 775um. HB for HBM keeps sliding from HBM4 to ′4E′; Hynix and Micron have gone quiet, only Samsung loudly promotes it —′typical for Samsung which often promotes the most aggressive technology... only to expectedly fail on execution.′

> *"All initial HBM4 will not use hybrid bonding, and we expect that to remain true for much longer than most would hope."*
> — The Memory Wall: Past, Present, and Future of DRAM (2024-09-03)

10. **'Memory-Parkinson': every HBM capacity bump is instantly eaten by bigger models, so memory is always the next bottleneck.**

    In inference, weights live permanently in HBM alongside a KVCache that grows every token; decode repeatedly reads both, so most LLM inference is bandwidth-bound. As reasoning models 'think' for tens of minutes (Deep Research) with 100k+ token contexts, capacity pressure explodes and serving at low batch sizes hurts economics. Each capacity jump (80GB/3.35TB/s H100 -> 192GB/8TB/s GB200) just resets the baseline for 'reasonable' model size. RL/inference is now the main driver of progress, permanently structural for memory demand.

    > *"as AI chips get more HBM, developers immediately build larger models to fill it, so memory is always the next bottleneck."*
    > — Scaling the Memory Wall: The Rise and Roadmap of HBM (2025-08-12)

11. **SRAM scaling is dead too —MediaTek's logic-based xBIT is a workaround, not a cure.**

    From N5 to N2, logic area fell 40% but 6T-HC SRAM bitcells shrank only 2% (8T-HC 18%); N3E's HD bitcell even regressed to N5 density. SRAM is ~120 F2 —20x less dense than DRAM's 6F2. MediaTek's 10-transistor xBIT (balanced 4/6 NMOS/PMOS) gets 22-63% higher density and 30%+ lower power vs a standard 8T cell, using pure logic rules. It shows how hard on-die cache scaling has become —reinforcing the case for offloading capacity to HBM/LPDDR tiers instead.

    > *"Despite logic area decreasing by 40% from N5 to N2, 8-transistor high-current SRAM bitcells have only decreased in area by 18%. 6-transistor high-current (6T-HC) bitcells are even worse, only decreasing by 2%."*
    > — ISSCC 2026: NVIDIA & Broadcom CPO, HBM4 & LPDDR6... (2026-04-15)

12. **NAND is a layer-count and bits-per-cell race —and SK Hynix is quietly losing the density war.**

    SanDisk/Kioxia's BiCS10 (332L, 3 decks) is the world's densest at 37.6 Gb/mm2 QLC, dethroning Hynix's 321L V9 (28.8 Gb/mm2 QLC / 21 Gb/mm2 TLC —30% behind). Micron matches Hynix's density with only 2 decks (lower cost). Since AI has choked cleanroom space, makers can only upgrade existing lines, so denser processes = more supply per constrained wafer. SK Hynix's exotic answers (Mo wordlines, 5-bits-per-cell multi-site cell) are R&D-stage, not yet cost-effective.

    > *"SanDisk and Kioxia demonstrated their BiCS10 NAND, with 332 layers and 3 decks. This is the highest reported NAND bit density, at 37.6 Gb/mm2, dethroning the previous champion, SK Hynix's 321L V9."*
    > — ISSCC 2026: NVIDIA & Broadcom CPO, HBM4 & LPDDR6... (2026-04-15)

13. **The real AI-memory bottleneck isn't the DRAM cell —it's CoWoS advanced packaging co-locating HBM and logic.**

    HBM3E needs 1,000+ wires between XPU and memory —undoable on PCB/substrate, so a silicon interposer (CoWoS-S) is mandatory. That capacity, not GPU dies, gated 2023-24 supply. The flow (TSV interposer -> chip-on-wafer flip-chip reflow -> wafer-on-substrate) spans ~28 upstream vendors; SemiAnalysis argues the market misprices which of them actually win. Because HBM is limited to the SOC's two shoreline edges (the other two are for off-package I/O), 'shoreline area' becomes a first-order design constraint.

> *"the primary limiting factor is the CoWoS advanced packaging to put the 5nm ASICs and HBM together."*
> — AI Expansion - Supply Chain Analysis For CoWoS And HBM (2023-07-26)

14. **China's HBM will come via custom, non-JEDEC designs —and via reclaimed HBM smuggled out of GPU packages.**

    US export controls (Dec 2024) ban raw HBM stacks into China, but chips containing HBM can ship if under FLOPS limits —so banned HBM is reexported (via CoAsia/Faraday/SPIL) and desoldered/reclaimed from GPU packages. Huawei runs its own HBM affiliates (XMC fab + SJSemi packaging, both entity-listed). Critically, Huawei and CXMT will use custom HBM that bypasses slow JEDEC standards/PHYs, letting them close the bandwidth gap without matching the incumbents' interface roadmap.

    > *"Huawei and CXMT will have custom HBM that is not based on the slow JEDEC standards and phys, so it will be able to close the bandwidth disadvantage."*
    > — China's CXMT Is Set to Challenge DRAM Incumbents (2026-06-23)

Article Deep-Dives

Scaling the Memory Wall: The Rise and Roadmap of HBM (2025-08-12)

The definitive HBM primer and roadmap. Explains why HBM's wide bus + vertical stack demands an interposer (1,000+ wires), why front-end TSV/PDN yield (not just stacking) is the real problem, and why it's still margin-accretive. Details the SK Hynix-Hanmi TC-bonder monopoly fight that nearly halted HBM shipments in April 2025, the shoreline-area constraint, KVCache offload to LPDDR/NVMe tiers, and the 'memory-Parkinson' dynamic. Covers HBM4's custom base die, JEDEC's 720->775um height relaxation (a blow to hybrid bonding), and China's CXMT/Huawei domestic HBM push. Nvidia still commands the lion's share of HBM demand through 2027 (Rubin Ultra = 1TB/GPU).

The Memory Wall: Past, Present, and Future of DRAM (2024-09-03)

The bucket's intellectual backbone. Traces DRAM from the 1T1C cell + sense-amp inventions to today's stagnation (capacitors at ~100:1 aspect ratio, ~40,000 electrons/cell, sense-amp sensing limits). Lays out the full scaling menu: near-term 4F2 + vertical-channel transistors (~30% theoretical density), medium-term compute-in-memory (banks have ~4 TB/s potential but interfaces tap ~1/16), custom HBM4 base dies, emerging FeRAM/MRAM (denser than DRAM but ship in MB), and long-term 3D DRAM. Key insight: DRAM's 'dumb' half-duplex interface wastes 95-99% of read/write energy —the opening for a new PHY/IP war.

China's CXMT Is Set to Challenge DRAM Incumbents (2026-06-23)

The definitive CXMT deep dive ahead of its STAR Market IPO (likely China's largest semiconductor listing). Traces its origins: licensed ~7,000 Qimonda patents + 2.8TB of docs + poached Qimonda/Korean/Taiwanese/US talent (Kuesters, Ping) + patient Hefei state-VC that absorbed ~RMB36.65B of accumulated losses over a decade. Verdict: explosive 1Q26 earnings ($7.3B, ~70% op-margin) are cycle/ASP-driven, not competitiveness —DDR5 cost/bit still >30% worse, HBM barely 5 kwspm and ~25% yield. Debunks the 'China floods and crushes pricing' fear this cycle. Flags the IPO's consolidation accounting (74% minority interest) and lack of any disclosed HBM project.

ISSCC 2026: NVIDIA & Broadcom CPO, HBM4 & LPDDR6, TSMC Active LSI, Logic-Based SRAM, UCIe-S and More (2026-04-15)

The frontier scorecard. Samsung's HBM4 (36GB 12-hi, 3.3 TB/s, 13 Gb/s/pin, VDDQ 0.75V) confirms it has closed the gap on Hynix technically, though it still trails on reliability and 1c yield (~50%). Samsung & SK Hynix LPDDR6 (up to 14.4 Gb/s, dual sub-channel efficiency mode), SK Hynix 1c GDDR7 (48 Gb/s). Samsung 4F2 COP DRAM cuts core-circuit area 17.0%->2.7% via hybrid-bonded cell-on-peri (but suffers floating-body leakage). SanDisk/Kioxia BiCS10 (332L) takes NAND density crown. MediaTek's logic-based xBIT SRAM cell and TSMC N16 MRAM round out the 'scaling is dead, find workarounds' theme.

Interconnects Beyond Copper, 1,000 CFETs, SK Hynix Next-Gen NAND, 2D Materials, and More (IEDM) (2026-01-13)

The NAND scaling deep dive amid the supercycle. With no cleanroom space for new capacity, makers can only densify existing lines: SK Hynix's 321L V9 packs +44% memory/wafer over V8 by adding a 3rd deck (etch limit ~120 layers/deck). But its 21 Gb/mm2 TLC trails Micron (2 decks, cheaper) and is dwarfed by BiCS10 (29 Gb/mm2 TLC, 37+ QLC). Samsung switches NAND wordlines W->Mo for performance and skips 3xx layers (286L->43xL). The standout: SK Hynix's multi-site cell splitting the channel to enable 5-bits-per-cell —R&D only, not yet cost-effective.

Key Data

DRAM density scaling	~2x in the last decade (vs 2x every 18 months in the golden age = ~100x/decade)	The single fact behind memory's structural scarcity and cyclicality.
HBM share of AI GPU manufacturing cost	H100 ~50%+ -> Blackwell ~60%+	Biggest absolute & relative BOM increase Ampere->Blackwell Ultra; a wealth transfer to SK Hynix.
HBM price premium vs DDR5	3x+ per GB	Customers pay it because no other DRAM makes a competitive accelerator.
HBM stack-yield compounding	99%/layer -> 92% at 8-hi, 87% at 12-hi	Illustrative; real yield degrades faster as non-critical defects accumulate.
Samsung HBM4 (ISSCC 2026)	36GB 12-hi, 3.3 TB/s, 2,048 IO, 13 Gb/s/pin, VDDQ 0.75V	>2x the JEDEC HBM4 spec (6.4 Gb/s, ~2 TB/s); 1c DRAM + SF4 base die.
HBM4 base-die foundry split	Hynix = TSMC N12; Samsung = SF4; Micron = internal CMOS	SF4 is the most expensive; Hynix/Micron chose lower-cost paths.
Samsung 1c HBM4 front-end yield (2025)	~50% (improving)	Samsung skipped 1b, jumping 1a->1c; a margin risk on HBM4.

DRAM energy split (interface vs cell)	DDR5 >99% interface; HBM ~95% interface / 5% cell	The waste that motivates compute-in-memory and modern base-die PHYs.
Bank potential vs delivered bandwidth	~4 TB/s/chip potential; HBM die delivers only ~1/16 (256-bit bus)	UCIe PHY could reach ~11 Tbps/mm (~12x HBM3E), energy/bit 2pJ->0.25pJ.
CXMT FY25 financials	Revenue ~$8.6B (+156% YoY), first net profit ~$1B, GM 37.8%	1Q26: $7.3B rev (~700% YoY), ~70% op-margin —ASP-driven, not share.
CXMT vs incumbents DRAM revenue (CY25)	CXMT ~$8.6B vs Samsung ~$72.3B, SK Hynix ~$52.1B, Micron ~$37.2B	Clear #4 globally, but still an order of magnitude behind the leaders.
CXMT DDR5 cost-per-bit gap	>30% higher than the big-3; ASP only 5-10% below leaders	Margin is pricing, not product competitiveness this cycle.
CXMT HBM3 8-hi modeled yield	~35% front-end x ~70% back-end ~= 25% overall	Struggling on 8-hi, greater challenges at 12-hi; likely skips HBM3 for HBM3E.
DRAM wafer capacity (end-2026E)	Samsung ~720k, SK Hynix ~595k, Micron ~385k, CXMT ~350k wspm	CXMT closing on #3 Micron by wafers; global DRAM share ~13%->17%.
SanDisk/Kioxia BiCS10 NAND	332 layers, 3 decks, 37.6 Gb/mm2 QLC (world's densest)	vs SK Hynix 321L V9 at 28.8 Gb/mm2 QLC —30% behind.
SK Hynix V9 321L density gain	+44% memory/wafer over 238L V8; +30% process steps, +20% etch steps	Layer count up ~35% —outrunning process-step growth, so still cost-down.

Emerging memory density	MRAM 0.49 Gb/mm2 > Micron D1b DRAM 0.435; ships in MB not GB	FeRAM matched D1b density at IEDM 2023 but too costly; not DRAM replacements.

Investment Angle

Winners

SK Hynix	Clear HBM leader on MR-MUF packaging and reliability; primary beneficiary of every HBM content increase in Nvidia's roadmap; highest gross margin of the majors (60.4% FY25) thanks to its HBM mix. TSMC partners on its HBM4 N12 base die.
Micron	Leapfrogged into HBM on TSV/power-delivery focus (claims ~30% lower power), matches Hynix NAND density with fewer/cheaper decks, and rides the broad DRAM shortage. Uses a lower-cost internal CMOS HBM4 base die. Also a FeRAM leader.
TSMC	Owns the CoWoS interposer that is the real AI packaging bottleneck AND the foundry logic node (N12) for HBM4 base dies —double exposure to HBM's growth without making DRAM. Also fabs custom base dies for OpenAI/Nvidia/AMD.
Hanmi (TC **bonders**) / **EVG** / **TEL**	Hanmi held a near-monopoly on HBM thermocompression bonders (100% at Hynix until recently). EVG/TEL gain from wafer-bonding tools needed for VCT/4F2 and hybrid-bonded cell-on-peri DRAM. Lam dominates cryo high-aspect-ratio NAND etch.
CXMT (**speculative**)	Rising #4 DRAM maker, ~13%->17% wafer share, riding the same pricing supercycle to a first-ever profit and a landmark IPO — but it's an ASP play, not a technology one, and still 30%+ behind on cost/bit.

Bottlenecks

- CoWoS interposer capacity —the real gate on AI accelerator supply; 1,000+ wires can't be routed on PCB, so silicon interposers are mandatory.
- HBM front-end TSV/PDN yield and 12-hi+ stacking losses —the binding constraint on effective HBM bit supply (why Rubin Ultra uses 12-hi not 16-hi HBM4E).
- TC-bonder supply —a single-vendor dependency (Hanmi) that nearly halted Hynix HBM in April 2025.
- GPU 'shoreline' area —HBM limited to 2 SOC edges, capping how many stacks fit per package.

- Fab construction lead times —too long to react, which is exactly why the shortage persists to ~2028.

Risks

- Cyclicality: memory is the most cyclical semi sub-industry; ASP-driven earnings (esp. CXMT) reverse violently when the cycle turns.
- Samsung wildcard: perennially over-promises aggressive tech (hybrid bonding), then fails execution —but is genuinely closing the HBM4 gap. Its bad HBM yield ironically tightens total DRAM supply.
- China oversupply overhang: overplayed for the next ~2 years per SemiAnalysis, but CXMT/Huawei custom non-JEDEC HBM + government self-sufficiency push are a longer-term structural threat.
- Interface/IP disruption: modern base-die PHYs (Eliyan UMI/NuLink, UCIe) could reshuffle where value accrues and even reduce HBM capacity needs —'someone is going to win big in the IP wars.'
- Export-control leakage: reclaimed/smuggled HBM and reexport loopholes blunt controls and quietly feed Chinese accelerators.

Catalysts

- CXMT STAR Market IPO —likely China's largest-ever semiconductor listing; a valuation and disclosure event for the whole China memory complex.
- HBM4 ramp (2026-27) —base-die-on-foundry-node transition; Samsung's competitive re-entry vs Hynix; custom base dies for OpenAI/Nvidia/AMD/Amazon.
- DRAM price doubling in 2026 —the 'once-in-four-decades shortage' spanning HBM/DDR5/DDR4/DDR3, potentially exceeding forecasts by year-end.
- Server DRAM + HBM crossing >50% of DRAM demand by end-2027 —mix shift that widens leader ASP over CXMT.
- Rubin Ultra (1TB HBM4E/GPU) and Amazon direct-HBM procurement —the next leg of per-chip capacity and a new buying model.

Open Questions

- When does hybrid bonding actually arrive for HBM? It keeps sliding from HBM4 to '4E'; nobody has a high-volume 16+ die HB solution, and JEDEC's height relaxation bought incumbents time to avoid it.
- When does 3D DRAM insert, and who wins? Roadmaps say ~2030-2035; the big-3 tellingly avoid publishing serious papers because it's the race that will reshuffle DRAM market share.
- Does a modern base-die PHY / compute-in-memory standard emerge from JEDEC or from fast-moving memory-GPU pairs —and does it reduce HBM capacity needs, undercutting the memory-content thesis?
- Can CXMT/Huawei's custom non-JEDEC HBM actually close the bandwidth gap at usable yield —and how much does export-control leakage (reclaimed/reexported HBM) matter in the interim?
- When does the supercycle break? Fab lead times keep supply tight to ~2028, but memory has cycled for 50 years —the ASP-driven earnings (esp. CXMT) are the most exposed on the turn.

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Chapter 4

Advanced Packaging

Transistors kept shrinking but IO didn't —packaging is where broken Moore's Law economics get repaired. Hybrid bonding is the biggest manufacturing shift since EUV, and it moves the whole game from OSATs into the fab.

Overview

SemiAnalysis's throughline across this bucket: transistor density doubled every ~2 years while chip IO data rates doubled only every ~4, so designs became pad-limited just as cost-per-transistor stopped falling (N7 ~$9.5k vs N5 ~$16k wafers). Chiplets fix yield but not IO —and aren't even always cheaper (reticle utilization can inflate wafer cost 26%; Intel re-aggregated Emerald Rapids back to 2 big dies). The escape is packaging density: from 150-200µm flip-chip bumps to fanout (~8x IO density), 2.5D interposers (~16x), 36µm micro-bump 3D (31x), 17µm hybrid-bonded V-Cache (138x) and Sony's 6.3µm image sensors (567x). Hybrid bonding —bumpless Cu-Cu at sub-10µm pitch with a roadmap to hundreds of nanometers —is 'the most transformative innovation since EUV', but it demands ISO 1-3 cleanrooms and fab tools (CVD/PVD/ECD/CMP), structurally excluding OSATs and handing the market to TSMC, Intel and front-end equipment vendors. The battlegrounds: wafer-on-wafer (sub-50nm alignment, but no known-good-die sorting) vs die-to-wafer (KGD, but dirty and slow —Besi's claimed 2,000 dies/hr collapses below 1,000 at 1µm accuracy); Intel's curious ~300-tool TCB fleet; fanout/bridges/coreless ABF eating silicon interposers from below; and hyperscalers (Graviton3, Dojo, AmpereOne) using packaging to commoditize incumbents' CPUs.

Core Knowledge

Advanced packaging (the <100µm definition) & the pitch ladder

SemiAnalysis defines 'advanced' as any packaging with bump pitch below 100µm (tool vendors calling all flip chip 'advanced' are ignored). The ladder vs standard flip chip (150-200µm): fanout at 90-60µm ≈8x IO density, 2.5D at 55-50µm ≈16x, 36µm micro-bump 3D ≈31x, 17µm hybrid-bonded TSVs ≈138x, Sony's 6.3µm CIS ≈567x. Each rung is a step-function in interconnect density and energy-per-bit.

Pad-limited design (the IO wall)

Transistors scaled 2x/2yr but IO data rates only 2x/4yr, and flip-chip bump pitch barely moved since the 1990s (AMD 200µm→130µm = only 2.35x more IO; Intel →100µm = 4x). Old designs ported to new nodes can't shrink because pads set the floor on die area —which is also why Gelsinger's 'move automotive chips to Intel 16' pitch fails on unit economics.

Chiplets & Known Good Die (KGD)

Splitting big chips beats defect math (a 2,000-step process at six-sigma per step still gives D0≈0.678 —semis are 'way better than six sigma'). AMD covers a huge market with 3 tape-outs vs Intel's 5. But chiplets add interface area, packaging yield risk, and power; the whole 3D-stacking economics question reduces to whether you can test and bond only known-good dies.

Hybrid bonding (bumpless Cu-Cu)

Dielectric-to-dielectric plus direct copper-to-copper bonding with no solder: pads recessed ~5nm below a SiCN/SiO surface, pressed flush, then annealed. Scales below 10µm pitch with a roadmap into hundreds of nanometers; kills solder resistance so energy/bit plummets. Requires 0.5nm dielectric / 1nm copper surface roughness, multiple precision CMP passes, and via-middle TSVs to feed power and signal through the stack.

W2W vs D2W (wafer-on-wafer vs die-to-wafer)

W2W bonds two whole wafers: sub-50nm alignment, cleaner, cheap, mature (Sony CIS, YMTC NAND, Graphcore Bow) —but both dies must be the same size and you stack defective dies onto good ones, so it loses on large low-yield logic. D2W picks and places known good dies: wins on big dies but is slower, dirtier (singulation particles), and less accurate. Collective D2W and self-assembly try to square the circle.

TCB (thermocompression bonding)

One tool places each die, applies force + heat (+ vibration) to reflow solder in place —eliminating batch-oven CTE warpage, gap variation and tilt. It packages ultra-thin dies (HBM stacks; SK Hynix 12-hi HBM3 dies thinned to 30µm) and mixed pitches on one package. The catch: ~$1.25M per tool at 500-1,000 dies/hr vs ~$450k flip-chip placers at 3,000-10,000/hr.

2.5D: silicon interposers (CoWoS) vs bridges (EMIB)

2.5D stacks active silicon on a passive routing die at 55-50µm pitch —Nvidia GPUs + HBM on TSMC CoWoS-S is the volume case. Interposers are expensive (TSV-heavy), so the industry substitutes: EMIB embeds a small silicon bridge in the ABF substrate; CoWoS-R uses organic RDL (R+ adds high-density IPD capacitors to carry 6.4Gbps HBM3, with a roadmap to 45x reticle vs CoWoS-S extending to only 4x).

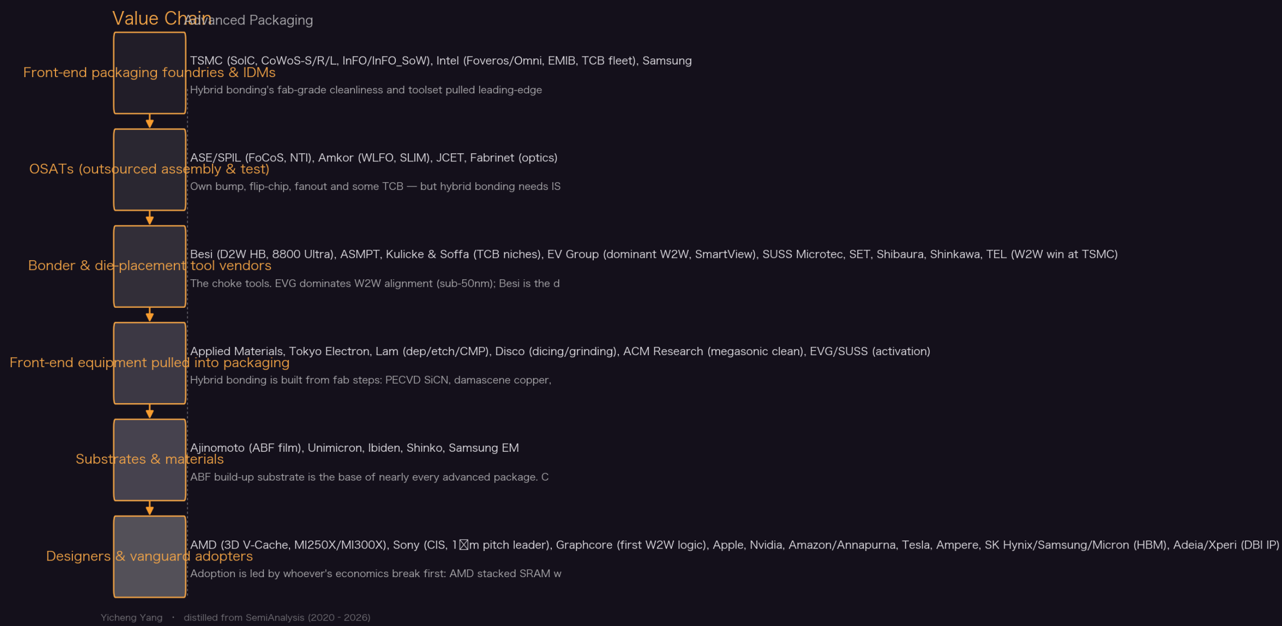
Fanout RDL & ABF build-up substrate

Fanout re-spreads a die's IO across a reconstituted wafer/panel (TSMC InFO in every Apple A/M chip; wafer-scale InFO_SoW in Tesla Dojo). ABF build-up substrate (Ajinomoto film) is the base of nearly everything —'hybrid substrates'. The L/S race: ABF 10µm, Cisco coreless 6µm, Unimicron coreless 3µm, EMIB 5µm→2µm, advanced fanouts 2µm, Amkor SLIM 0.4µm —improving ABF keeps cannibalizing fanout from below.

Reticle limit & reticle utilization

A scanner exposes a 26mm x 33mm field (mask 104x132mm at 4x reduction); Nvidia/Intel datacenter dies have sat near this limit for 5+ years. Less obvious: lithography is ~1/3 of processed-wafer cost, and a chiplet that tiles the field badly forces more scan passes —SemiAnalysis's example needs 1.875x the steps, repricing the wafer from $17,000 to $21,364, shrinking the chiplet's silicon saving to just $26 —and very likely making the 'cheaper' chiplet dearer than monolithic once packaging costs are added.

Value Chain



Key Theses

1. **The root cause: IO scaling runs at half the pace of transistor scaling, so the industry hit a pad-limited wall that only packaging can fix.**

   Density of real chips grew ~2x/3yr against 2x/yr transistor density partly because IO, power delivery and SRAM stalled. Flip chip stayed ~150-200µm from 2000 to 2021; AMD's move to 130µm bought just 2.35x more IO against orders-of-magnitude more transistors. Designers first coped by bloating dies with cache (Infinity Cache cut a 384-bit GDDR6 bus to 256-bit) —but Nvidia and Intel have been at reticle limit for 5+ years. Packaging density is the only lever left.

   > *"Moore's law has had the industry increasing transistor densities roughly 2x every 2 years, but the rate of IO data rates have only been 2x every 4 years."*
   > — Advanced Packaging Part 1 –Pad Limited Designs (2021-12-15)

2. **Cost per transistor is rising, not falling —the deflationary engine of the industry broke at 7nm/5nm, and that reversal is what funds the packaging renaissance.**

   AMD's own data shows cost per yielded mm2 inflecting sharply at 7nm/5nm. SemiAnalysis's receipts: N7 wafers ~$9,500 vs N5 ~$16,000 while Apple's die size barely fell —and they mock Morgan Stanley's TSMC downgrade chart claiming 5nm transistors were cheaper than 7nm. When shrink no longer buys cost, heterogeneous integration and packaging become the only route to more capability per dollar.

   > *"Cost per transistor stalled with the introduction of FinFET nodes, 7nm completely plateaued, and with 5nm they are higher than ever before."*
   > — Advanced Packaging Part 1 –Pad Limited Designs (2021-12-15)

3. **Hybrid bonding is the biggest paradigm shift in packaging since flip chip —SemiAnalysis rates it above EUV in design impact.**

   Everything from wire bond to TCB still uses solder bumps, which bottom out around 20μm pitch. Hybrid bonding is bumpless Cu-Cu: sub-10μm today, a roadmap into the hundreds of nanometers, lower resistance, latency 'in some cases shorter than global routing on chip'. It reshapes IP ecosystems, cell design and manufacturing flows —designers must start thinking in 3D. In 2024 it shipped only in AMD chips, CMOS sensors and some 3D NAND, which is exactly the opportunity.

   > *"Hybrid bonding is going to be the most transformative innovation to semiconductor manufacturing since EUV."*
   > — Hybrid Bonding Process Flow - Advanced Packaging Part 5 (2024-02-09)

4. **Hybrid bonding is a front-end process —its cleanliness physics structurally lock OSATs out and hand packaging to TSMC, Intel and fab-tool vendors.**

   A particle just 1μm tall creates a 10mm bond void; surface roughness must stay within 0.5nm (dielectric) / 1nm (copper). That demands ISO 3 / class 1 cleanrooms or better —TSMC and Intel are going to ISO 2 or ISO 1 —plus fab tools OSATs barely own: CVD, PVD, ECD, CMP, plasma activation, megasonic cleans. ASE and Amkor would need to build new fab-grade cleanrooms; TSMC and Intel just reuse old fabs. The margin pool migrates accordingly.

   > *"It is very difficult for OSATs to pursue hybrid bonding given this upgrade in cleanliness requirement."*
   > — Hybrid Bonding Process Flow - Advanced Packaging Part 5 (2024-02-09)

5. **W2W vs D2W is decided by one variable: known good die. W2W is cheaper and more accurate until die size and yield losses flip the curve.**

   W2W separates alignment from bonding, achieves sub-50nm accuracy, and dominates high-yield small-die volume (CIS, NAND). But it cannot sort dies, so defective dies get bonded to good ones —its cost curve steepens brutally with die size, while D2W (5 bond steps per AMD V-Cache part) stays flat by only placing KGD. That's why AMD's D2W V-Cache was first to productize big-die stacking while W2W logic waited for a cheap top die (Graphcore's passive capacitor wafer).

   > *"The ability to test and bond only known good die (KGD), instead of risking defects stacking and wasting good silicon, is critical and why die-on-wafer (D2W) is the first to be productized."*
   > — Hybrid Bonding Process Flow - Advanced Packaging Part 5 (2024-02-09)

6. **Intel's ~300-tool TCB bet looks economically irrational and is actually a moat: in high-margin silicon, yield beats tool amortization.**

   A TCB tool costs ~$1.25M and places 500-1,000 dies/hr vs ~$450k and 3,000-10,000/hr for flip-chip —yet Intel owns nearly 300 TCB tools and its $7B Malaysia facility will double that, using TCB even where flip chip would do. TCB kills CTE warpage, gap variation and tilt, enables 30μm-thin HBM dies and mixed 55μm/100μm pitches on one Sapphire Rapids package, and a decade of co-development means TSMC and Samsung can't instantly follow into Foveros-Omni-class assembly.

   > *"Given Intel's heavy share in high power and high margin applications, the yield loss and reliability concerns far outweigh the miniscule, amortized cost of the tool per unit packaged."*
   > — Advanced Packaging Part 3 –Intel's Curious Bet on Thermocompression Bonding (2022-01-19)

7. **Chiplets aren't automatically cheaper —reticle utilization and Intel's Emerald Rapids re-aggregation both prove packaging intensity is a knob that turns both ways.**

   Two SemiAnalysis counterexamples to chiplet dogma. (1) Lithography is ~1/3 of wafer cost and scanners expose fixed 26x33mm fields: a 2-chiplet design that tiles the field badly needs 1.875x the scan steps, repricing the wafer $17,000→$21,364 and cutting the chiplet's saving from $136 to $26 —very likely more expensive than the 800mm2 monolithic die once packaging costs are counted. (2) Emerald Rapids went from four 394mm2 dies back to two 763mm2 dies: interface silicon fell 16.2%→5.8% of die area, EMIBs 10→3, core-area utilization 50.7%→62.7% —yet EMR still costs more per CPU than SPR (34 vs 37 CPUs/wafer) because big rectangles tile round wafers badly.

   > *"If your yields are good, why waste area on redundant IO and chiplet interconnects when you can just use fewer, larger dies?"*
   > — Intel Emerald Rapids Backtracks on Chiplets (2023-05-03)

8. **Graphcore Bow showed hybrid bonding is 'free' performance: +40% clocks from a capacitor wafer, with zero architecture, node or software change.**

   TSMC's first W2W SoIC product bonded a deep-trench-capacitor wafer under the same 7nm IPU: smoothing transient spikes lifted clocks from 1.325GHz to 1.85GHz (+40%) at +16% perf/W and equal cost. W2W TSVs run ~1/10th the pitch of chip-on-wafer —10,000 connections where D2W fits 100. The same physics later let TSMC's CoWoS-R+ IPDs carry 6.4Gbps HBM3. Power delivery, not logic, is often the binding constraint 3D stacking relaxes first.

   > *"In the area that chip-on-wafer hybrid bonding is able to offer 100 TSVs, the wafer-on-wafer technology can offer 10,000."*
   > — Graphcore Announces World's First 3D Wafer On Wafer Hybrid Bond Processor (2022-03-03)

9. **Packaging is the hyperscalers' commoditization weapon: Graviton3 shipped 55µm micro-bump chiplets, DDR5 and PCIe 5.0 before Intel or AMD.**

   Amazon's 7-die design (282mm2 compute + tiny DDR5/PCIe tiles) used ≤55µm bumps while every x86 CPU sat at ≥100µm, avoided AMD's ~100W IO-die power tax, dropped sockets for BGA to pack 3 CPUs per NIC per 1U —and SemiAnalysis exclusively identified the packaging vendor as 'a certain blue company' (Intel). AmpereOne extends the pattern: chiplets under a full carrier-wafer of structural silicon acting as CTE-matched heat spreader with direct-die cooling. System-level packaging choices, not core IPC, are what terrify merchant silicon.

   > *"The size of the micro-bumps connecting each die are <=55um whereas every CPU from Intel and AMD is still at >=100um."*
   > — Amazon Graviton 3 Uses Chiplets & Advanced Packaging To Commoditize High Performance CPUs (2021-12-02)

10. **Tesla Dojo made the package the unit of scale: a wafer-scale InFO_SoW tile with 36TB/s off-tile bandwidth and vertical 15kW power delivery.**

    25 known-good D1 dies (645mm2, 400W each) are packaged into one fanout training tile —SemiAnalysis called the TSMC InFO_SoW packaging before Tesla's announcement. Each tile: 9 PFLOPS BF16, 36TB/s off-tile via a proprietary connector, ~10kW of silicon inside a ~15kW tile, with custom voltage regulators reflowed directly onto the fanout wafer so power enters vertically from below while heat exits above. It beat Cerebras on off-wafer bandwidth and previewed today's system-in-package direction for AI.

> *"This tile far surpasses anything from Nvidia, Graphcore, Cerebras, Groq, Tenstorrent, SambaNova, or any other AI training geared start up in per unit performance and scale up capabilities."*
> — Tesla Dojo - Unique Packaging and Chip Design (2021-08-20)

11. **The 2.1D-2.5D lines are blurring: bridges, chip-last fanout and coreless ABF are eating the expensive silicon interposer from below.**

    EMIB and fanout-embedded bridges (MI250X, M1 Ultra) deliver most interposer benefits without full TSV-laden silicon; TSMC's CoWoS-R reaches 3x reticle with a 45x roadmap while CoWoS-S extends to just 4x. The supply chain logic favors chip-last flows that ship testable known-good substrates. Meanwhile ABF keeps densifying —10μm L/S standard, 6μm Cisco coreless, 3μm Unimicron on panels, versus 2μm advanced fanout and 0.4-0.5μm Amkor SLIM/ASE NTI first layers. Nothing here is a stable monopoly except the substrate base itself.

    > *"ABF substrates are catching up and surpassing fanout RDLs for many use cases."*
    > — The Future Of Packaging Gets Blurry –Advanced Packaging Part 4 (2022-11-01)

12. **The bonder market runs on a throughput-vs-accuracy lie: rated specs halve in the real world, and that gap is SemiAnalysis's 'very non-consensus view on BESI'.**

    TSMC's 3μm-pitch SoIC data: 98% bond yield below 0.5μm misalignment, resistance blowing out from 0.5-1μm, and 60% yield beyond 1μm. Besi's 8800 Ultra claims <200nm accuracy, but SemiAnalysis hears ~0.5μm with wide variance even at half rated throughput; a claimed 2,000 dies/hr drops under 1,000 at 1μm accuracy. Meanwhile Intel/CEA-LETI's capillary self-alignment hit 150nm mean misalignment with faster placement —a technology that could reset the D2W tool landscape entirely.

    > *"Despite Besi's claims of 2,000 dies placed per hour, to reach even 1-micron accuracy, the throughput drops below 1,000 dies placed per hour."*
    > — Packaging Developments From ECTC 2022 (2022-06-08)

13. **TSV formation is the hidden gate on the AI supply chain —it bottlenecks both HBM and CoWoS, and avoiding TSVs is itself a product strategy.**

    Via-middle TSVs need deep reactive ion etch, insulation/barrier CVD, seed PVD, copper ECD fill and backside reveal —slow, tool-intensive steps. SemiAnalysis reports TSV formation bottlenecking HBM and CoWoS output, and notes customers switching from silicon interposers to CoWoS-R specifically to skip interposer TSV cost. Stacking economics compound: HBM dies thin to 30μm at 12-hi, forcing TCB today and pushing SK Hynix toward W2W hybrid bonding at 16-hi.

    > *"We understand that TSV formation is a step that is bottlenecking HBM and CoWoS production."*
    > — Hybrid Bonding Process Flow - Advanced Packaging Part 5 (2024-02-09)

14. **The endgame beyond copper: optical interposers (Lightmatter Passage) promise 768Tbps per tile —but SemiAnalysis flags it may be vaporware.**

    Passage 3D-packages up to 48 customer ASICs on a wafer-scale photonic interposer (GF Fotonix 45CLO), stitching waveguides across reticles at 0.004dB loss per crossing, reconfiguring the whole fabric in <1ms with 2ns max chip-to-chip latency, and feeding 700W per tile through TSVs. It's ~40x the interconnect density of fiber-attach co-packaged optics. SemiAnalysis's caveats: the MZI fabric is one-in-one-out (much of the 768Tbps idles), and customers must trust an unproven partner with their most expensive silicon.

> *"This could just be vaporware, or it could be the future for high-end leading edge disaggregated server designs."*
> — Beyond Advanced Packaging: Lightmatter Passage Chiplets Co-Packaged On Optical Interposer (2022-08-22)

Article Deep-Dives

Hybrid Bonding Process Flow - Advanced Packaging Part 5 (2024-02-09)

The definitive process-flow primer. Walks TSV formation (via-middle DRIE + CVD/PVD/ECD + backside reveal, currently bottlenecking HBM/CoWoS), hybrid-bond layer fabrication (PECVD SiCN, damascene copper, ~5nm pad recess, multi-slurry CMP), probing/singulation particle management (laser/plasma over blade dicing —Disco), N2 plasma activation, and megasonic DI cleans. Frames the W2W vs D2W cost crossover around known good die, details collective D2W (AMD V-Cache = 5 bond steps) and Intel's QMC reconstituted W2W, and lists the full tool ecosystem (Besi, EVG, TEL, AMAT, ASMPT, SUSS, SET, Shibaura) with a teased 'very non-consensus view on BESI' and firm-level adoption modeling through 2030.

Packaging Developments From ECTC 2022 (2022-06-08)

The conference distillation that quantified the frontier: TSMC gen-4 SoIC at 3μm pitch (100k pads/mm2, 98% yield <0.5μm misalignment); Sony's 1μm-pitch hybrid bond with a contrarian protruding-copper CMP process delivering orders-of-magnitude better contact resistance; Intel/CEA-LETI water-droplet self-alignment (150nm mean misalignment, 98% bond yield) as a possible D2W disruptor; SK Hynix W2W hybrid bonding aimed at 16-hi HBM; Samsung's MCM/2.5D/3D power-area overhead study; AMD V-Cache's hidden 5th support die and its 17μm production pitch (vs 9μm marketing); plus ASE probing co-packaged optics and CoWoS-R+ carrying 6.4Gbps HBM3 via high-density IPDs.

Advanced Packaging Part 1 –Pad Limited Designs, Breakdown Of Economic Semiconductor Scaling, Heterogeneous Compute, and Chiplets (2021-12-15)

The series manifesto. Chains the argument: real chip density lags transistor density (SRAM death, power, IO); flip-chip pitch stagnated for 20 years creating pad-limited designs (why Gelsinger's 'port automotive chips to Intel 16' fails); cost-per-transistor now rises (N7 $9.5k vs N5 $16k wafers); die sizes hit reticle limit; chiplets fix defect math (AMD's 3 tape-outs vs Intel's 5; a 2,000-step six-sigma process would still yield D0=0.678) but not IO. Conclusion: 'we are in the midst of the semiconductor design renaissance which is being pushed forward by advanced packaging.'

Advanced Packaging Part 3 –Intel's Curious Bet on Thermocompression Bonding, ASM Pacific, Kulicke and Soffa, and Besi TCB Tool Landscape (2022-01-19)

Explains why TCB exists (batch reflow's CTE warpage, gap variation, tilt —AMD Fiji's field failures) and why Intel bought ~300 tools despite brutal economics ($1.25M/500-1,000 dph vs $450k/3,000-10,000 dph): in high-margin, high-power silicon, yield and reliability dwarf tool amortization, and one tool platform covers standard, 2.5D and 3D packages (SPR's mixed 55μm EMIB + 100μm pitches; Foveros Omni at 130/100/36μm is 'nigh on impossible' with standard flip chip). HBM's 30μm-thin 12-hi stacks make TCB mandatory for Samsung/SK Hynix/Micron. ASMPT, K&S and Besi each hold distinct niches; Intel's co-developed platforms are the largest orders.

The Future Of Packaging Gets Blurry –Fanouts, ABF, Organic Interposers, Embedded Bridges –Advanced Packaging Part 4 (2022-11-01)

Maps the 2.1D-2.5D middle ground where most volume will live. Four base classes (3D, 2.5D, fanout

RDL, ABF build-up) blur via hybrid substrates: EMIB bridges in ABF cavities, MI250X's fanout-embedded bridges atop ABF, chip-first vs chip-last flows. Key supply-chain insight: the PC/DC industry runs on matching known-good substrates with known-good dies, so chip-last (testable substrate before die attach) is preferred if cost allows. The L/S race (ABF 10µm, coreless 6µm/3µm, EMIB 5→2µm, fanout 2µm, SLIM 0.4µm) means improving ABF keeps cannibalizing fanout —no packaging technology's moat is safe below the silicon interposer.

Die Size And Reticle Conundrum –Cost Model With Lithography Scanner Throughput (2022-06-19)

The free-article cost model that punctures chiplet dogma. Because a scanner exposes fixed 26x33mm fields (mask 104x132mm, 4x reduction) and lithography is ~1/3 of processed-wafer cost, a chiplet that tiles the field poorly forces 1.875x the scan steps; a margin-preserving foundry reprices the wafer from $17,000 to $21,364, shrinking the chiplet's silicon saving from $136 to just $26 per product ($541 vs $567); once packaging costs are added, the monolithic die is very likely cheaper. Caveats acknowledged (most real chiplet architectures improve utilization; High-NA halves the scan), but the discipline stands: die-size choice must be co-optimized with the reticle grid.

Key Data

Transistor vs IO scaling	Transistor density 2x every 2 years; IO data rates 2x every 4 years	The decades-long divergence that created pad-limited designs and the entire advanced-packaging market.
Advanced packaging definition	Bump pitch < 100µm (standard flip chip = 150-200µm, essentially unchanged 2000→2021)	SemiAnalysis's line in the sand; tool vendors calling all flip chip 'advanced' are ignored.
IO density ladder vs standard flip chip	Fanout (90-60µm) ~8x → 2.5D (55-50µm) ~16x → 3D µbump (36µm) 31x → HB V-Cache (17µm TSV) 138x → Sony CIS (6.3µm) 567x	Each packaging generation is a step-function in interconnect density; hybrid bonding roadmap extends to 100s of nm.
Wafer prices, N7 vs N5	~$9,500 (N7) vs ~$16,000 (N5)	Evidence cost-per-transistor is rising; SemiAnalysis's rebuttal to Morgan Stanley's TSMC downgrade.

Hybrid bonding particle sensitivity	A 1µm-tall particle → ~10mm-diameter bond void	Why HB requires ISO 3/class 1 cleanrooms or better (TSMC/Intel going ISO 2-1) —and why OSATs are locked out.
HB surface-roughness spec	≤0.5nm dielectric, ≤1nm copper pads; Cu recessed ~5nm pre-anneal	Maintained through the whole flow via multi-pass CMP; even wafer-sort probing must be polished out afterwards.
Bond alignment accuracy, W2W vs D2W	W2W sub-50nm; Besi 8800 Ultra claims <200nm but 'more like 0.5µm with wide variance' at half rated throughput	The spec-vs-reality gap behind SemiAnalysis's non-consensus BESI view.
TSMC SoIC gen-4 (3µm pitch) bond yield	100,000 pads/mm2; 98% yield <0.5µm misalignment, 60% beyond 1µm	Defines the accuracy bar every D2W bonder must clear as pitches shrink from 9µm to 3µm.
D2W bonder throughput cliff	Claimed 2,000 dies/hr → <1,000/hr at 1µm accuracy	Throughput-accuracy tradeoff is the central battle of the hybrid-bonding tool market.
TCB vs flip-chip tool economics	TCB: ~$1.25M, 500-1,000 dies/hr. Flip-chip placer: ~$450k, 3,000-10,000 dies/hr	6-20x throughput gap at ~1/3 the price — yet Intel chooses TCB for yield/reliability in high-margin parts.
Intel TCB fleet	~300 tools, doubling with the $7B Malaysia packaging facility	Far exceeds Intel's advanced-packaging needs; a decade of co-development = a hard-to-copy moat.
Reticle utilization cost penalty	Bad field tiling → 1.875x scan steps → wafer repriced $17,000 → $21,364 (litho ≈ 1/3 of wafer cost)	The worked example where the 2-chiplet design's silicon advantage collapses from $136 to $26 ($541 vs $567) —and very likely flips to a loss once packaging costs are included.

Emerald Rapids re-aggregation	4 dies → 2; chiplet interface 16.2% → 5.8% of die area; EMIBs 10 → 3; core-area utilization 50.7% → 62.7%; but 37 → 34 CPUs/wafer	Fewer chiplets bought back area yet still raised cost per CPU —big rectangles tile round wafers badly.
Graphcore Bow uplift from W2W hybrid bonding	+40% clocks (1.325 → 1.85GHz), +16% perf/W, same node/architecture/cost; W2W TSV pitch ~1/10th of chip-on-wafer	First 3D W2W-bonded logic processor; the top die is just deep-trench capacitors for power delivery.
AMD 3D V-Cache production reality	17µm production pitch (TechInsights) vs the 9µm AMD used in marketing comparisons; 5 bond steps per chip; a 5th structural support die on top	First and (through early 2024) only D2W hybrid-bonded logic product; claimed 3x interconnect efficiency and 16x density vs 36µm µbump.
HBM die thinning	30µm per die in SK Hynix 12-hi HBM3 (< half a human hair)	Forces TCB today; SK Hynix presented W2W hybrid bonding for 16-hi HBM at ECTC 2022; TSV formation bottlenecks HBM & CoWoS.
Substrate/RDL line-space race (L/S)	ABF 10µm → Cisco coreless 6µm → Unimicron coreless 3µm (panel) | EMIB 5µm → 2µm | advanced fanout 2µm | Amkor SLIM 0.4µm / ASE NTI 0.5µm (first layer)	Cheaper organic technologies keep climbing into silicon-interposer territory; CoWoS-R roadmap = 45x reticle vs CoWoS-S 4x.

Samsung's packaging-overhead study (450mm2 HPC design)	vs monolithic: MCM +2.1% power/+5.6% area; 2.5D +1.1%/+2.4%; 3D +0.04% power/+2.4% area	Ideal-case numbers, but they quantify why 3D stacking is the endgame: near-zero power overhead.

Investment Angle

Winners

TSMC	Owns every rung of the ladder —InFO (Apple), CoWoS-S/R/R+ (Nvidia, roadmap to 45x reticle), SoIC CoW (AMD V-Cache/MI300) and SoIC WoW (Graphcore) —and hybrid bonding's fab-grade requirements pull even more of the package into its fabs. Vanguard customers de-risk each new flow before volume.
EV Group (EVG)	Dominant in W2W fusion/hybrid bonding with patented SmartView alignment (sub-50nm). W2W is the mature, high-volume path (every Sony CIS, YMTC NAND, Graphcore Bow) and the likely vehicle for 16-hi HBM hybrid bonding.
Besi (with a caveat)	De-facto D2W hybrid-bonding bonder at TSMC (8800 Ultra picks and places every V-Cache die). But SemiAnalysis repeatedly flags real accuracy ~0.5µm at half rated throughput and holds 'a very non-consensus view' —the D2W moat may be narrower than the consensus multiple assumes, especially if collective/self-assembly flows mature.
Front-end tool & process incumbents (AMAT, TEL, Disco, ACM)	Hybrid bonding converts packaging capex into front-end tool spend: PECVD, PVD/ECD, precision CMP, plasma dicing, megasonic cleans. TEL won TSMC's W2W bonder socket (the Graphcore flow); Disco leads clean dicing and 'has more than tripled' since SemiAnalysis's write-up.
Intel (packaging franchise)	The one area where Intel is genuinely differentiated: a decade of co-developed TCB (~300 tools, $7B Malaysia doubling), EMIB bridges at 55µm ramping in every Xeon, Foveros/Omni bringing 36µm 3D to mass-market client chips, and QMC research pointing at reconstituted W2W. Packaging is Intel's credible foundry hook.
ASMPT & Kulicke and Soffa	The TCB order book rises structurally with HBM stack heights (30µm dies leave no alternative) and with OSAT/mobile adoption; each of the three TCB vendors 'excels in different areas' and owns a defensible niche.

Bottlenecks

- TSV formation —the deep-etch/CVD/ECD sequence bottlenecking both HBM and CoWoS output; customers switch to CoWoS-R partly to dodge interposer TSV cost.
- D2W bonder throughput-vs-accuracy: 2,000 dies/hr claims collapse below 1,000 at 1µm accuracy, while 3µm pitches demand <0.5µm placement for 98% yield.
- ISO 1-3 cleanroom + front-end toolset capacity —hybrid-bonding supply can't come from OSAT brownfield; it must be built to fab standards.
- KGD testing without wrecking the bond surface: probe marks violate the 1nm roughness spec, forcing extra CMP passes and constraining wafer-sort strategies.

Risks

- The chiplet cost fallacy cuts both ways: reticle utilization and Emerald Rapids show packaging-intensive designs can lose to monolithic —advanced-packaging TAM is not a one-way ratchet.
- Hybrid-bonding adoption slippage: through early 2024 AMD remained the only D2W logic adopter; the cost barrier is real and each pitch shrink resets the yield learning curve.
- OSAT disintermediation: as value migrates to fab-grade processes, ASE/Amkor risk being stuck with commoditizing bump/fanout while foundries take the premium layers.
- Reliability/CTE physics: warpage and thermal cycling killed early interposer products (AMD Fiji); chip-first fanout still burns known-good dies on packaging yield; every new hybrid substrate re-opens these failure modes.

Catalysts

- Hybrid bonding breaking out beyond V-Cache: MI300-class D2W volume, SK Hynix W2W for 16-hi HBM, and TSMC N2-era customer-specific SoIC flows would inflect the whole tool chain.
- CoWoS/HBM capacity expansions and the CoWoS-R/R+ substitution wave (45x-reticle roadmap) as AI package sizes balloon into the kilowatt range.
- Intel's packaging ramp: $7B Malaysia facility, EMR/Meteor-Arrow Lake EMIB+Foveros volume, and TCB fleet doubling —a read-through for ASMPT/K&S/Besi order books.
- Panel-level fanout and coreless ABF hitting cost crossover; co-packaged optics / optical interposers (Lightmatter Passage, Ayar Labs) getting a first volume design win.

Open Questions

- When does D2W hybrid bonding get cheap enough to spread beyond AMD —into mobile, client PCs and AI accelerators? SemiAnalysis's adoption model runs to 2030, but through early 2024 the answer was 'one customer'.
- Can pick-and-place bonders economically reach sub-200nm accuracy, or do collective flows and capillary self-assembly (Intel/CEA-LETI's 150nm) reset the tool landscape —and with it Besi's presumed moat?
- Do OSATs ever climb into hybrid bonding, or is leading-edge packaging permanently annexed by TSMC and Intel, leaving ASE/Amkor the commoditizing layers?

- In HBM, when does hybrid bonding displace TCB? 16-hi stacks with 30μm dies strain solder physics, yet TCB keeps being extended —the crossover timing moves billions in tool orders.
- Where is the true chiplet/monolithic frontier once reticle utilization, interface overhead and packaging yield are all priced —will more designs ′do an Emerald Rapids′ and re-aggregate as nodes mature?

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Chapter 5

Semiconductor Equipment

The picks-and-shovels of AI: whoever owns EUV, etch, deposition and metrology owns the physical bottleneck of Moore's Law —and, increasingly, the cost curve is decided by process tricks, not just by ASML's next scanner.

Overview

Wafer fabrication equipment (WFE) is a ~$100B/year oligopoly split by process step, not by node. ASML holds a hard monopoly on EUV lithography (each low-NA tool >$200M, each High-NA EXE:5000 ~$400M and 150 metric tons); Applied Materials and Lam Research dominate deposition and etch; Tokyo Electron owns 100% of EUV photoresist coat/develop tracks; KLA owns process control. SemiAnalysis's core, repeated, non-consensus thesis is that lithography is NOT destiny: as 2D scaling dies, value migrates from the scanner to complementary patterning (AMAT Sculpta pattern-shaping, directed self-assembly, metal-oxide/dry resist) and to etch+deposition-driven vertical scaling (3D NAND, gate-all-around, 3D DRAM). Their most contrarian call —that High-NA EUV single-patterning is actually MORE expensive than low-NA double-patterning through the 1nm node —cuts directly against ASML's marketing and reframes the whole roadmap around cost-per-wafer, not resolution. The memory downturn/AI-driven upturn cycle, Chinese equipment substitution, and export-control gaps are the macro overlay.

Core Knowledge

WFE (Wafer Fabrication Equipment)

The front-end toolset that turns bare silicon into patterned chips: lithography, deposition, etch, CMP, ion implant, and process control (metrology/inspection). A ~$100B/year market segmented by step, each step an oligopoly.

EUV / High-NA EUV

Extreme ultraviolet lithography at 13.5nm wavelength —ASML monopoly. High-NA (0.55 aperture, EXE:5000) images smaller features but halves the exposure field, forcing die-stitching. The critical debate: is it worth ~2x the cost?

Multi-patterning (LELE/SADP/SALELE)

Using multiple exposures + etch/deposition to print features finer than one exposure allows. Doubles/triples litho cost but keeps each exposure at a cheaper, higher-dose point on the exponential dose-vs-CD curve.

Pattern shaping (AMAT Sculpta)

A directional (angled ribbon-beam plasma) etch that elongates features in one axis after a single exposure, replacing a second litho pass. Can remove EUV double-patterning on some layers; ~$4.5B potential EUV demand reduction.

DSA (Directed Self-Assembly)

Block copolymers (PS-b-PMMA) that self-organize into ~9-20nm lines guided by a coarse EUV template. ′Heals′ line-edge roughness, letting EUV run at 50%+ lower dose —Intel′s proposed ′magic bullet′ to make High-NA economical at 14A.

CAR vs MOR vs Dry Resist

Photoresist platforms. CAR (organic, TEL/JSR/TOK incumbent); MOR (metal-oxide, Inpria/JSR, better etch selectivity for thin High-NA films); Dry resist (Lam CVD-deposited metal, ~2x throughput, ~50% dose cut). A $5B+ battle.

ALD & Batch vs Single-Wafer

Atomic Layer Deposition lays down one atomic layer at a time —highest conformality for high-aspect-ratio 3D structures. Batch (Kokusai ~70% share) processes 100s of wafers for memory; single-wafer (ASMI/AMAT/Lam) for precise logic films.

HAR Etch (High Aspect Ratio)

Etching channel holes 70x deeper than wide through 3D NAND stacks (>250 layers, ~7 microns). Lam′s crown jewel; Tokyo Electron′s cryogenic etch threatens >$1B of it. The gating step for NAND vertical scaling.

Process Control (Metrology/Inspection)

Measuring/inspecting wafers between steps to catch defects before yield is destroyed (a single fab excursion cost TSMC $550M in 2019). KLA-dominated; ~10-15% of WFE. AI (e.g. Lynceus) aims to cut this capex via virtual metrology.

Value Chain

Key Theses

1. **High-NA EUV single-patterning is MORE expensive than low-NA double-patterning —all the way through the 1nm node.**

   Dose requirements rise exponentially as critical dimension shrinks. Low-NA double-patterning splits a layer into two exposures with ~2x-larger features each, sitting far down the exponential dose curve, so the scanner runs at full stage-limited throughput. High-NA single-exposure is dose-limited even at a hypothetical 1kW source. The throughput advantage is so strong that despite 2x the wafer passes, low-NA double-patterning is cheaper. IBM's own SPIE data confirmed it: a single High-NA exposure costs ~2.5x a single low-NA exposure, so High-NA only wins when it replaces 3+ low-NA masks (e.g. 4-mask SALELE).

   > *"Our lithography models show that despite reducing complexity, high-NA EUV single patterning costs significantly more than double-patterning using existing low-NA machines for upcoming technology nodes including 1.4nm/14A."*
   > — ASML Dilemma: High-NA EUV is Worse vs Low-NA EUV Multi-Patterning (2023-12-11)

2. **Lithography is not destiny —as 2D scaling dies, value migrates from ASML's scanner to complementary patterning and to etch/deposition-driven vertical scaling.**

   NAND broke the mold in 2013: it stopped relying on lithography and went vertical, so density improved ~30%/year and cost/bit fell ~21%/year while the value shifted to HAR etch and ALD. NAND litho-intensity is only ~10-12% of WFE. The same physics now hits logic (GAA nanosheets, then CFET) and DRAM (3D DRAM), raising etch/deposition intensity. The inflection to 'stacking not shrinking' is exactly where SemiAnalysis places High-NA's natural, cost-effective insertion —the 2030+ 1nm/7A era, not 14A.

   > *"This is because NAND flipped the formula and stopped relying on lithography for patterning smaller cells. Instead, NAND has relied on a different architecture, 3D NAND... density has improved by 30% per year at a very consistent rate."*
   > — NAND Flash Monopoly Broken? Tokyo Electron Moly Dep + Cryo Etch Takes On Lam Research (2023-07-16)

3. **Applied Materials' Sculpta 'pattern shaping' is real, works, and could cut EUV double-patterning demand by ~$4.5B —not the immature science project critics dismissed.**

   Sculpta fires a highly directional angled ribbon-beam plasma (beam angle controllable 0-70 degrees) to elongate a feature on one axis after a single exposure, replacing a second litho pass for tight tip-to-tip, corner-to-corner via, and stochastic-defect repair (>90% defect reduction). AMAT claims per-layer savings of ~$250M capex per 100k wafers/month and ~$50/wafer opex versus adding an EUV loop. It's not vaporware: AMAT has published since 2015, the first customer engaged in ~2017, and it inserts on a 2nm-class node in late-2024/2025. But it can't be used everywhere —it shapes existing features, it doesn't add routing density.

   > *"According to Applied Materials, the Sculpta tool could be used to reduce the use of EUV lithography by as much as HALF for some layers. If true, this would reshape the cost structure of the industry."*
   > — EUV Requirements Halved? Applied Materials' Sculpta Redefines Lithography And Patterning (2023-03-06)

4. **Directed Self-Assembly (DSA) is Intel's 'magic bullet' —the only thing that could make High-NA economical at 14A by cutting EUV dose 50%+.**

   Because High-NA's cost problem is the exponential dose-vs-CD curve, the fix is to relax the EUV exposure and 'heal' it. DSA block copolymers (PS-b-PMMA) self-assemble into ~9nm lines aligned to a coarse, low-dose EUV guide pattern; they align to the average of the guide, so the guide's line-edge roughness can be poor. That relaxes dose by 50%+ (Intel R&D shows 25 mJ/cm2 = a 3-4x cut). This is why Intel —alone among chipmakers —bet on High-NA in HVM at 14A while TSMC/Samsung ordered only R&D tools. The catch: DSA has been stuck in research for a decade because of defectivity risk.

   > *"Relaxing image quality requirements for the EUV exposure means the dose can be reduced by 50% or more."*
   > — Intel's 14A Magic Bullet: Directed Self-Assembly (DSA) (2024-04-18)

5. **The NAND monopoly is breakable: Tokyo Electron's cryogenic HAR etch + molybdenum deposition can move >$1B of revenue away from Lam Research.**

   NAND vertical scaling has hit the HAR-etch wall (6-7 micron / 70:1 channel holes; ~40nm minimum cell thickness). Two manufacturing changes threaten Lam's crown-jewel etch dominance:

cryogenic etch (etching at very low temperature to push depth/aspect ratio further, where TEL is strong) and a tungsten-to-molybdenum wordline material change (lower resistance, thinner, enabling more layers). SemiAnalysis frames both as >$1B of share potentially changing hands —a rare crack in one of WFE's tightest oligopolies.

> *"a potential big shift in market share from Lam Research to Tokyo Electron (TEL) due to two manufacturing changes representing over $1 billion of revenues potentially changing hands."*
> — NAND Flash Monopoly Broken? Tokyo Electron Moly Dep + Cryo Etch Takes On Lam Research (2023-07-16)

6. **Nanoimprint lithography will NOT replace EUV —the mask lasts ~50 wafers vs. 100,000+ for a photomask, and defectivity kills the economics.**

   On paper NIL is seductive: a 4-cell tool costs ~1/10th of an EUV scanner, ~1/4 the cost-per-wafer, ~90% less power, and can theoretically match EUV resolution with no stochastics. In practice the nanometer-scale 3D stamp is fragile —Canon's templates last ~50 wafers (customers dispute Canon's higher claims), and any repeater defect is copied to every field, so inspecting every template would consume 'roughly the entire annual output of mask-inspection tools worldwide.' Kioxia and Micron, who WANT it to work, keep repeating the same complaints. Its real use is niches (metalenses, MEMS). Note the export-control gap: a Japan-sold NIL tool could be re-exported to China.

   > *"How long does that stamp last? Right now, about 50 wafers... Compare that to a photolithography mask which has a lifetime of well over 100,000 wafers."*
   > — Nanoimprint Lithography: Stop Saying It Will Replace EUV (2025-10-26)

7. **Metal-oxide resist's time has finally come —and it arrives bundled with High-NA at ~A10, not because the industry chose it but because thin-film physics forces it.**

   High-NA's depth of focus shrinks with the square of NA, forcing very thin resist. Organic CAR resist that thin lacks etch resistance to survive pattern transfer; metal-oxide resist (MOR) has far better etch selectivity plus better resolution-LER-sensitivity. The crossover is ~30nm-pitch vias, ~2 nodes out at A10 —the same point SemiAnalysis models High-NA becoming economical. Most SPIE 2025 High-NA data already used MOR over CAR. A parallel war rages over wet (TEL spin-coat) vs dry (Lam CVD) application of that resist, with winners now emerging in research.

   > *"It will probably make sense to adopt MOR at the same time as high-NA for a small number of critical layers like metals and vias. Most SPIE papers this year with high-NA exposure data used MOR rather than CAR."*
   > — High-NA is Here (for R&D), EUV Cost, Pattern Shaping Gaining Share... (2025-04-14)

8. **ASML's own long-term guidance is questionable —its 20 High-NA tools/year by 2028 target is 'impossible,' and its wafer model bakes in 10-20% 'inefficient' capacity.**

   SemiAnalysis repeatedly pokes holes in first-party equipment guidance. ASML's investor-day math assumed 600 DUV + 90 EUV tools/year by 2025 and 20 High-NA/year by 2028 —the last of which they call flatly impossible given the cost problem. ASML's demand model leans on 'technological sovereignty' regionalization adding ~10% inefficient capacity (rising to ~20% of incremental 2025-30 wafers), which would drag utilization toward ~72%. Their fab-count slide got 'a 4 out of 10.' The takeaway: buy-side/sell-side that just plug ASML's guidance are ignoring a large long-term risk.

   > *"the most lofty, and in our view, impossible to hit target of theirs is the plan to have 20 High-NA EUV tool annual shipments by 2028."*
   > — ASML Dilemma: High-NA EUV is Worse vs Low-NA EUV Multi-Patterning (2023-12-11)

9. **Export controls have gaping holes: DUV, photoresist, components and service —not just EUV —must be blocked, or China rebuilds 7nm from existing tools.**

   The 2023 trilateral (US/Japan/Netherlands) deal was 'extremely limited,' covering mainly ArFi tools. But SAQP with ArFi (k1=0.391) reaches 5nm-class 28nm minimum-metal-pitch, and even dry-ArF or advanced-KrF can reach 7nm/14nm pitches at ~19% higher wafer cost —trivial for a state subsidizing at nuclear-plant scale. SMIC alone has 100+ DUV tools and could reach >100k wafers/month of 7nm. The un-restricted photoresist supply chain (~75% Japanese) and tool service/parts are the biggest holes: any nation can be cut off from all lithography if Japan restricts resist. Separately, ASML declared it 'can continue to ship all non-EUV systems to China.'

   > *"As a European-based company with limited US technology in our systems, ASML can continue to ship all non-EUV lithography systems to China out of the Netherlands."*
   > — The Gaps In The New China Lithography Restrictions (2023-01-29)

10. **Batch ALD is a hidden monopoly niche: Kokusai Electric owns ~70% of batch ALD, wins critical 3D NAND deposition at all top-5 makers, and its intensity rises with GAA.**

    WFE isn't one market —it's dozens of niches each with a hidden leader. Kokusai (KKR-owned, IPO'd 2023) holds ~70% of batch ALD (~46% of all batch deposition), and is the process-of-record for 3D NAND's blocking oxide, charge-trap nitride and tunnel oxide at ALL top-5 NAND makers. AMAT tried to buy it for $2.2B, raised the offer 59% to $3.5B, then aborted after two years awaiting China antitrust approval —an endorsement of the moat. Batch beats single-wafer above 64-layer NAND (depth loading flips the throughput math), and GAA needs several more ALD steps, so KE's TAM structurally grows; TSMC is already its 2nd-largest customer.

    > *"KE dominates the batch ALD process with ~70% market share... For the blocking oxide, charge trap nitride and tunnel oxide, KE's batch ALD is the process tool of record among ALL of the top 5 NAND players."*
    > — Going Vertical: Gate All Around, 3D DRAM, 3D NAND - Kokusai Electric IPO (2023-10-15)

11. **Fixed costs are exploding —a 3nm mask set approaches $40M —which paradoxically drives MORE equipment demand at trailing nodes as designs proliferate.**

    Mask-set cost scales far faster than wafer cost: ~$100Ks at 90-45nm, >$1M at 28nm, >$10M at 7nm, ~$40M at 3nm. Combined with soaring verification/validation cost (Intel needed 12 steppings for Sapphire Rapids), only high-volume designs can amortize leading-edge economics —culling AI-chip startups. The flip side, per ASML, is a 40%+ increase in >28nm mask-set starts (48% per eBeam Initiative): not old chips, but NEW designs on mature nodes. That, plus regionalization, is what keeps trailing-edge DUV, deposition and etch demand —and Nikon/Canon/SMEE competition — structurally alive.

    > *"At 28nm it moves beyond $1M. With 7nm, the cost increases beyond $10M, and now, as we cross the 3nm barrier, mask sets will begin to push into the $40M range."*
    > — The Dark Side Of The Semiconductor Design Renaissance (2022-07-24)

12. **AI is coming for metrology capex: virtual/inline AI process control (e.g. Lynceus) can cut defect-detection time 90% and metrology sampling 4x.**

    Metrology/inspection is pure overhead —floorspace and cycle time that doesn't advance the wafer —yet indispensable: a 2,000-step process at 6-sigma per step still yields D0=0.678 (>half the die defective), and one undetected excursion cost TSMC $550M in 2019. Lynceus feeds tool FDC/test data into AI models to predict quality inline, achieving (in two HVM fabs) a 90% cut in time-to-detect (4 days to 3 hours) and a 4x reduction in metrology/inspection sampling. The structural implication: AI virtual metrology is a headwind to the KLA/Onto/Nova sampling-capex growth story even as complexity rises.

> *"The results are astounding with a 90% reduction in time to detect a defect from 4 days to 3 hours. Furthermore, they were able to achieve a 4x reduction in metrology and inspection sampling frequency."*
> — Lynceus: Inline, Real-time, AI Based Process Control Monitoring (2022-07-28)

Article Deep-Dives

ASML Dilemma: High-NA EUV is Worse vs Low-NA EUV Multi-Patterning (2023-12-11)

The keystone contrarian piece (co-authored by an ex-ASML analyst). Builds a bottom-up litho-cost model proving High-NA single-patterning is MORE expensive than low-NA double-patterning from 3nm through 1nm, because dose rises exponentially with shrinking CD and low-NA runs at full stage-limited throughput. Documents how ASML quietly shifted its own marketing from 'lower cost' (2020) to merely 'less complex' (2021+) —and calls the 20-High-NA-tools-by-2028 target impossible. Places natural High-NA insertion at 2030+ (1nm/7A), when scaling shifts from shrinking to stacking and CD stops falling.

EUV Requirements Halved? Applied Materials' Sculpta Redefines Lithography And Patterning (2023-03-06)

Deep dive on AMAT's Centura Sculpta 'pattern shaping' —an angled ribbon-beam plasma etch that elongates features on one axis to replace a second litho pass. Explains the three use cases (tight corner-to-corner vias, narrow tip-to-tip trenches, >90% stochastic-defect repair), the metal-interconnect insertion at a 2nm-class node in 2024-25, and quantifies a potential ~$4.5B EUV-demand reduction. Rebuts the finance/industry dismissal that it's 'just fancy RIE' —noting AMAT published since 2015 with a first customer since ~2017. Maps ramifications across ASML, Lam, TEL, JSR, KLA, Onto and the resist/mask supply chain.

Going Vertical: Gate All Around, 3D DRAM, 3D NAND - Kokusai Electric IPO (2023-10-15)

The definitive map of the deposition landscape, framed around the Kokusai Electric IPO. Explains ALD's self-limiting conformality, why batch (Kokusai ~70%) beats single-wafer above 64-layer NAND (depth loading), and where each vendor (Kokusai, ASMI, TEL, AMAT, Lam) wins in 3D NAND, DRAM capacitors and GAA. Key structural call: GAA and 3D DRAM sharply raise ALD/etch intensity while cutting litho intensity —the value migration thesis in one company. AMAT tried to buy Kokusai for $2.2B, raised the offer 59% to $3.5B, then aborted after two years awaiting China antitrust approval; TSMC is its 2nd-largest customer.

NAND Flash Monopoly Broken? Tokyo Electron Moly Dep + Cryo Etch Takes On Lam Research (2023-07-16)

Maps the four avenues of NAND scaling (logical/vertical/lateral/architectural) and shows most are tapped out, leaving HAR etch and deposition as the battleground. Two manufacturing shifts —TEL's cryogenic etch and a tungsten-to-molybdenum wordline change —threaten to move >$1B from Lam. Also delivers the memory-cycle call: 2023-24 a NAND glut (utilization ~60%, worst since 1997) setting up a 2025 shortage, with YMTC cut off by export controls removing China's counter-cyclical buffer.

High-NA is Here (for R&D), EUV Cost, Pattern Shaping Gaining Share, Metal Oxide & Dry Resist, Hyper-NA (2025-04-14)

The SPIE 2025 field report —the freshest read on the roadmap. Intel ran 30,000 wafers on two EXE:5000 tools with source power at 110% of target and 0.6nm overlay, declaring both 'in production' (running proven processes, not commercial products). IBM's cost talk confirmed SemiAnalysis's model: High-NA single exposure ~2.5x a low-NA one, only winning vs 3+ masks. Metal-oxide resist arrives with High-NA at ~A10; the wet(TEL) vs dry(Lam) resist war and pattern shaping (Sculpta / TEL Acrevia) gain share; ASML pushes the 6x12″ mask to 'early next decade' and teases hyper-NA for the CFET era.

Nanoimprint Lithography: Stop Saying It Will Replace EUV (2025-10-26)

A definitive myth-buster on Canon's J-FIL nanoimprint. On paper NIL wins on cost (~1/10 tool price, ~1/4 cost/wafer, ~90% less power) and resolution. In practice the fragile nanometer-scale stamp lasts ~50 wafers (vs >100,000 for a photomask), every repeater defect prints to every field, and inspecting every template would consume nearly the entire world's mask-inspection output. Kioxia and Micron —who want it to work —keep repeating defect/lifetime/cost complaints. Real use is niches (metalenses, MEMS). Flags a serious export-control gap: a Japan-sold NIL tool can be re-exported to China, potentially handing EUV-level capability to SMIC/Huawei.

Key Data

Low-NA EUV tool price	>$200M (NXE:3800E)	Cost per wafer is dominated by throughput, so the whole High-NA cost debate turns on this.
High-NA EUV tool price / weight	~$400M / 150 metric tons (EXE:5000)	~2x a low-NA tool; depreciates >$150,000/day.
High-NA vs low-NA single-exposure cost	~2.5x (IBM SPIE data)	So High-NA only wins where it replaces 3+ low-NA masks; 4-mask SALELE is 1.7-2.1x a single High-NA pass.
Lithography share of a 3nm node cost	~35%	The prize everyone wants to shrink via pattern shaping / DSA / vertical scaling.
High-NA source power at first shipment	110% of target	First EUV system ever to exceed initial target (vs NXE:3300 at 15%, NXE:3400B at 50%); reliability ~85%.
Intel High-NA wafers exposed	30,000 across 2 EXE:5000 tools	Intel is years ahead; TSMC/Samsung only ordered R&D tools. Overlay 0.6nm to a low-NA tool.
6x12″ mask throughput uplift	+23-50%	Would eliminate High-NA stitching and swing economics —but ASML says only 'early next decade,' and 2 normal-mask tools = $200-300M more ASML revenue.
Sculpta per-layer savings	~$250M capex / 100k WSPM + ~$50/wafer opex	AMAT's claim vs adding an EUV loop; potential ~$4.5B EUV demand reduction at full adoption.

3D NAND density / cost-per-bit scaling	+30%/yr density, -21%/yr cost/bit	Since 2013; slowing to low-to-mid-teens cost decline as HAR etch hits the 6-7 micron / 70:1 wall.
Kokusai batch-ALD share	~70% (batch ALD); ~46% (all batch deposition)	AMAT tried to buy it for $3.5B (raised 59%) before China antitrust blocked it.
TEL EUV coat/develop track share	100%; ~$5B+ annual run-rate	Monopoly on the track around every ASML tool, pulling in adjacent etch/clean wins.
Lam dry-resist revenue target	~$1.5B over 5 yrs; multi-$B by decade end	Higher gross margin than Lam's ~46% corporate average; JSR bought Inpria for $514M to defend MOR.
NIL mask lifetime vs photomask	~50 wafers vs >100,000	The defectivity killer; NIL throughput ~100 wph (4-cell) vs 220 (EUV)/330 (DUV).
3nm mask set cost	~$40M (vs >$10M at 7nm, >$1M at 28nm)	Fixed-cost explosion; a leading-edge chip tape-out runs $50-75M all-in at 7nm.
Lynceus AI process-control gains	90% faster defect detect (4d to 3h); 4x less sampling	In two HVM fabs; a structural headwind to metrology sampling capex.
Lam Malaysia manufacturing shift	>1/3 of future capacity; +50% total footprint (~845k sq ft)	Cost-driven offshoring of high-value tool manufacturing from the US.
ASML 2030 revenue / WFE market	$44-60B ASML implies ~$150-280B WFE	SemiAnalysis: guidance is optimistic and bakes in 10-20% 'inefficient' regionalized capacity.

Investment Angle

Winners

ASML	Absolute EUV/High-NA monopoly and DUV leader —the hardest bottleneck in tech. But the SemiAnalysis nuance is a double-edge: High-NA ramp is slower and less economic than guided, and a customer-friendly 6x12″ mask (which ASML resists because 2 normal tools = more revenue) is the swing factor. Own the moat, discount the guidance.

Applied Materials (AMAT)	#2 in WFE and the broadest portfolio (deposition, etch, PVD, implant, plus Sculpta pattern shaping and panel-level packaging PVD). Structurally positioned for the value migration away from pure litho toward materials-engineering and vertical scaling; Sculpta is a new litho-adjacent TAM.
Lam Research (LRCX)	HAR-etch and deposition leader —the engine of 3D NAND/DRAM vertical scaling —plus the dry-resist disruptor (~$1.5B target, above-corporate margin). Risk: TEL cryo etch + moly change contest >$1B of NAND. Malaysia offshoring lifts gross margin.
Tokyo Electron (TEL)	100% of EUV coat/develop tracks (~$5B+ run-rate monopoly), a strong batch-ALD #2, and the cryo-etch/moly challenger poised to take NAND share from Lam. Defends resist via wet-develop innovation; JSR/Inpria alliance backs MOR.
KLA	Dominant in process control/metrology —a toll booth on every fab across 1000+ steps, rising with complexity. Watch the AI/virtual-metrology headwind (Lynceus-type) to sampling capex, but the moat in physical inspection is deep.
Kokusai Electric	Hidden batch-ALD monopoly (~70%), process-of-record at all top-5 NAND makers, TAM growing with GAA. AMAT's aborted $3.5B buyout (a $2.2B offer raised 59%, then abandoned after two years awaiting China antitrust approval) is the third-party validation of the moat.
Enablers: Veeco, EV Group, BE Semi, Lasertec, JSR, Entegris	Niche monopolies riding 3D/heterogeneous integration: Veeco (EUV mask-blank deposition, laser anneal, GaN); EV Group/BE Semi (wafer/hybrid bonding); Lasertec (actinic EUV mask inspection); JSR/Entegris (resist & materials).

Bottlenecks

- EUV source power —CO2-laser-on-tin-droplet EUV tops out; dose requirements outrun ASML/Trumpf's power roadmap, capping throughput and forcing multi-patterning or DSA.
- EUV mask ecosystem —the 6″x6″ reticle standard (since the 1980s) and actinic mask inspection are chokepoints; a larger 6x12″ mask needs the whole blank/write/inspect chain rebuilt.
- HAR etch depth —6-7 micron / 70:1 channel-hole wall limits NAND vertical scaling; cryo etch and string-stacking are the only ways up.
- Fixed-cost explosion —$40M 3nm mask sets + verification (Intel's 12 Sapphire Rapids steppings) restrict the leading edge to high-volume designs.

Risks

- WFE cyclicality —a memory glut (2022-23 NAND utilization ~60%) or fab-buildout delays (TSMC/Intel/Samsung push-outs) can cut equipment revenue sharply and quickly.

- High-NA over-investment —being too early with a billion-dollar fleet on a technology that's not yet cheaper than low-NA (Intel's bet) could be as damaging as being late.
- China substitution + export-control blowback —SMEE/AMEC/Naura rising at trailing edge; controls that only cover ArFi leave DUV/resist/service holes, and cut off a huge China TAM for Western toolmakers.
- Value migration mis-bet —backing the wrong patterning path (High-NA vs multi-patterning vs Sculpta vs DSA vs MOR/dry resist) strands R&D and capacity.

Catalysts

- AI capex super-cycle —hyperscaler datacenter/AI spend drives leading-edge logic + HBM/DRAM WFE; larger dies and more designs lift both advanced and trailing WFE.
- GAA + 3D DRAM ramp —nanosheet logic and 3D DRAM sharply raise ALD/etch intensity, reshuffling deposition/etch share (Kokusai, ASMI, Lam, TEL, AMAT).
- 2025 memory shortage —the 2023-24 NAND glut + capex delays set up a sharp WFE recovery as supply/demand rebalance.
- 6x12″ mask decision + High-NA/DSA insertion at 14A/A10 —an ASML commitment to large masks, or Intel proving High-NA+DSA economical, would re-rate the whole litho roadmap.

Open Questions

- Does High-NA ever become genuinely cheaper than low-NA multi-patterning —i.e. does the 6x12″ mask arrive, and can DSA/MOR cut dose enough to move insertion from A10/1nm forward to 14A?
- Who wins the resist war: TEL/JSR wet MOR (upgrade-only tracks) or Lam dry resist (new CVD tools, ~2x throughput)? The answer decides billions in coater/developer/etch share.
- Can Tokyo Electron actually pry >$1B of HAR-etch and wordline deposition from Lam via cryo etch + molybdenum —or does Lam defend its NAND crown?
- How far and fast do Chinese toolmakers (SMEE, Naura, AMEC) climb, and will Japan/Netherlands close the DUV/resist/service/re-export loopholes before SMIC scales 7nm/5nm on reconfigured DUV?
- Does AI virtual metrology (Lynceus-style) meaningfully shrink KLA/Onto/Nova sampling capex, or does rising process complexity keep physical inspection growing faster than AI can offset?

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Chapter 6

Networking, Interconnect & Optics

Bandwidth, not FLOPs, is the wall —copper reaches its 448G ceiling, and light takes over scale-up.

Overview

In the AI datacenter, networking is the second-largest capex and power bucket behind the accelerators themselves —roughly 15% of a GB300 NVL72 cluster's cost on a 3-layer InfiniBand network, rising to 18% on 4 layers, with the back-end (scale-out) fabric alone accounting for ~85% of networking cost and ~86% of networking power. SemiAnalysis frames the sector around two distinct fabrics with opposite economics. Scale-up (NVLink, UALink, Broadcom SUE) is the ultra-high-bandwidth, low-latency, all-to-all mesh that fuses dozens of GPUs into one 'world size'; NVLink 5 already moves 7.2 Tbit/s (900 GB/s) per GPU —9x the 800 Gbit/s of scale-out —and doubles again in Rubin. Scale-out (InfiniBand vs. Ethernet) links thousands of racks. The central tension: copper is cheap, cool and reliable but capped at ~2 m reach and is running out of SerDes runway (Nvidia is delivering '448G' in Rubin only via a bi-directional trick, not true 448G), while pluggable optics carry ~60% of scale-out networking cost and burn a DSP that eats ~50% of an 800G module's power. Co-packaged optics (CPO) is SemiAnalysis's marquee thesis: it kills the DSP and the long-reach SerDes, cuts optical-link energy ~65-73%, and —critically —removes the reach limit so scale-up world sizes can span multiple racks. But CPO's TCO win in scale-out is modest (only ~3-7% cluster cost, 2-4% cluster power), so the real prize is scale-up, arriving with Rubin Ultra / Feynman and hyperscaler ASICs (AWS Trainium 4) at the end of the decade. Around this sit the merchant-silicon wars —Broadcom vs. Marvell vs. Nvidia in switch ASICs, DSPs and custom XPUs —the Ethernet-vs-InfiniBand standards fight (UEC, UALink, SUE), Google's optical circuit switching, and a China variant (Huawei CloudMatrix) that throws power and 100%-optics at the problem because it lacks silicon, not electricity.

Core Knowledge

Scale-up vs Scale-out

Scale-up = the ultra-fast, low-latency, all-to-all fabric fusing GPUs into one shared-memory 'world' (NVLink, UALink, SUE); ~9x the per-GPU bandwidth of scale-out. Scale-out = the back-end fabric linking racks/pods (InfiniBand or Ethernet). Scale-up demands are so much higher that scale-up interconnect TAM already dwarfs scale-out.

Co-Packaged Optics (CPO) / NPO / OBO

Placing the optical engine on the same package as the switch/XPU ASIC, eliminating the DSP and the 15-30 cm electrical run to a front-panel transceiver. NPO = optical engine socketed on a nearby substrate (detachable); OBO = engine on the system PCB ('worst of both worlds'). CPO cuts link energy 65-73% and removes copper's reach limit.

DSP / LPO / DR/FR optics

The DSP retimes and error-corrects the electrical signal inside a pluggable transceiver; it is the single most power-hungry and costly module component (~50% of an 800G SR8's power, 20-30% of BoM). LPO (linear pluggable) tries to delete the DSP by driving optics straight from switch SerDes; DR (parallel single-mode, ~500m-2km) and FR (WDM, 2-10km) are the datacenter reach classes.

SerDes & the 224G→448G wall

SerDes serialize parallel data onto a differential pair; Nvidia/Broadcom lead at 224G. Higher speeds raise insertion loss and only survive over very short reach. True 448G uni-directional needs PAM6/8 and is uncertain; Nvidia's Rubin '448G per lane' is actually bi-directional 224G+224G sharing one wire —a clever dodge, not a real doubling.

Modulators: MZM vs MRM vs EAM

MZM (Mach-Zehnder): easiest, thermally stable, but huge (~12,000 µm2). MRM (micro-ring): tiny (25-225 µm2), built-in WDM, but 10-100x more temperature-sensitive —a 2°C drift shifts resonance past collapse. EAM (electro-absorption): compact, thermally tolerant to ~80°C, Celestial AI's differentiator but stuck in C-band. Nvidia shocked the industry by shipping 200G-per-lane MRMs.

TSMC COUPE

TSMC's 'Compact Universal Photonic Engine' —the emerging standard for heterogeneously integrating a PIC (photonic IC, N65) with an EIC (electronic IC, N6) via bumpless SoIC hybrid bonding, giving >23x the bandwidth density of bumped integration at iso-power. Ships first in Nvidia's CPO; Broadcom, Ayar, Lightmatter all pivoting to it. Locks customers to TSMC-made PICs.

World size & rail-optimized topology

World size = the number of GPUs in a single all-to-all scale-up domain (GB200 jumped from 8 to 72). Rail-optimized = each server's 8 GPUs connect to 8 different leaf switches so any GPU reaches distant GPUs in one hop —great for MoE all-to-all, but forces optics (vs. cheap copper in a middle-of-rack design). Max hosts = k^L (ports^layers).

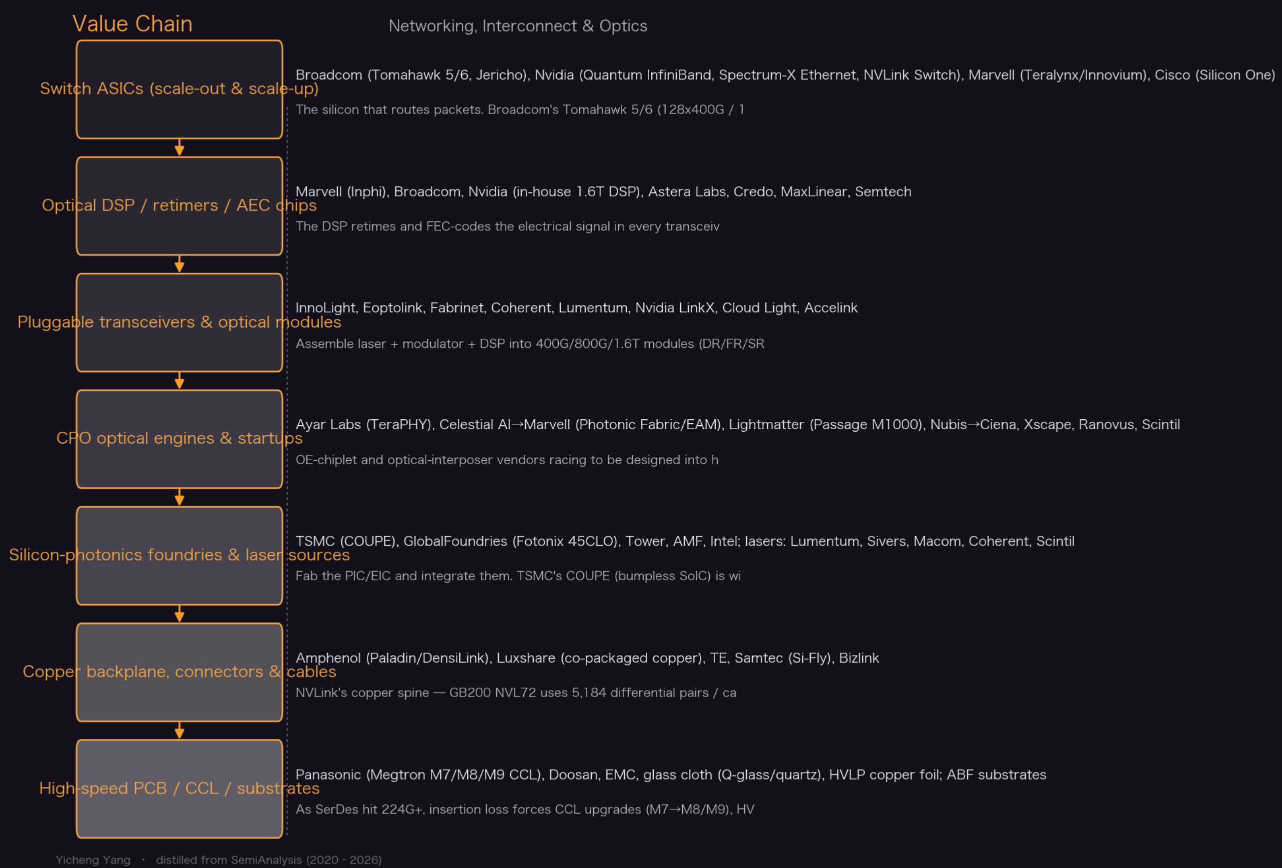
UEC / UALink / SUE

The open-standard answers to Nvidia's proprietary stack. UEC (Ultra Ethernet, 565-page spec) hardens Ethernet for scale-out with LibFabric, packet spraying and proper congestion control (deprecating RoCE/DCQCN). UALink and Broadcom's Scale-Up Ethernet (SUE, a lean 20-page spec) target scale-up only —single switch layer, up to 1024 ports —and will ship as IP blocks inside host chips.

Optical Circuit Switch (OCS)

A MEMS-mirror switch that redirects light without decoding packets —bandwidth- and wavelength-agnostic, 'set-and-forget' (reconfig takes seconds). Google's in-house Apollo/Palomar 136x136 OCS uses ~108 W vs. ~3,000 W for an equivalent electrical switch, saves >$3B, and lets Google reuse the switch across optics-speed upgrades. Used in all TPU v4/v5 pods.

Value Chain



Key Theses

1. **CPO's killer app is scale-up, not scale-out —because in scale-out the TCO win is diluted to almost nothing.**

   CPO cuts scale-out transceiver power ~84% and cost ~86%, but networking is only ~9% of cluster power to begin with, so a 3-layer CPO network saves just ~2% total cluster power and ~3% cluster cost (2-layer: ~4% power, ~7% cost). Against that sit lost transceiver-vendor bargaining power, blast-radius fear and field-service pain. Scale-up is where the reach limit and SerDes wall actually bite, so that is the real prize.

   > *"CPO for scale-up is now not a matter of if and why, but when and how."*
   > — Co-Packaged Optics (CPO) Book –Scaling with Light (2026-01-01)

2. **Copper's SerDes runway is nearly gone; Nvidia's Rubin '448G' is a bi-directional trick, and that erosion of the NVLink moat lets AMD and hyperscalers catch up.**

   NVLink bandwidth grew 11x from v1 to v5 almost entirely via SerDes speed (20G→200G), not lane count. NVLink 6 stays on 200G and doubles bandwidth only by simultaneous bi-directional signaling (224G TX + 224G RX on one differential pair with echo cancellation), avoiding a doubling of the ~5,000 backplane cables to ~10,000. True 448G uni-directional needs PAM6/8 and is uncertain. Once SerDes stalls, Nvidia's scale-up moat narrows.

> *"For Nvidia, whose NVLink scale up fabric is an important moat, this roadblock could make it easier for competitors such as AMD, and the hyperscalers to catch up."*
> — Co-Packaged Optics (CPO) Book –Scaling with Light (2026-01-01)

3. **The DSP is 'public enemy number one' of optical cost and power —which is why the industry is trying to delete it.**

   The DSP is ~50% of an 800G SR8 module's power and 20-30% of its BoM. An 18k-GPU GB300 cluster's back-end alone would burn ~480 kW of DSP power (~1.8 kW per rack). LPO (linear pluggable) attacks it first by driving optics straight from switch SerDes, but LPO hasn't taken off as Marvell's Loi Nguyen predicted; CPO finishes the job by co-packaging the engine so no DSP is needed at all.

   > *"So –accounting for 50% of the power draw and 20-30% of the BoM of a typical transceiver –some regard DSPs as public enemy number one of cost and power efficiency."*
   > — Co-Packaged Optics (CPO) Book –Scaling with Light (2026-01-01)

4. **TSMC's COUPE has quietly won CPO integration —even taking Broadcom, the CPO pioneer, off its own packaging.**

   COUPE's bumpless SoIC hybrid bonding gives >23x the bandwidth density of bumped integration at iso-power. Broadcom's own FOWLP (via SPIL) can't scale past 100G/lane due to through-mold-via parasitics, forcing Broadcom to move future CPO to COUPE —and to switch from edge-coupling+MZM to grating-coupling+MRM, essentially starting fresh. Adopting COUPE locks the customer into TSMC-made PICs, since TSMC won't package others' SiPho wafers.

   > *"This highlights TSMC's technological edge, enabling them to secure wins even in optics, a domain where they have historically been considered weaker."*
   > — Co-Packaged Optics (CPO) Book –Scaling with Light (2026-01-01)

5. **The 'optical boogeyman' that actually cuts transceiver demand is copper NVLink inside the rack —not the NVL72 itself.**

   When Jensen said NVL72's copper NVLink saved 20 kW of transceivers, optical investors panicked that scale-out optics demand would fall. SemiAnalysis: false —the NVL72 still has 72 back-end OSFP ports (1 per GPU), the same transceiver-to-GPU ratio as H100, so scale-out optics scale with GPU count. Using optics for the 900 GB/s NVLink spine would have needed 648 1.6T transceivers at ~$550k/rack of BoM (~$2.2M at 75% margin) —which is exactly why Nvidia used 5,184 copper cables.

   > *"For the back end scale out network –the NVL72 rack showcased at GTC still has 72 OSFP ports at 400G / 800G –one for each GPU... As GPU network sizes scale, the number of optical transceivers required also scales."*
   > — Nvidia's Optical Boogeyman (2024-03-25)

6. **Reliability, not TCO, is CPO's real gate —80% of returned pluggable modules are 'no trouble found', and Meta's 15 CPO switches over 11 months isn't enough proof.**

   A ~1M-link pluggable cluster sees dozens of link flaps/day, and 80% of returned modules are 'no trouble found' —CPO's determinism is the draw. Meta/Broadcom (Bailly 51.2T) showed 2.6M-hour MTBF for CPO vs 0.5-1M for 2xFR4 pluggables, zero UCWs to 4M port-device-hours. But 15M '400G port-device-hours' is only ~15 switches $\times$ 11 months in a lab; a 0.06% unserviceable failure has a 64-port blast radius. SemiAnalysis calls for far larger field tests before billions commit.

> *"80% of optical modules that are returned due to some link failure are 'no trouble found'."*
> — Co-Packaged Optics (CPO) Book –Scaling with Light (2026-01-01)

7. **Ethernet is clawing back share from InfiniBand; Nvidia's own Spectrum-X now out-ships Quantum InfiniBand in Blackwell.**

   Ethernet lost early ground to InfiniBand but is returning on cost and flexibility. Spectrum-X SN5600 and Broadcom Tomahawk 5 both offer 128x400G vs Quantum-2's 64x400G, letting a 100k cluster be 3 tiers not 4 (1.33x fewer transceivers). Hyperscalers pick Tomahawk 5 to dodge the 'Nvidia tax' and mix any transceiver —the cost of admission is patching NCCL yourself. UEC standardizes these gains across the industry.

   > *"Even Nvidia recognizes the dominance of Ethernet and with the Blackwell generation, Spectrum-X Ethernet is out shipping their Quantum InfiniBand by a large amount."*
   > — The New AI Networks | Ultra Ethernet | UALink vs Broadcom SUE (2025-06-11)

8. **Optical circuit switching gives Google a >$3B structural cost and power edge —and threatens Broadcom's spine-switch business.**

   Google's in-house Apollo/Palomar 136x136 OCS (MEMS mirrors, O-band 1310nm) draws ~108 W vs ~3,000 W for an equivalent electrical switch, and because it's bandwidth/wavelength-agnostic it survives multiple optics upgrades —so Google eliminates Broadcom switches in the spine and reuses OCS across generations. Google claims +30% throughput, -40% power, -30% capex, 50x less downtime, and used it to train PaLM; OCS underpins every TPU v4/v5 pod.

   > *"This custom networking stack enables them to save at least $3 billion versus the industry standard implemented by competitors such as Amazon and Microsoft."*
   > — Google OCS Apollo: The >$3 Billion Game-Changer (2023-03-17)

9. **Broadcom is the silent #2 AI chip company —its networking + custom-silicon franchise (Google TPU, Meta MTIA, now OpenAI) is the anti-Nvidia bet.**

   Networking is the second-largest AI infra spend bucket after accelerators, and Broadcom leads merchant switching (Tomahawk/Jericho) plus optical components. Its custom-silicon arm designs Google's TPU and Meta's MTIA and grew from <20% of LSI revenue to $2-3B+. In CPO, Broadcom shipped the first production systems (Bailly, Humboldt) and its CPO experience made it OpenAI's ASIC design partner —precisely because CPO is on OpenAI's roadmap.

   > *"Broadcom is the second largest AI chip company in the world in terms of revenue behind NVIDIA, with multiple billions of dollars of accelerator sales."*
   > — Broadcom's Google TPU Revenue Explosion (2023-08-30)

10. **Marvell's DSP moat is under 'death by a thousand cuts' —and the CPO era favors rivals who are ahead in productization.**

    Marvell (via Inphi) leads 112G PAM4 DSPs/TIAs and coherent optics, and had 100% of Nvidia's H100-gen DSP. But Broadcom, Nvidia, Arista, Microsoft, Meta and Macom are all designing Marvell out —Broadcom broke into Nvidia's GB200 DSP, and in CPO, Broadcom/Intel/Nvidia/Ayar are ahead in productization timelines. Marvell's counter is custom XPUs and its multi-billion-dollar Celestial AI buy —with a $2.25B revenue-contingent earn-out —to own scale-up CPO for AWS Trainium 4.

> *"All the other titans in networking including Broadcom, Nvidia, Arista Networks, Microsoft, Meta, Macom, and many others are rallying toward designing Marvell out."*
> — Marvell's DSP Dilemma? Networking's Tectonic Shift (2023-03-08)

11. **Huawei's CloudMatrix 384 shows a system-level win —beating GB200 NVL72 on aggregate compute by throwing 100% optics and ~4x the power at the problem.**

    CM384 wires 384 Ascend 910C chips all-to-all across 16 racks (100% optics, 0% copper, ~6,912 400G LPO transceivers, ~18 per chip). It delivers ~300 PFLOPs dense BF16 (~2x GB200 NVL72), 3.6x memory capacity and 2.1x bandwidth —but at 4.1x the power and 2.5x worse power/FLOP. Because China is silicon-constrained not power-constrained, forgoing copper density for optical scale-up is the rational trade. It resembles Nvidia's abandoned DGX H100 NVL256 'Ranger'.

    > *"Huawei is a generation behind in chips, but its scale-up solution is arguably a generation ahead of Nvidia and AMD's current products on the market."*
    > — Huawei AI CloudMatrix 384 (2025-04-16)

12. **Rubin's networking innovation is evolutionary, but content growth is real —2x NICs per GPU, ~2.3x high-end PCB area, and cableless design.**

    VR NVL72 doubles scale-out to 1.6T per GPU by putting two 800G ConnectX-9 packages per GPU (not one faster NIC), enabling multi-plane deployments; CX-9 adds 800G Ethernet on 4x200G SerDes. NVLink Switch count doubles to 36/rack. Cableless design (no flyover cables except switch-to-SMM) forces PCB upgrades to M8/M9 CCL, HVLP4 foil and (debated) quartz cloth, growing high-end PCB area ~2.3x vs GB300. VR NVL72 costs ~45% more per GPU than GB300.

    > *"NVLink 6 used in Vera Rubin doubles NVLink bandwidth by implementing bi-directional signaling over the same number of copper cables - effectively delivering 4 Lanes of 200G per NVLink."*
    > — Vera Rubin –Extreme Co-Design (2026-02-25)

13. **High-radix 'switch-in-a-box' lets Nvidia flatten networks and hide the shuffle —a real selling point that also raises switching cost.**

    Nvidia's CPO switches present as very high port count —Quantum 3450 = 144x800G, Spectrum 6800 = 512x800G —by internally shuffling across 4 ASICs (a multi-plane 'topology in a box'). Spectrum 6800 can connect 131,072 GPUs on 2 layers vs 8,192 for the 6810, letting customers drop from 3 layers to 2 and skip patch panels/octopus cables. Because max hosts = k^L, doubling logical ports 4x's the cluster size —the magic of high radix.

    > *"the Spectrum 6800 at 512 ports of 800G can connect 131,072 GPUs."*
    > — Co-Packaged Optics (CPO) Book –Scaling with Light (2026-01-01)

Article Deep-Dives

Co-Packaged Optics (CPO) Book –Scaling with Light for the Next Wave of Interconnect (2026-01-01)

The sector's defining 24k-word treatise. Five parts: (1) TCO —CPO's scale-out savings dilute to 2-7% at cluster level, so scale-up is the killer app; (2) the tech —DSP/LPO/CPO evolution, SerDes hitting the 224G→448G wall, Wide I/O (UCIe ~10 Tbit/s/mm), co-packaged copper as a 448G bridge; (3) bringing it to market —TSMC COUPE bumpless SoIC (>23x density), edge vs grating coupling, and the

MZM/MRM/EAM modulator trade-offs (Nvidia ships 200G MRMs); (4) products —Nvidia Quantum-X/Spectrum-X, Broadcom Bailly/Davisson, Intel/MediaTek roadmaps, and deep profiles of Ayar Labs, Nubis (→Ciena), Celestial AI (→Marvell, EAM/Trainium 4), Lightmatter (M1000 optical interposer), Xscape, Ranovus, Scintil; (5) Nvidia's CPO supply chain (paywalled). Verdict: scale-out CPO ships only 10-15k units in 2026 as a pipe-cleaner; the real deployment is scale-up at Rubin Ultra/Feynman.

Vera Rubin –Extreme Co-Design: An Evolution from Grace Blackwell Oberon (2026-02-25)

Full teardown of Nvidia's Rubin platform's six co-designed silicon products and the VR NVL72 rack. Networking core: NVLink 6 doubles bandwidth via bi-directional SerDes (224G TX+224G RX per differential pair with echo cancellation over ~1m), keeping backplane cable count flat but doubling NVLink Switch chips to 36/rack. ConnectX-9 stays at 800G but doubles NICs per GPU to give 1.6T scale-out; Spectrum-6 CPO (102.4T/409.6T) enables larger scale-out. The rack goes cableless —only one flyover cable (switch-to-SMM) remains —forcing PCB upgrades to M8/M9 CCL, HVLP4 copper, 32-layer NVLink6 boards, and (debated) quartz cloth, growing high-end PCB area ~2.3x. VR NVL72 is ~45% more expensive per GPU than GB300, and is the first Nvidia generation to see CPO in the scale-out back-end.

GB200 Hardware Architecture - Component Supply Chain & BOM (2024-07-17)

The definitive GB200 rack BoM across 50+ subcomponents and four form factors (NVL72, NVL36x2, Ariel, x86 Miranda). Networking essence: four fabrics (front-end Ethernet, back-end IB/Ethernet, NVLink scale-up, OOB); NVLink uses 5,184 copper differential pairs on Amphenol Paladin backplane —most cost in connectors, not wire —and copper beats optics decisively (optical spine = ~$550k/rack BoM, 20 kW). Debunks the '$3k/GPU NVLink' investor myth. CX-7→CX-8 drives 400G SR4 → 800G DR4 optics and 1.6T DR8; Nvidia expands transceiver supply to Eoptolink (800G LPO/1.6T DSP) beyond Fabrinet/InnoLight, and Broadcom breaks Marvell's DSP monopoly.

100,000 H100 Clusters: Power, Network Topology, Ethernet vs InfiniBand, Reliability (2024-06-17)

Anatomy of frontier training clusters. A 100k H100 cluster needs ~150 MW and swallows 1.59 TWh/yr. Network design follows parallelism (tensor within node/NVLink, pipeline within island, data across islands). Rail-optimized vs middle-of-rack is a copper-vs-optics choice: a non-rail design converts 98,304 transceivers to cheap DAC copper (25-33% of the GPU fabric). Reliability dominates: optics link failures give ~26 min to first job failure, forcing memory-reconstruction recovery. Spectrum-X vs Tomahawk 5: TH5 dodges the 'Nvidia tax' and needs its own NCCL work; Spectrum-X's first gen needs power-hungry BlueField-3 (+5 MW/100k).

The New AI Networks | Ultra Ethernet UEC | UALink vs Broadcom Scale Up Ethernet SUE (2025-06-11)

In-depth read of the open-standard stack challenging Nvidia. UEC RC1 (565 pages) hardens Ethernet for scale-out: hardware-accelerated LibFabric, the 'Job' abstraction, packet spraying with entropy-based load balancing, time-based congestion control (sub-500ns) that deprecates RoCE/DCQCN/PFC, and post-quantum security. UALink and Broadcom SUE (a lean 20-page spec) target scale-up only —single switch layer, ≤1024 ports —and will ship as IP blocks inside host chips (like NVLink). SUE delegates spraying/load-balancing to software (a win for software-strong firms); UALink is heavier with credit-based flow control. UEC lags on referencing 200G links.

Google OCS Apollo: The >$3 Billion Game-Changer in Datacenter Networking (2023-03-17)

How Google's in-house optical circuit switching rewires datacenter economics. Apollo replaces electrical packet switches in the spine with MEMS-mirror OCS that redirect light without decoding packets —bandwidth/wavelength-agnostic and 'set-and-forget'. The Palomar 136x136 OCS (moved to O-band 1310nm to cut crosstalk) draws ~108 W vs ~3,000 W for an equivalent EPS, is reusable across optics-speed upgrades (~70% of EPS capex), and eliminates Broadcom spine switches. Google claims +30% throughput, -40% power, -30% capex, 50x less downtime; OCS underpins all TPU v4/v5 systems and helped train PaLM.

Key Data

Networking share of AI cluster cost	15% (3-layer IB) → 18% (4-layer)	Second-largest capex after the server itself, for a GB300 NVL72 cluster.
Back-end fabric share of networking cost/power	~85% cost / ~86% power	Scale-out back-end dominates; transceivers alone ~60% of 3-layer networking cost, ~45% of power.
CPO optical-link power saving vs DSP transceiver	65-73% (up to 80% aspired)	800G DSP transceiver ~16-17W vs CPO OE+ELS ~4-5.4W per 800G.
DSP share of transceiver power / BoM	~50% power / 20-30% BoM	For an 800G SR8; ~480 kW of DSP power in an 18k-GPU GB300 back-end.
NVLink 5 per-GPU scale-up bandwidth	7.2 Tbit/s (900 GB/s), 9x scale-out	Rises to 14.4 Tbit/s in Rubin; scale-out per GPU is 800 Gbit/s on CX-8.
Copper scale-up reach limit	~2 m at 200G/lane	Caps NVLink world size to 1-2 racks; optics removes this to scale across racks.
GB200 NVL72 copper NVLink cabling	5,184 differential pairs / cables	Optics for the same spine would need 648 1.6T transceivers, ~20 kW, ~$550k/rack BoM.
NVL576 optical scale-up BoM penalty	>$5.6M/rack (~$9.7k/GPU)	~$38.8k/GPU to customer at 75% margin —why optical scale-up (and DGX H100 NVL256) never shipped.
CPO scale-out cluster power / cost saving	2-4% power / 3-7% cost	Networking is only ~9% of cluster power, so CPO's big networking savings dilute at cluster level.
Meta/Broadcom CPO reliability (Bailly 51.2T)	2.6M-hr MTBF vs 0.5-1M for 2xFR4	Zero UCWs to 4M port-device-hours across 15 switches over ~11 months.
TSMC COUPE bandwidth-density gain	>23x (SoIC bumpless vs bumped, iso-power)	PIC on N65, EIC on N6; interposer-integrated OEs could reach ~4 Tbit/s/mm.
Modulator sizes (MZM vs MRM vs EAM)	~12,000 vs 25-225 vs ~250 µm2	MRM ~50-500x smaller than MZM but 10-100x more temperature-sensitive.
Nvidia CPO MRM line rate milestone	200G/lane PAM4 MRM in production	Matches fastest MZMs; disproves the notion MRMs are NRZ-limited.

Ethernet vs InfiniBand switch radix	128x400G (TH5/SN5600) vs 64x400G (Q-2)	Higher radix lets a 100k cluster run 3 tiers not 4 (1.33x fewer transceivers).
Google Apollo OCS power	~108 W vs ~3,000 W (136-port EPS)	Bandwidth/wavelength-agnostic; used in all TPU v4/v5 pods; saves >$3B.
Huawei CloudMatrix 384 optics	6,912 400G LPO transceivers, ~18/chip	100% optics / 0% copper; ~300 PFLOPs BF16 but 4.1x GB200 NVL72 power.
Time to first job failure (100k H100)	~26.3 min (single 400G) → ~42 min (Cedar-7)	Optics failures dominate; drives memory-reconstruction fault recovery over checkpointing.
Celestial AI link efficiency	~2.5 pJ/bit E-O-E (+0.7 laser) vs ~10 pJ/bit copper	Marvell buy: $500M run-rate exiting FY28, guided to $1B by CY28-end; Trainium 4 target.

Investment Angle

Winners

TSMC	COUPE bumpless SoIC is winning next-gen CPO integration from GF/Tower —even Broadcom, Ayar and Lightmatter are pivoting to it, and using COUPE locks customers into TSMC-made PICs.
Broadcom (AVGO)	The silent #2 AI chip company: merchant switch leader (Tomahawk 5/6), first to ship production CPO (Bailly/Humboldt/Davisson), and custom-XPU designer for Google TPU, Meta MTIA and now OpenAI.
Nvidia (NVDA)	Owns the scale-up moat (NVLink), bundles Spectrum-X/Quantum with NCCL, and is pipe-cleaning the CPO supply chain first —Spectrum-X already out-ships Quantum in Blackwell.
Marvell (MRVL) + Celestial AI	Leads 112G/224G PAM4 DSPs and coherent optics; the multi-billion-dollar Celestial AI buy (EAM optical interposers, with a $2.25B revenue-contingent earn-out) targets scale-up CPO in AWS Trainium 4 with $1B run-rate guided by CY28-end.
Amphenol / connector & copper vendors	NVLink's 5,184-cable copper backplane (Paladin) plus co-packaged copper (Luxshare) keep growing as copper persists in-rack even into Rubin; connectors are the value, not wire.
CCL / high-speed PCB materials (Panasonic, Doosan)	Rubin's cableless design forces M8/M9 CCL, HVLP4 foil and (debated) quartz cloth, growing high-end PCB area ~2.3x vs GB300 —a quiet content-growth story.

Transceiver assemblers (InnoLight, Eoptolink, Fabrinet)	Optics scale with GPU count (the 'boogeyman' fear is wrong); CX-8 drives 800G DR4/1.6T DR8; supply broadens as Nvidia adds Eoptolink for LPO/1.6T DSP.

Bottlenecks

- External Laser Sources (ELS) —the top CPO failure point; production ramp of multi-wavelength lasers (Sivers, Lumentum, Lightmatter GUIDE) gates scale-up CPO.
- Fiber attach / FAU alignment —still largely manual, yield drops with each fiber; automation (Ficontec) is low-throughput. Nubis' 36-fiber 2D array is the densest shipping.
- MRM thermal control —a 2°C drift shifts resonance past collapse; managing it under a hot XPU (Lightmatter/Celestial placing modulators beneath the die) is unproven at scale.
- CPO supply-chain immaturity —only ~10-15k scale-out CPO switches ship in 2026 as a pipe-cleaner; Rubin Ultra CPO timeline (late 2027) is 'too ambitious', pushing real scale-up to Feynman.
- Copper's 448G wall —true 448G uni-directional SerDes needs PAM6/8 with uncertain timing; co-packaged copper (twinax on substrate) is the interim bridge.

Risks

- CPO's scale-out TCO is underwhelming (2-7% cluster cost) —adoption could stall if reliability/serviceability fears outweigh a modest saving; Google refuses CPO on reliability grounds.
- Vendor lock-in loss of bargaining power —CPO collapses a multi-transceiver-vendor market into a few switch vendors, a key reason hyperscalers hesitate.
- Startup crowding vs incumbents —Ayar/Celestial/Lightmatter/Nubis must out-execute Nvidia/Broadcom/Marvell who already have proprietary solutions; consolidation is underway (Nubis→Ciena, Celestial→Marvell).
- Marvell DSP moat erosion —Broadcom, Nvidia and in-house hyperscaler efforts are designing Marvell out of DSPs; its thesis now rides on custom XPU + Celestial CPO execution.
- Ethernet share shift away from Nvidia InfiniBand —hyperscalers adopting Tomahawk 5 + UEC erodes Nvidia's networking attach, though Nvidia counters with Spectrum-X.

Catalysts

- Nvidia scale-out CPO ramp (Quantum-X 2H25 IB, Spectrum-X 2H26 Ethernet) as the supply-chain pipe-cleaner; watch 10-15k unit 2026 volume.
- AWS Trainium 4 with Celestial/Marvell scale-up CPO ramping late 2027 —the first high-volume scale-up CPO deployment; Amazon warrants vest on Photonic Fabric purchases.
- Larger-scale Meta/Broadcom CPO field reliability data —the industry needs more than 15 lab switches before committing billions to scale-up CPO.
- UEC RC1 productization + interoperability plugfests, and UALink/SUE IP appearing inside hyperscaler host chips.
- Rubin Ultra 'Kyber' (144 GPU packages / 576 dies, 4x NVL72 density) and the eventual copper-to-optics scale-up transition at Feynman.

Open Questions

- When does copper actually hit the wall on scale-up? Nvidia insists on copper (co-packaged copper, bi-directional SerDes) through Rubin Ultra 2027-28 —will 448G force optics at Feynman, or can CPC stretch copper further?
- Which modulator wins scale-up CPO —Nvidia's MRM (proven at 200G but thermally fragile), Nubis' MZM (interoperable), or Celestial's EAM (thermally tolerant but C-band-locked)? COUPE's PDK biases toward MRM.
- Does 'fast and narrow' (fewer, faster fibers) or 'slow and wide' (many slow fibers + WDM) win the OE scaling roadmap to 12.8T+? Fiber-attach yield and laser wavelength counts decide it.
- Can Ethernet (UEC) + open scale-up (UALink/SUE) genuinely erode Nvidia's networking lock-in, or does NCCL + first-class Spectrum-X support keep the attach rate high?
- How much of the CPO startup field survives independently vs gets absorbed (Nubis→Ciena, Celestial→Marvell)? Bookended vs standards-based (interoperable) architecture may decide who lives.

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Chapter 7

Datacenter, Power & Cooling

AI's real bottleneck is no longer chips but electrons, and the datacenter is being redesigned from the grid substation down to the sub-1V rail to squeeze out every token per watt.

Overview

SemiAnalysis frames this sector through one lens: cost per token is set by how much compute you can pack densely, power, and cool. That forces three simultaneous revolutions. (1) Power supply: the US grid is 'sold out' —roughly a terawatt of load requests are queued, ELCC headroom goes negative by 2027, and interconnection now runs ~5-7 years —so behind-the-meter (BTM) onsite gas, fuel cells and engines are set to power over half of new US datacenter capacity by 2028. (2) Electrical distribution: rack density is climbing toward 600kW+ (Kyber) and eventually ~2,300W chips, breaking 48-54V distribution and forcing an 800VDC transition that migrates conversion content from grey space to white space and enables a Solid-State Transformer (SST) endgame. (3) Cooling: >1000W chips end the air-cooled era, pushing direct-to-chip liquid cooling (DLC), CDUs and quick-disconnects, while hyperscalers already run 1.08-1.15 PUE via free cooling. Layered on top are grid-stability risk from gigawatt synchronous AI loads, the political fight over household electricity bills, water-footprint myths, and the speculative frontier of space datacenters. SemiAnalysis's edge is a bottom-up, building-by-building Datacenter/Industrials/Energy model tracking 6,000+ datacenters and 40,000 generators via satellite imagery, letting it call winners and 'fake losers' before the market.

Core Knowledge

PUE / WUE (Power & Water Usage Effectiveness)

PUE = total facility power / IT power; industry-average PUE is ~1.6 (Uptime survey excluding hyperscalers), but Google/Meta run ~1.1 and Microsoft/AWS ~1.15. WUE (liters/kWh) can exceed 2 for water-cooled chillers; Meta ~0.20, Microsoft ~0.3 company-wide but >2 in dry Phoenix. Non-IT power is 60-80% cooling, 15-30% electrical losses. Fan Law: a 10% airflow cut yields ~27% fan-energy cut (cubic).

Behind-the-Meter (BTM) / Bring Your Own Generation (BYOG)

Generating power onsite (gas turbines, engines, fuel cells) rather than waiting years for a grid connection. 'Speed is the moat': an AI cloud nets ~$10-13M revenue per MW annually, so bringing 200MW online six months early is worth ~$1B+. BTM is expected to power over half of new US datacenter capacity by 2028, up from <7% in 2025.

800VDC / ±400VDC & the Power Rack (Sidecar)

Distributing ~800V DC to the rack instead of 48-54V. A 600kW rack at 54V needs ~12,500A vs ~750A at 800V (~16.7x less current, I^2R losses ~220-278x lower). The 'power rack' (OCP Diablo 400, co-authored by Google/Meta/Microsoft) disaggregates AC-DC rectification, BBUs and supercaps into a dedicated cabinet (~$400-500k ASP, ~10x a standard AC power rack). ±400V bipolar leverages the mature EV supply chain (650V GaN, 400V caps).

Solid-State Transformer (SST)

Replaces the iron-core transformer + rectifier with high-frequency (20kHz+) semiconductor conversion, going directly from medium-voltage AC (13.8-45kV) to 800VDC. Infineon claims ~40x weight and ~14x size reduction; targets >97% efficiency (best public benchmark: ETH Zurich 98% at 400kW). Depends on 3,300V+ SiC MOSFETs still in limited production. The ′800VDC endgame′ (Phase 4, ~2029+).

Direct-to-Chip Liquid Cooling (DLC) & CDU

Cold plates carry coolant directly against the silicon; the GB200 NVL72 (120kW, 72 GPUs) is DLC-exclusive. A Coolant Distribution Unit (CDU, typically >1MW) exchanges heat between the IT loop and facility water. Nvidia's ramp is causing shortages of seemingly simple parts like Quick Disconnects. Rear-Door Heat Exchangers (RDHx, 30-50kW) are a bridge; xAI Memphis combined RDHx + DLC.

ELCC (Effective Load Carrying Capability) & Grid Headroom

ELCC = the ′true′ firm-capacity value of a plant to the system, discounting intermittent solar/wind steeply and thermal forced-outage risk. Incremental 4hr BESS adds little marginal ELCC once <4hr risk is saturated. Grid headroom = accredited supply − peak demand − required reserves; on a UCAP basis it turns red (negative) across a growing set of US subregions by 2027, an aggregate ~40GW deficit by 2030.

Load Fluctuations, Inertia & LVRT

Synchronous GPU training swings power ~15x more than a cloud datacenter (Google: 1.5MW→15MW), with MW-scale spikes sub-second. Spinning generators supply inertia; inverter-based solar/BESS do not. A low-voltage ride-through (LVRT) is a transient voltage sag (30ms-5s); if datacenters trip during one, ERCOT modeled 1.5-2.5GW of West Texas load disconnecting near-simultaneously —echoing the April 2025 Iberian blackout (collapse in 27 seconds).

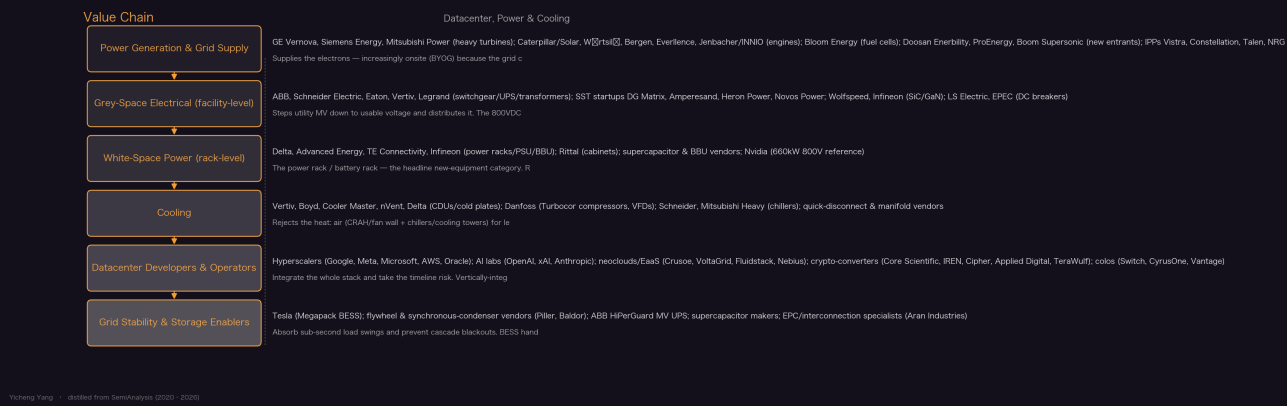
Capacity Auction (PJM BRA) vs Energy-Only (ERCOT)

PJM pays plants to stay on standby via a forward capacity auction (Base Residual Auction), priced two years out off a simulated demand curve (VRR). The 2025/26 clearing jumped 9.3x to $270/MW-day (some zones ~$450), driving ~15% higher household bills. ERCOT has no capacity market —real-time scarcity pricing only —and Texas power prices stayed roughly flat despite equivalent AI buildout. SemiAnalysis's verdict: the bill shock is government/market-design driven, not AI.

Aeroderivative / IGT / RICE / SOFC (onsite generation types)

Aeroderivatives = jet engines bolted down (30-60MW, 5-10min ramp, $1,700-2,000/kW). Industrial Gas Turbines (IGTs, 5-50MW, ~20min ramp, $1,500-1,800/kW). Reciprocating engines (RICE, 3-20MW, 10min ramp, $1,700-2,000/kW) —Jenbacher J624 is a favorite. Solid-oxide fuel cells (Bloom Energy, no combustion, $3,000-4,000/kW, near-instant permitting). Heavy-duty CCGTs are most efficient (>60%) but slowest (4-6yr COD).

Value Chain



Key Theses

1. **The US grid is 'sold out' —headroom goes negative by 2027 and BTM powers half of new datacenters by 2028.**

   Roughly a terawatt of load requests are queued against a 745GW US peak; barely ~15GW of net-new ELCC capacity is added per year, and less than 10GW/yr of gas in 2026-27. Interconnection now runs ~5 years (7 in North Virginia). On a UCAP basis, grid headroom turns red across a growing set of subregions by 2027. Since a utility's timeline can slip from 2027 to 2029 with no penalty, the marginal buyer goes behind the meter for speed and certainty; DC BTM equipment TAM is set to cross 50GW/year by 2029.

   > *"BTM will power well over half of new US datacenters in 2028+, and the Total Addressable Market (TAM) for DC BTM equipment to cross 50GW/year by 2029."*
   > — US Grid Constraints: Towards 40GW+ of Behind-The-Meter Datacenter by 2028? (2026-06-25)

2. **Speed is the moat: onsite gas is worth billions because six months of earlier revenue dwarfs the extra cost.**

   An AI cloud nets ~$10-13M per MW annually, so getting 200MW online six months early is worth ~$1-1.2B (an NPV of ~$400-500M even discounted). xAI proved it —Colossus built in ~122 days by bypassing the grid with rented, truck-mounted Solar/Caterpillar turbines and VoltaGrid engines. OpenAI/Oracle then placed the largest-ever onsite order (2.3GW in Texas, Oct 2025), and 12 different suppliers now each hold >400MW of US onsite-gas orders. Power as a % of AI TCO is 'mostly insignificant,' so any power secured is worth billions.

   > *"AI cloud revenue can net $10-12M per MW annually, meaning that getting 200 MW of datacenter powered and online even six months earlier can net $1-1.2 billion in revenue."*
   > — How AI Labs Are Solving the Power Crisis: The Onsite Gas Deep Dive (2025-12-30)

3. **800VDC is inevitable —physics, not preference —because 600kW+ racks break 48-54V distribution.**

   Resistive losses scale with current squared. A 600kW Kyber-class rack draws ~12,500A at 54V but only ~750A at 800V —~16.7x less current and, at fixed resistance, ~219-278x lower I^2R heating. A

1MW rack at 48-54V needs ~200kg of copper busbar; at GW scale that's hundreds of tons. Today's NVL72 already uses up to 8 power shelves; at Kyber power a 48V approach would need ~64U of power hardware —an entire rack, leaving no room for compute. 800VDC also cuts facility power ~5% (~50MW saved at 1GW IT load).

> *"800VDC is the physics enabler for 2,300W TDP chips and 600kW racks, and those 600kW racks are the direct consequence of the push for density, because density is what drives cost per token down."*
> — Inside the 800VDC Revolution –Part 1 (2026-05-26)

4. **The 800VDC transition migrates content from grey space to white space and ends with the Solid-State Transformer.**

   SemiAnalysis models four phases: (1/2) 2026-28 white-space retrofit via the ~$400-500k power rack (sidecar TAM peaks ~$11B in 2028); (3) 2028/29 grey-space DC distribution with centralized rectifiers, eliminating the central UPS ($1.2M) and AC switchboards/PDUs; (4) 2029+ SSTs collapsing MV transformer + rectifier into one device (SST TAM ~$32B by 2030 at ~$1.25M/MW). Total electrical content per MW stays in a $3.6-4.8M band —it's a content migration, with cumulative path efficiency rising from ~82% (baseline AC) to ~87.4% by Phase 4.

   > *"Total electrical content per MW stays in a $3.6-4.8M band across four of the five architectures we model. The main headline is a content migration from grey space to white space."*
   > — Inside the 800VDC Revolution –Part 1 (2026-05-26)

5. **>1000W chips end the air-cooled era; liquid cooling is the fastest-evolving, highest-risk datacenter system.**

   The GB200 NVL72 (120kW, 72 GPUs) is DLC-exclusive and is the highest-volume Blackwell SKU; chip TDP is climbing toward 1500W+. Cooling is already the 2nd-largest capex after electrical and carries the steepest learning curve and execution/obsolescence risk. Nvidia's ramp is causing shortages of components 'that on first glance look simple such as Quick Disconnects.' Demand for liquid cooling is underestimated, driving inefficient 'bridge' solutions like RDHx (used with DLC at xAI Memphis) because there won't be enough liquid-ready datacenters.

   > *"Nvidia's massive ramp is causing shortages on components that on first glance look simple such as Quick Disconnects!"*
   > — Datacenter Anatomy Part 2 –Cooling Systems (2025-02-13)

6. **Hyperscalers already run 1.08-1.15 PUE with air cooling —debunking the 'air can't be efficient' myth.**

   The industry-average 1.6 PUE (Uptime survey) excludes hyperscalers; Google/Meta run ~1.1, Microsoft/AWS ~1.15. They win via free cooling (air/water economizers), running server inlets above 30°C (sometimes >40°C) with custom servers, and deep CFD. Meta's 'H' hits 1.08 PUE / 0.20 WUE but takes ~2 years to build a shell (2-3x rivals) —so Meta scrapped in-progress 'H' shells for a faster AI-ready design. Google trades energy for water (PUE 1.10 but WUE >1.0 via evaporative towers).

   > *"Hyperscalers have largely proved wrong the popular narrative that air cooling is not an energy efficient technology."*
   > — Datacenter Anatomy Part 2 –Cooling Systems (2025-02-13)

7. **Gigawatt synchronous AI loads threaten cascade blackouts —an Iberian-Peninsula scenario in Texas.**

   ERCOT modeled a fault on a West Texas 345kV line: in all four scenarios ≥1.5GW of datacenter load disconnects near-instantly; on a duck-curve day with datacenters not equipped for LVRT, ~2.5GW (all of West Texas) trips at once. If >2.6GW system-wide (or 2.0GW in West Texas) drops, frequency exceeds ERCOT's 60.4Hz danger zone and risks cascade —exactly the April 28, 2025 Iberian blackout, where 2.2GW tripped and the grid collapsed in 27 seconds. Synchronous condensers help but still leave 1.3-1.9GW at risk and cost $10-20M per GW.

   > *"if 2-2.5 GW of datacenter load tripped off the grid in short succession, then similar voltage and frequency fluctuations could cause cascade failures through Texas···All this, potentially started by a squirrel stepping on the wrong power line."*
   > — AI Training Load Fluctuations at Gigawatt-scale - Risk of Power Grid Blackout? (2025-06-25)

8. **AI is being blamed for household electric bills, but the fault is market design (PJM), not AI.**

   PJM's capacity auction cleared 9.3x higher ($29→$270/MW-day, ~$450 in some zones), adding $25-30/month to bills and driving ~15% higher household costs across 67M residents —off a simulated demand curve (VRR) that PJM itself has repeatedly miscut (a methodology change made 14GW of gas capacity disappear overnight). ERCOT, an energy-only market with no capacity auction, saw equivalent AI buildout yet roughly flat prices. Winter Storm Fern proved the point: PJM's $270/MW-day 'bought' a 21GW generation failure (Dominion spiked to $1,800/MWh) while ERCOT held.

   > *"In PJM, plants get paid no matter what, even if they fail when needed···PJM's 9.3x capacity price increase was supposed to buy reliability. It did not."*
   > — Are AI Datacenters Increasing Electric Bills for American Households? (2026-03-03)

9. **BESS is the most flexible fluctuation fix but not a cheap one —a GW-scale system costs ~$1B.**

   Tesla's Megapack can charge/discharge hundreds of MW within seconds, managing fluctuations at MW/ms, MW/s and MW/min —more flexible than capacitor banks, diesel gensets or grid resources. But per Lazard, a 100MW/2hr BESS runs $38-80M and 4hr $76-157M, so a GW-scale system is 'close to a billion dollars' and is additive to (not a replacement for) UPS and gensets. BESS also unlocks demand response: 20-90hrs/yr of curtailment could free 6.5-14.7GW of new ERCOT load with no upgrades —but utility DRMS and incentives remain immature.

   > *"a BESS suitable for a GW-scale datacenter would cost close to a billion dollars, and for that price, Tesla would not consider BESS a replacement for a UPS or a diesel generator."*
   > — AI Training Load Fluctuations at Gigawatt-scale - Risk of Power Grid Blackout? (2025-06-25)

10. **'Half of 2026 datacenter capacity is canceled' is wrong —the panic is built on a flawed denominator.**

    The viral claim traces to a Bloomberg/Sightline figure that only tracks large, publicly-announced megaprojects (~12GW basis, of which ~5GW 'confirmed'), skewing toward speculative builds by inexperienced developers that were never landing in 2026 anyway. SemiAnalysis's building-by-building satellite tracking shows those slip to 2028+, not cancel, and its 2026 outlook hasn't moved. The real signal: over a terawatt now sits in the US large-load queue, and leading operators are adept at navigating permitting, grid and equipment constraints.

> *"the data sources behind these claims of '50% of 2026 datacenters are delayed' are essentially uninformed vibe-coded datacenter forecasts that take announcements at face value."*
> — Stop Saying Half of 2026 US Datacenter Capacity Is Canceled (2026-06-18)

11. **Space datacenters stay ~4x costlier than Earth until ~2040 —terrestrial power isn't exhausted yet.**

    A 30.5kW B300 cluster costs $8.64/hr/GPU in space vs $2.37 terrestrial (LCOC $10.91 vs $2.49 after redundancy), largely because launch is $1.6M of the $3.1M datacenter capex and space useful life is only ~5yr vs 15yr (levelized DC capex ~18x higher). SemiAnalysis maps five terrestrial supply layers (grid → converted/crypto sites → BTM → industrial production → the semiconductor ceiling) that must exhaust before orbit makes sense. The binding constraint today is actually silicon: AI consumes ~60% of TSMC N3 in 2026, ~86% in 2027 —a limit space cannot solve.

    > *"In our base case scenario, the cost difference between space and terrestrial datacenters starts at over 4x in 2026, before narrowing to parity in ~2040."*
    > — To Boldly Go: The Case for Space Datacenters (2026-06-03)

12. **Heavy-duty turbines are the tightest supply chain; the industry is scarred by two boom-bust cycles.**

    GE Vernova, Siemens Energy and Mitsubishi are order-booked into 2028-29 with nonrefundable reservations beyond, yet respond with caution —GE targets just 24GW/yr (its 2007-16 level), Siemens ~30GW by 2028-30, all without expanding factory footprint. The bottleneck is turbine blades: exotic single-crystal nickel alloys (rhenium, cobalt, tantalum, yttrium —yttrium is under Chinese export control) made by just four Western firms (PCC, Howmet, CPP, Doncasters). Heavy cores are 300-500-ton logistics nightmares needing 24-30 months. Aeros, IGTs and RICE are far less constrained.

    > *"few manufacturers appear fully 'AGI-pilled' despite surging demand···much of it reflects PTSD from 30 years of boom-bust cycles in gas generation."*
    > — How AI Labs Are Solving the Power Crisis: The Onsite Gas Deep Dive (2025-12-30)

13. **±400V bipolar won the 800VDC voltage war by borrowing the EV supply chain, but no standard has settled.**

    OCP Diablo 400 (Google/Meta/Microsoft) defines ±400V bipolar as base config so each rail sits only 400V from the grounded midpoint, letting operators use mature EV-grade 650V GaN FETs and 400V-class caps —Google: '400VDC allows us to leverage the supply chain established by electric vehicles.' But there's no one-size-fits-all: Nvidia sits outside Diablo 400 with a monopolar 660kW 800V design; Meta runs 600-800kW, Google pushes 900kW, Amazon lands 800kW. Full NEC code support only targets 2029, so pre-2029 sites need custom AHJ approvals —a barrier for colos, not hyperscalers.

    > *"selecting 400 VDC as the nominal voltage allows us to leverage the supply chain established by electric vehicles, for greater economies of scale, more efficient manufacturing, and improved quality and scale."*
    > — Inside the 800VDC Revolution –Part 1 (2026-05-26)

14. **The datacenter water panic targets the wrong problem —Colossus 2 uses ~2.5x one In-N-Out store.**

SemiAnalysis modeled xAI's 400MW Colossus 2 (PUE 1.15, 70% utilization, ~30% wet operation) at ~346M gallons/yr (0.9M gal/day, WUE ~0.51 L/kWh). A single In-N-Out store (burgers only) consumes ~147M gallons/yr —so the whole gigawatt-class datacenter is only ~2.5:1 versus one burger joint. With 400+ In-N-Out locations and hundreds of thousands of burger restaurants, the water argument for slowing datacenters is misdirected. Cooling architecture (dry vs wet) and onsite gas generation water use dominate the honest accounting.

> *"Colossus 2's blue water footprint is around 346 million gallons per year, while an average In-N-Out store (yes, burgers only) comes in at around 147 million gallons. That's roughly a ~2.5 : 1 ratio."*
> — From Tokens to Burgers: A Water Footprint Face-Off (2026-01-15)

Article Deep-Dives

Inside the 800VDC Revolution –Part 1 (2026-05-26)

The definitive map of the rack-power transition. As racks climb toward 600kW+ (Kyber) and chips toward 2,300W, 48-54V distribution physically breaks (I^2R losses scale with current squared), forcing 800VDC. SemiAnalysis models four phases: (1/2, 2026-28) white-space retrofit via the ~$400-500k HVDC power rack (OCP Diablo 400, co-authored by Google/Meta/Microsoft; ±400V bipolar to reuse the EV supply chain); (3, ~2028/29) grey-space DC distribution with centralized rectifiers, killing the central UPS ($1.2M) and AC switchboards; (4, 2029+) Solid-State Transformers collapsing MV transformer + rectifier into one device. Sidecar TAM peaks ~$11B (2028); SST TAM ~$32B by 2030. Total electrical content per MW stays $3.6-4.8M —a migration from grey to white space —while path efficiency rises 82%→87.4%. Winners: Delta, Infineon, Wolfspeed (10kV SiC), DG Matrix, Amperesand, Heron Power. Barriers: NEC 2029 code, DC arc-flash safety, grounding fragmentation, and a 3,300V+ SiC supply gate.

How AI Labs Are Solving the Power Crisis: The Onsite Gas Deep Dive (2025-12-30)

The BYOG bible. With the grid 'sold out' (Texas approves barely >1GW/yr against tens of GW/month of requests) and interconnection ~5 years, AI labs bypass the grid entirely. xAI pioneered rented, truck-mounted 16MW Solar turbines + VoltaGrid engines to build Colossus in ~122 days; OpenAI/Oracle then placed a 2.3GW Texas order. Full taxonomy of generation: aeroderivatives (jet engines bolted down, 30-60MW, 5-10min ramp, $1,700-2,000/kW), IGTs ($1,500-1,800/kW), reciprocating engines (RICE, Jenbacher J624), and Bloom SOFC fuel cells ($3,000-4,000/kW, no combustion, near-instant permitting). Deployment configs: bridge power, islanded, N+1+1 redundancy, gas+battery hybrids, Energy-as-a-Service (VoltaGrid). Load surges need synchronous condensers, flywheels or BESS. The binding constraint is heavy-duty turbine blades —single-crystal nickel alloys from 4 Western firms —and 30 years of boom-bust PTSD keeping GE/Siemens/MHI cautious.

Datacenter Anatomy Part 2 –Cooling Systems (2025-02-13)

The comprehensive cooling primer. Modern datacenters generate >50x an office's heat/sq-ft; cooling is 60-80% of non-IT power and the 2nd-largest capex. Walks the full heat path —server fans, CRAC/CRAH/fan walls (500-600kW), RDHx (30-50kW), CDUs (>1MW), water- vs air-cooled chillers (COP ~7), evaporative vs dry cooling towers, WUE (>2 L/kWh for wet). Debunks the myth air can't be efficient: hyperscalers hit 1.08-1.15 PUE via free cooling, >30°C server inlets, custom servers, and CFD (Fan Law: -10% airflow = -27% energy). The GenAI shift: the GB200 NVL72 (120kW, DLC-exclusive) forces direct-to-chip liquid cooling; Nvidia's ramp is shorting simple parts like Quick Disconnects. Case studies of Microsoft (direct evaporative), Meta ('H', 1.08 PUE but slow), Google (energy-for-water), AWS.

AI Training Load Fluctuations at Gigawatt-scale - Risk of Power Grid Blackout? (2025-06-25)

Why synchronous GPU training is uniquely dangerous for the grid. AI loads swing ~15x more than a cloud datacenter (Google: 1.5MW→15MW) with MW-scale sub-second spikes from batch cycles, checkpointing and AllReduce. Inverter-based solar/BESS lack the inertia spinning generators provide. ERCOT modeled a West Texas 345kV fault: ≥1.5GW of datacenter load disconnects near-instantly; on a duck-curve day, 2.5GW (all of West Texas) trips at once. Beyond 2.6GW system-wide (2.0GW West Texas), frequency exceeds the 60.4Hz danger zone and risks cascade —mirroring the April 2025 Iberian blackout (2.2GW tripped, 27-second collapse). Solutions: synchronous condensers ($10-20M/GW, still leave 1.3-1.9GW at risk), and BESS —the most flexible fix (MW/ms→MW/min) but ~$1B at GW-scale, additive to UPS/gensets. Demand response could free 6.5-14.7GW in ERCOT but utility DRMS is immature.

US Grid Constraints: Towards 40GW+ of Behind-The-Meter Datacenter by 2028? (2026-06-25)

Models the exact grid shortfall BTM must fill. From 40,000 tracked generators, only ~15GW/yr of net-new ELCC capacity is added; <10GW/yr of gas in 2026-27 as slow CCGTs dominate utility orders and transformer/turbine lead times stretch to 3-4 years. On a UCAP basis, headroom turns red across a growing set of subregions by 2027 (PJM's 2027/28 auction cleared ~6.5GW short of its 20% target). BTM wins on speed and certainty —utility timelines slip with no penalty (Switch posted a multi-billion LOC facility) while AI labs accept lower redundancy (Meta targets 2 nines, no gensets). Details ERCOT's Batch Zero co-location framework (NMA, BYOG, WLPUN, PCLR) codifying hybrid grid+onsite structures, with ~2,885MW of announced co-location deals (Crusoe, AWS/Comanche Peak, CyrusOne).

Are AI Datacenters Increasing Electric Bills for American Households? (2026-03-03)

A market-design autopsy. PJM's capacity auction (Base Residual Auction) cleared 9.3x higher ($29→$270/MW-day, ~$450 some zones), adding $25-30/month to 67M residents' bills —off a simulated VRR demand curve PJM keeps miscutting (a methodology change made 14GW of gas 'disappear' overnight). The IMM found removing all datacenters would cut capacity payments $9.33B (64%). But ERCOT, an energy-only market, saw equivalent AI buildout and roughly flat prices. Winter Storm Fern (Jan 2026) was the proof: PJM's record capacity price 'bought' a 21GW generation failure (Dominion $1,800/MWh) while ERCOT held. Verdict: the bill shock is government/market-design driven, not AI —and ERCOT's single-state regulatory agility (SB6) is 'durable and investable' vs PJM's 13-state FERC gridlock.

Key Data

US datacenter buildout	+21GW (2026) → +84GW (2030)	Record pace; global tracked capacity (ex-China) ~89GW (2026) → up to 338GW (2030).
US grid load-request queue	~1 terawatt	Against a 745GW US peak; ERCOT alone has ~108GW of large-load requests (with duplicates).
Net-new ELCC grid capacity added	~15GW/yr (rising to 20GW+)	Headroom turns negative in 2027, ~40GW aggregate deficit by 2030.

AI cloud revenue per MW	~$10-13M / MW / year	Why 'speed is the moat'; 200MW six months early ≈ $1B+ revenue / ~$400-500M NPV.
BTM share of new US DC capacity	<7% (2025) → >50% (2028)	DC BTM equipment TAM crosses 50GW/year by 2029; 26GW cumulative confirmed BTM by 2030.
800VDC current/loss reduction	~16.7x less current, ~219-278x less I^2R	600kW rack: ~12,500A at 54V → ~750A at 800V; enables ~2,300W chips.
HVDC power rack ASP	$400-500k/unit (~$0.5M/MW)	~10x a standard $40k AC power rack; sidecar TAM peaks ~$11B in 2028.
SST TAM by 2030	~$32B (@ ~$1.25M/MW)	>$320M flowed into SST startups in the year to Mar 2026; Eaton bought Resilient Power.
Electrical-path efficiency	~82% (AC baseline) → ~87.4% (Phase 4)	Phase 2 (86.5%) via UPS elimination = ~58MW saved at 1GW IT; Nvidia cites ~5% (~50MW).
Onsite generation capex	$1,500-4,000/kW	IGT $1,500-1,800; aero/RICE $1,700-2,000; Bloom SOFC $3,000-4,000.
Hyperscaler PUE / WUE	PUE 1.08-1.15; WUE 0.20 to >2.0	Meta 1.08/0.20; Google 1.10 but WUE >1.0; Microsoft ~0.3 avg but >2 in Phoenix.
GB200 NVL72 rack	120kW, 72 GPUs, DLC-only	Highest-volume Blackwell SKU; chip TDP heading to 1500W+ then ~2,300W.
PJM capacity auction (2025/26)	9.3x jump to $270/MW-day	~$450 in some zones; +$25-30/month on bills; total ~$16B / ~$120k/MW; 2026/27 capped $329.
Winter Storm Fern (Jan 2026)	PJM lost ~21GW; Dominion $1,800/MWh	15% of cleared fleet failed; DOE tapped ~35GW of DC/industrial backup; ERCOT held (~$300/MWh).
ERCOT cascade limits	>2.6GW system / >2.0GW West TX	Load-loss beyond this risks 60.4Hz danger zone; Iberian blackout (2.2GW, collapse in 27s).
BESS cost (Lazard)	100MW: $38-80M (2hr) / $76-157M (4hr)	GW-scale ≈ ~$1B, additive to UPS/gensets; synchronous condenser $30-60k/MVAr.
Space vs terrestrial compute cost	$8.64 vs $2.37 /hr/GPU (B300, 2026)	LCOC $10.91 vs $2.49; parity ~2040 base case (early 2030s in 'Elon' case).

AI share of TSMC N3	~60% (2026) → ~86% (2027)	Silicon, not power, is the binding constraint; AI ~0.3% of global generation today → ~5% by 2030.

Investment Angle

Winners

Bloom Energy	First called out as the biggest BTM beneficiary in Dec 2024; SOFC fuel cells (no combustion, near-instant EPA permitting, deployable in weeks) are taking share fast. Claims 2GW/yr capacity by end-2026. Premium ($3,000-4,000/kW) offset by speed and siting flexibility (usable in population centers).
GE Vernova / Siemens Energy / Mitsubishi Power	The turbine 'Big Three' —order-booked into 2028-29 with nonrefundable reservations. But SemiAnalysis flags risk of 'peak' orders and secondary-market turbine supply surging, so the durable winners may be fuel cells and RICE, not just the majors.
Delta Electronics	Leading the 800VDC transition —power racks, PSU shelves, 110kW power shelves with 80kW BBU, the first 2.4MW 800VDC In-Row CDU (GTC 2026), and 800VDC busway. Straddles power electronics AND liquid cooling, the two content-growth engines.
Infineon / Wolfspeed (power semis)	The silicon under 800VDC/SST. Infineon claims ~40x/14x SST weight/size reduction and a 12kW BBU roadmap (99.5% efficiency); Wolfspeed's 10kV SiC MOSFET (bare die since Mar 2026) opens direct MV rectification. ±400V bipolar rides the mature EV 650V-GaN supply chain.
SST startups: DG Matrix, Amperesand, Heron Power, Novos Power	>$320M funded in the year to Mar 2026. DG Matrix (ABB-backed, Infineon SiC) is the only SST in Nvidia's MGX reference; Heron building a 40GW US plant; Amperesand targets 30MW deployed in 2026. Eaton acquired Resilient Power for SST expertise.
Tesla (Megapack BESS)	The default answer to gigawatt load fluctuations, LVRT ride-through and demand response —pairs with turbines at xAI. GW-scale ~$1B TAM per site, plus the emerging role as a firm-capacity and demand-response asset class.
Liquid-cooling ecosystem: Vertiv, Boyd, nVent, Danfoss	DLC ramp shorts CDUs, cold plates, manifolds and Quick Disconnects. Danfoss Turbocor compressors already run on 700-813V DC —a rare cooling component ready for the DC-native future.
Neoclouds / EaaS: Crusoe, VoltaGrid	VoltaGrid runs full Energy-as-a-Service (e.g. 2.3GW for a 1.4GW Vantage campus, 64% overbuild) bundling RICE + synchronous condensers + BESS. Crusoe books bridge power (Abilene) and 1.2GW of Boom Superpower turbines.

Bottlenecks

- Firm dispatchable power —<10GW/yr of gas added in 2026-27, ~15GW/yr net-new ELCC vs a ~terawatt queue; grid headroom negative by 2027.
- Heavy-duty turbine blades —single-crystal nickel alloys (rhenium, cobalt, yttrium under Chinese export control) from just 4 Western firms; 300-500-ton cores need 24-30 months.
- Grain-oriented electrical steel (GOES) & large power transformers —longest lead times in the stack (3-4 years); ~80% of US high-power transformers are imported.
- 3,300V+ SiC MOSFETs —gate the MV-input SST; still in limited production, forcing LV-input SST as the earlier-to-market variant.
- Quick Disconnects, CDUs & liquid-ready datacenter shells —Nvidia's ramp shorts simple parts; too few DLC-ready halls force inefficient RDHx 'bridge' solutions.
- Grid interconnection & transmission —~5-7 year queues; conversion (not queue depth) now binds in PJM (24GW of executed agreements terminated since 2020).

Risks

- Peak gas turbine orders —SemiAnalysis flags risk that current turbine order surge is temporary; secondary-market turbine availability is surging, pressuring OEM pricing.
- Datacenter delays overstated but real —SemiAnalysis debunks '50% canceled' yet flags specific slips (STACK/Oracle to 2029 on gas pipeline); demand is fine, execution slips.
- Grid stability / regulatory clampdown —NERC issued a Level 3 alert (May 2026) on large loads; ERCOT ride-through mandates (NOGRR282) and a Computational Load Entity registration raise the compliance bar.
- 800VDC safety/code gap —full NEC support only ~2029; DC arc-flash has no IEEE 1584/NFPA 70E PPE table; grounding fragmentation makes each site a custom engineering project.
- AI-bubble / demand risk —1GW of AI capacity implies ~$12-13B annual revenue; if token demand disappoints, the whole power/cooling capex thesis unwinds. OEMs remember 30 years of boom-bust.
- Silicon ceiling caps everything —AI consumes ~60% of TSMC N3 in 2026, ~86% in 2027; power/DC headroom can't be used without chips.

Catalysts

- 800VDC-native compute (Kyber/Vera Rubin Ultra) shipping late-2026/2027 —the inflection where 800VDC becomes mandatory, not optional, spiking penetration.
- NEC 2029 code + OCP/UL standards (year-end 2026 white paper, UL 857 Ed.15 to 1500VDC) unlocking colo/smaller-builder 800VDC adoption.
- SST UL certification —DG Matrix targets end-Q2 2026; first certification unlocks the ~$32B facility-level TAM (~2029 scale adoption).
- ERCOT Batch Zero / co-location rules (NPRR1325, effective Jul 2026) + FERC's large-load rulemaking (RM26-4) codifying hybrid grid+BTM structures.

- Behind-the-meter TAM inflection —DC BTM equipment crossing 50GW/year by 2029; new-entrant order flow (Boom Superpower, Doosan DGT6, ProEnergy retrofits).
- Load-flexibility / demand response reform —20-90hrs/yr curtailment could unlock 6.5-14.7GW in ERCOT; PJM's Expedited Interconnection Track (June 2026).

Open Questions

- Which 800VDC grounding/voltage topology wins —±400V bipolar (EV supply chain) vs single-ended 800V monopolar (Nvidia)? The choice is a vendor-ecosystem commitment, not just technical, and no consensus exists.
- Do SSTs displace MV rectifiers, and can they hit datacenter-grade reliability? No vendor has field-reliability data at scale (longest deployment is a 2011 Swiss railway PETT); ETH Zurich found a line-frequency transformer + SiC rectifier can match SST efficiency.
- Does BESS/demand-response actually get deployed at scale, or do immature utility DRMS and weak incentives ($160-300k/month for 20MW = a rounding error) keep it niche outside Texas cryptominers?
- Is the current gas-turbine order surge a durable buildout or a repeat boom-bust? Secondary-market turbine availability is already surging and SemiAnalysis flags 'peak gas turbine orders.'
- When (if ever) does terrestrial power exhaust enough to justify space datacenters —and can chip reliability (3-6% annual GPU failures needing human intervention) be solved in orbit? Parity is ~2040 base case, but silicon, not power, binds first.

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Chapter 8

Hyperscalers & AI Economics

The trillion-dollar buildout, decoded: who pays for GPUs, who captures the margin, and why a token is now the unit of economic value.

Overview

SemiAnalysis treats the AI buildout not as a hardware story but as an economics story stacked on top of hardware. Their unifying frame is the 'AI Token Economic Stack' —Energy → Datacenter → Accelerator → Neocloud/Hyperscaler → Model Lab → Application —with a token as the atomic unit that ties watts to revenue. Two proprietary engines anchor almost every note: the Datacenter Industry Model (building-by-building MW tracking, the best leading indicator of capex) and the Tokenomics / AI Cloud TCO Model (every major compute contract, its cost, margin, ROIC and RPO). The core, repeated theses: (1) headline GPU $/hr is a bad proxy for true cost —real TCO is dominated by goodput (useful work), reliability, storage, networking, setup and debugging, which SemiAnalysis quantifies via ClusterMAX and a public Goodput calculator; (2) value capture has violently rotated —2023-25 all the money went to infra (Nvidia, power, memory), but from ~December 2025 'agentic AI began to really work' and the model labs suddenly captured the value, with Anthropic inference gross margins going from 38% to >70% and ARR from $9B to $30-44B in months; (3) the hyperscalers diverge on business model —AWS's token-as-a-service (Bedrock) + vertical silicon (Trainium/Graviton) is structurally higher-margin than Azure's/GCP's IaaS mix, while Microsoft's 'Big Pause' handed Oracle ~$150B of OpenAI gross profit; (4) Nvidia and TSMC deliberately under-price to scarcity, acting as the 'central bank of AI' to keep the ecosystem expanding rather than extract maximum rent. Underneath sit older but foundational pieces (GPU cloud economics, the AWS 'cloud crisis', LLM search cost, DeepSeek tokenomics) that established the vocabulary the market now uses.

Core Knowledge

AI Token Economic Stack

SemiAnalysis's master framework: Energy → Datacenter → Accelerator (GPU/ASIC) → IaaS (bare-metal rental) → PaaS/Token-as-a-Service → Models → Applications. A token is the atomic unit that flows through and lets you convert watts into dollars at every layer. Whoever controls the scarce layer at a given moment captures the margin.

Goodput (vs Throughput)

The share of raw GPU throughput that is actually useful work. A cluster can burn full power while producing garbage —GPU fell off the bus, NCCL stalled, an OOM at the next checkpoint, or a job restart eating 10-15 minutes plus all compute since the last checkpoint. 'Goodput expense' is the hidden % tax you pay for a less reliable provider; it dominates large-job TCO.

Cluster TCO ≠ GPU $/hr

True cost = GPUs + Storage + Networking + Control Plane + Support uplift + Goodput expense + Setup expense + Debugging expense, with setup amortized over contract term. Two clouds at the same $/GPU-hr can differ 10-60% in TCO once you count EFA tuning weeks, poor storage tiers, and downtime.

Token-as-a-Service (TaaS) vs IaaS mix

Two ways a hyperscaler sells AI. IaaS = rent bare-metal GPUs on multi-year deals (Azure/GCP heavy). TaaS = sell tokens via a managed endpoint (AWS Bedrock, Azure Foundry, GCP Vertex/Gemini). When you distribute someone else's model (e.g. Claude on Bedrock) you earn both an infra fee AND a revenue-share/distribution fee —the mix that lifts margin.

RPO / Pre-leasing as leading indicators

Remaining Performance Obligation (contracted-but-unbilled backlog) and datacenter pre-leasing MW are SemiAnalysis's earliest, hardest signals of a hyperscaler's true capex intent —visible quarters before the P&L. Microsoft's pre-leasing dwarfed all rivals combined in 2023-24; the collapse in that signal called the 'Big Pause'.

Server depreciation / useful life

Hyperscalers extended assumed useful life from 3-5yr (2020) to 5-6yr, boosting reported earnings. Michael Burry called it artificial; SemiAnalysis disagrees, arguing HPC supercomputers (Summit ran 6.5yr, V100 still rented) prove GPUs stay economically useful far past the 2-3yr product cycle. AWS runs the shortest (5yr) yet still shows rising margins.

Tokenomics: price is an OUTPUT

For any model, $/token is not fixed but solved for by tuning latency (TTFT), interactivity (tok/s/user), context window and batch size against your hardware. This is why DeepSeek's $0.55/$2.19 sticker didn't win share —third-party hosts served the same weights faster; and why 'blended' agentic prices (Opus at ~$0.99/MTok effective vs $5/$25 sticker) look nothing like the list price.

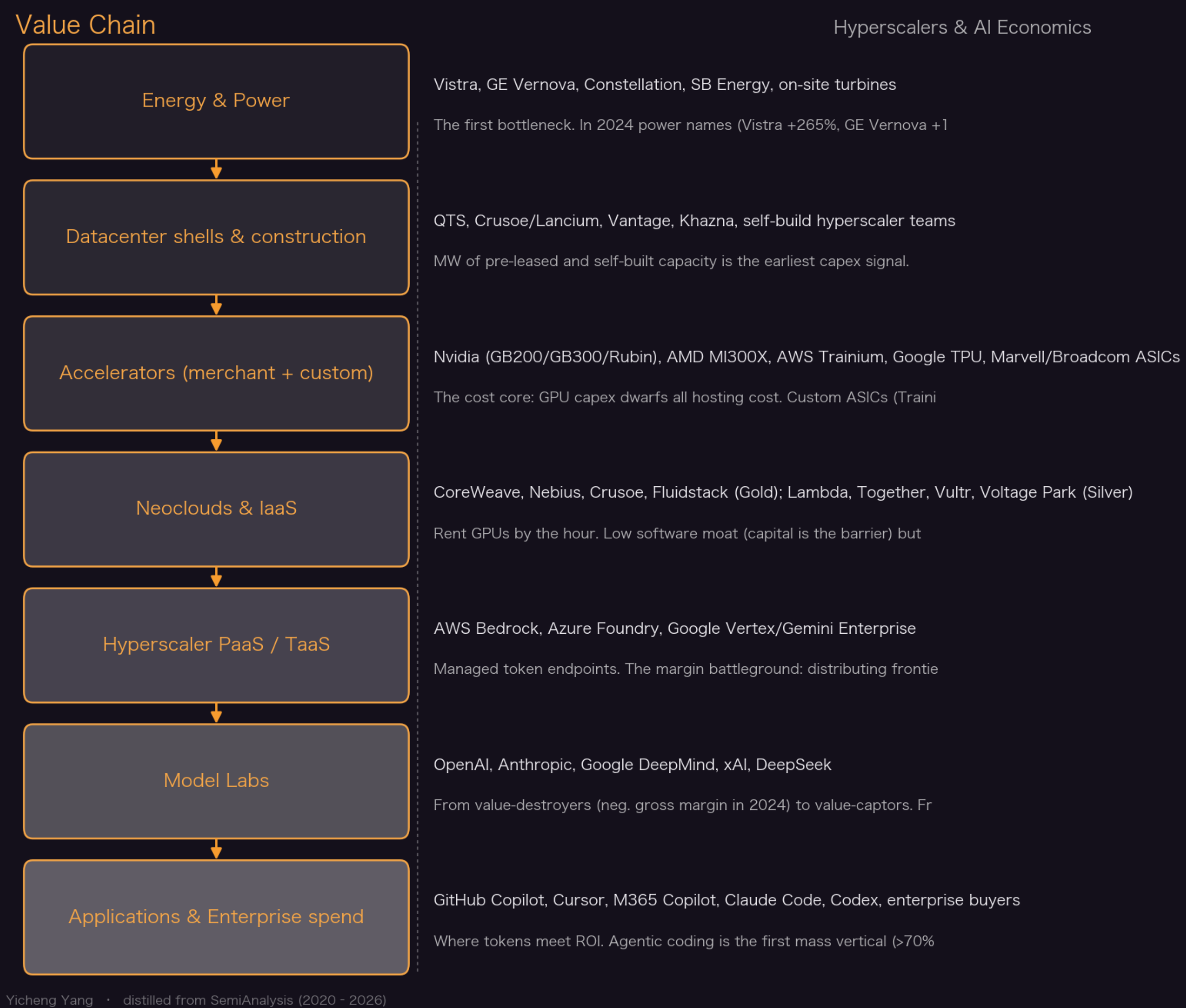
Neocloud / ClusterMAX tiers

'Neoclouds' are GPU-pure-play clouds (CoreWeave, Nebius, Crusoe, Lambda...). SemiAnalysis ranks all providers with ClusterMAX (Gold/Silver/Bronze) based on hands-on testing of 80+ clouds —networking, storage, reliability, support. Gold-tier commands a real price premium because its lower goodput/setup tax makes it cheaper on TCO even at higher $/hr.

Nvidia/TSMC as 'central bank of AI'

Despite controlling the scarcest layers (GPUs, N3 wafers) at full utilization, both deliberately hold pricing below scarcity-clearing levels. Motive: avoid antitrust attention, keep customers from diversifying, and let downstream labs/neoclouds stay profitable so total demand keeps expanding. They vent value downstream on purpose.

Value Chain



Key Theses

1. **GPU $/hr is a trap: real cluster TCO is 10-60% higher once you count goodput, setup and storage.** Holding GPU pricing equal, SemiAnalysis's TCO+Goodput calculators show a Gold-tier cloud is ~5-15% cheaper on TCO than Silver-tier across large training workloads, and Hyperscalers run >10% more expensive (support + EFA tuning). In a large LLM pretrain, goodput expense alone was 6.14% (Gold) vs 10.53% (Hyperscaler) vs 20.91% (Silver). But for fault-tolerant single-node inference the gap collapses to near zero —reliability only matters when your job is big relative to the cluster.

   > *"even in scenarios where pricing per GPU-hour is equal, there are always hidden costs across Storage, Network, Control Plane, Support, Goodput, Setup, and Debugging expenses."*
   > — How Much Do GPU Clusters Really Cost? (2026-04-20)

2. **Value capture violently rotated from infra to the model labs in December 2025 when agentic AI 'began to really work'.**

   2023-25 all AI money went to infra (Nvidia +25% AH in May 2023; power names led 2024; memory led 2025). Then agentic AI crossed an inflection: tasks worth thousands of person-hours now cost a few dollars of tokens. Anthropic ARR exploded \$9B→\$44B and inference gross margins went 38%→>70% in the same window. The labs went from capturing almost none of the value to capturing all of it.

   > *"The age of low gross margins for frontier model providers is over. Real agentic AI has permanently increased the market-clearing price per token, and there's no going back."*
   > — AI Value Capture - The Shift To Model Labs (2026-05-01)

3. **Anthropic's Bedrock deal structure —not just volume —is what inflected AWS margins above every other CSP.**

   On Bedrock, Anthropic is seller-of-record (books full token revenue) while AWS collects BOTH an infra fee AND a distribution/revenue-share fee —a margin stack IaaS can't match. AWS EBIT margins rose 213bp Q/Q while Oracle, CoreWeave and Azure margins fell, despite AWS running the shortest (5yr) depreciation. Bedrock is a ~\$5.5B run-rate business (80-90% Anthropic) and 37% of AWS AI revenue vs IaaS still 80%+ of Azure/GCP AI.

   > *"Our AWS Trainium chips, designed in-house for AI workloads, now power more than 50% of Amazon Bedrock token usage."*
   > — Anthropic Growth and Bedrock Mix Drive AWS Margins Higher While Peers Lag (2026-05-27)

4. **Microsoft's 'Big Pause' handed Oracle ~\$150B of OpenAI gross profit —a self-inflicted competitive gift.**

   Microsoft drove >60% of all new leased turnkey capacity Q1'23-Q2'24, then froze —walking away from >2GW of non-binding LOIs and 1.5GW of near-term self-build. OpenAI diversified to Oracle, CoreWeave, Nscale, Amazon, Google. Oracle signed >\$420B of OpenAI contract value (~\$150B gross profit); at ~\$30B/yr that would have lifted Microsoft's \$194B FY25 gross profit by >18%. SemiAnalysis frames it as partly deliberate (OpenAI would be ~50% of Azure at worse ROIC) but mostly a fumble.

   > *"They potentially have just allowed a competitor to fund their own entry into the AI factory business!"*
   > — Microsoft's AI Strategy Deconstructed - From Energy to Tokens (2025-11-12)

5. **Nvidia and TSMC deliberately under-price to scarcity —acting as the 'central bank of AI'.**

   With compute demand far above supply, both could reprice hard but don't. N3 is the tightest node (Nvidia, Broadcom, Annapurna, MediaTek, AMD all fighting for wafers) yet pricing is stable; Nvidia won't fully reprice Rubin to capture its performance/memory-cost gains. Motive: avoid antitrust scrutiny, prevent customer diversification, and keep downstream labs profitable so total demand keeps growing. They 'take the oxygen out of the room' to remain the protagonist.

   > *"By taking the oxygen out of the room –Nvidia aims to ensure it remains the main protagonist in the AI era for the foreseeable future."*
   > — AI Value Capture - The Shift To Model Labs (2026-05-01)

6. **A Blackwell GB300 NVL72 delivers ~17x (FP8) to 32x (FP4) the throughput of an H100 for only ~70% more TCO —the real driver of falling token cost.**

   Token production cost has plummeted because accelerator price increases are more than offset by throughput gains. On the same B300, software alone (wideEP + disagg + MTP) takes DeepSeek R1 from ~1k to ~14k tok/s/GPU —a 14x software-only swing. Stack hardware on top and GB300 NVL72 hits ~17x an H100 in FP8, 32x in FP4 (Hopper lacks native FP4), while costing only ~70% more per GPU. This is why labs can cut sticker prices (Opus 4.5 at $5/$25, down from $15/$75) and still expand margins.

   > *"One can 14x throughput with software improvements alone."*
   > — AI Value Capture - The Shift To Model Labs (2026-05-01)

7. **Sticker price per token is meaningless: blended agentic price is a fraction of it because of caching and input/output ratios.**

   SemiAnalysis estimates the true blended price for running Opus on agentic tasks at ~$0.99/MTok despite a $5/$25 sticker. Agentic workloads have ~300:1 input:output ratios and 90%+ cache hit rates; cached input costs only $0.50/MTok, so most tokens land in the cheapest tier. Analysts constantly get this wrong (I/O ratios, cached tokens, pricing). This is also why DeepSeek's cheap sticker ($0.55/$2.19) didn't commoditize the market —token price is an output of KPI tuning, not a fixed input.

   > *"a model's price per token is an OUTPUT of these KPI decisions which can be tuned based on the model providers' hardware and model setup."*
   > — DeepSeek Debrief: >128 Days Later (2025-07-03)

8. **AWS's 'cloud crisis' (2023) and 'AI resurgence' (2025) are the same call: business-model, not chips, decides who wins the GPU cloud era.**

   In 2023 SemiAnalysis warned AWS would lose the next compute wave despite owning the most servers —culture and business choices, not tech, would hamper it (Titan/Olympus models failed, Trainium1 uncompetitive). By Sept 2025 they made the out-of-consensus reversal: Anthropic (revenue 5x YTD to $5B annualized) + Trainium's best-perf/TCO-per-memory-bandwidth for inference/RL + token-as-a-service would reaccelerate AWS beyond 20% YoY. Both calls pivot on the same axis: does the provider have the model access and margin structure, not just the silicon?

   > *"Technological prowess, culture, and/or business decisions will hamper them from capturing the next wave of cloud computing like they have the last two."*
   > — Amazon's Cloud Crisis: How AWS Will Lose The Future Of Computing (2023-03-20)

9. **Oracle out-competes on structural cost, not software: ~20% capex advantage from networking, ODM sourcing and investment-grade debt.**

   Oracle's OCI, latecomer to cloud, wins large GPU contracts via three cost levers: (1) optimized RoCEv2 networking (built on Sun/Mellanox HPC heritage) cutting cluster CapEx ~20% vs a neocloud giant; (2) buying servers ODM-direct from Foxconn, bypassing Dell/Supermicro margin; (3) the market's lowest cost of capital via investment-grade debt vs neoclouds' double-digit debt. The ByteDance relationship made Johor, Malaysia the world's #2 AI hub. Larry's >$130B booking / >$420B RPO surge validated SemiAnalysis's early capex ramp call.

10. **Trainium is a targeted weapon, not a Nvidia-killer: it wins on perf/TCO where memory bandwidth is king (high-batch inference, RL).**

    Trainium2 (~500W, 667 TFLOP/s BF16, 96GB HBM3e) course-corrected after uncompetitive Trainium1/Inferentia AWS's 400k-Trainium2 'Project Rainier' cluster for Anthropic is effectively funded by Amazon's own $4B+ Anthropic investment circling back. SemiAnalysis is blunt: Amazon is a 'distant second' to Google in custom silicon, Trainium is unproven for frontier training, and most volume will be inference. But its natural offtake via Bedrock (where hardware is abstracted, porting takes days) makes it structurally valuable.

    > *"Amazon's new $4 billion investment in Anthropic is effectively going to find its way back into the Project Rainier 400k Trainium2 cluster, and there are no other major customers yet."*
    > — Amazon's AI Self Sufficiency | Trainium2 Architecture & Networking (2024-12-03)

11. **'Tokenmaxxing' headlines are overblown; enterprise token budgets are real but the adoption s-curve is still early.**

    After Meta's leaderboard (60T+ tokens/30 days, top user ~280B) and Uber's $1,500/mo cap made news, SemiAnalysis talked to 50+ enterprises. Reality: even Meta (~$50k/yr/employee at list) is only a 3-5% Anthropic customer; the 90th+ percentile drives most revenue and is at little risk. Ramp data: 99th pct spends ~$90k/yr/employee, 90th ~$7,300, but the MEDIAN Ramp customer just $136 and the median Fortune 500 <$100. Budgets range wildly ($250 aerospace $\rightarrow$ $2,000 Stripe/Workday $\rightarrow$ tens of thousands). Coding is the first vertical (>70% of OpenAI+Anthropic ARR); cyber and white-collar are next.

    > *"Even Meta, who was burning through 70T tokens per month in February... is only a 3-5% customer for Anthropic per our estimates."*
    > — TokenBudgeting: Our Conversations with Enterprises on Token Spend (2026-06-30)

12. **Extended server depreciation is defensible, not accounting fraud —GPUs stay economically useful past the product cycle.**

    Michael Burry alleged hyperscalers boost earnings by stretching useful life from 3-5yr to 5-6yr on a 2-3yr Nvidia cycle. SemiAnalysis calls this a 'fatal flaw': supercomputers prove longevity (IBM Summit ran 6.5yr; Fugaku/Sierra still on Top500; V100 from 2017 still rented on AWS/Lambda). Warranties now extend 6-7yr; the driver is reliability + incentives, not fiction. Notably AWS runs the SHORTEST depreciation (5yr) of all CSPs yet shows the only rising margin trend —the opposite of what an earnings-management story would predict.

    > *"Dr. Burry's claim is predicated on an assumption that the NVIDIA product cycle is now 2-3 years... We believe this is a fatal flaw in the argument."*
    > — Microsoft's AI Strategy Deconstructed - From Energy to Tokens (2025-11-12)

13. **Fault-tolerant training is a moat: it's a 'secret sauce' available only to frontier labs or those who pay for a software license.**

    At 1k+ GPU scale you can't rely on the provider to catch every failure; job restarts cost 10-15 min plus lost compute. The three fault-tolerant frameworks all carry tradeoffs: TorchFT (open, but GLOO comms cost >10% perf and whole-replica-group blast radius), AWS HyperPod Checkpointless (recovery in 1m45s vs 15min, but ~5% memory overhead, AWS/Megatron-only), Clockwork TorchPass (zero perf overhead but licensed + idle-node cost). Because none is both free and overhead-free, reliable large-scale training stays concentrated among well-resourced players.

> *"This leaves fault tolerant training as a secret sauce available to frontier labs or those willing to pay for a software license."*
> — How Much Do GPU Clusters Really Cost? (2026-04-20)

14. **Selling enterprise tokens is still tiny and hard —even Google's flagship disclosure implies <0.5% of GCP.**

    Converting tokens to revenue is riddled with errors (I/O ratios, cached tokens, pricing). Sundar Pichai's Q3'25 flex —~150 Google Cloud customers each processed ~1 trillion tokens over 12 months —actually implies Gemini token sales to those 150 enterprises are less than 0.5% of GCP. The TaaS enterprise business is nascent; the frontier-model access (Claude/GPT/Gemini) that AWS/Azure/Google monopolize matters far more than model-library breadth or headline throughput.

    > *"This disclosure suggests that sales of Gemini tokens to these 150 enterprises is less than 0.5% of GCP's business."*
    > — Microsoft's AI Strategy Deconstructed - From Energy to Tokens (2025-11-12)

Article Deep-Dives

How Much Do GPU Clusters Really Cost? (2026-04-20)

SemiAnalysis's definitive methodology for cluster TCO, turning gut-feel into dollars. It decomposes cost into GPUs + Storage + Networking + Control Plane + Support + Goodput + Setup + Debugging, then introduces the 'Grand Unifying Theory of Goodput' —quantifying downtime as an implicit % tax that scales with job size / cluster MTBF. Across three scenarios (large LLM pretrain, multimodal RL, inference endpoints) with equal $/GPU-hr, Gold-tier is baseline, Hyperscaler runs 1.10-1.61x and Silver 1.15x. It benchmarks three fault-tolerant frameworks (TorchFT, AWS Checkpointless at 1m45s recovery, Clockwork TorchPass) and ships free public TCO/Goodput calculators, plus a ClusterMAX 2.1 tier update.

Microsoft's AI Strategy Deconstructed - From Energy to Tokens (2025-11-12)

A full-stack teardown of Microsoft across the AI Token Economic Stack (Apps → Models → PaaS → IaaS → Chips). Narrative arc: 2023-24 all-in (Fairwater, world's largest DCs for OpenAI) → the 'Big Pause' → a 2025 return that finds MSFT out of near-term capacity and forced to rent from neoclouds and resell at thin margin. Key claims: MSFT stepped away from ~$150B of OpenAI gross profit (Oracle won it); Azure risks a ClusterMAX downgrade (weak CycleCloud/AKS for managed clusters); but MSFT still holds 100% of OpenAI API inference compute through 2032 and OpenAI IP rights via Foundry. Also defends extended depreciation against Michael Burry, and dissects GitHub Copilot's eroding moat and the struggling Maia ASIC.

AI Value Capture - The Shift To Model Labs (2026-05-01)

The single most important framing piece: where AI profit pools sit and where they're rotating. 2023-25 infra captured everything; from Dec-2025 agentic AI works and the labs capture it. Anchored by two mechanisms —tokens getting cheaper to produce (GB300 ~17-32x an H100 at only ~70% more TCO; 14x from software alone) and rising market-clearing token value (Opus repriced down to $5/$25 yet margins UP by shifting volume to pricier SKUs like Mythos $25/$125). Argues profits won't be competed away (frontier pricing power + compute scarcity). Introduces 'Nvidia as the central bank of AI' —Nvidia/TSMC deliberately under-pricing scarcity —and the 'One Chart to Rule Them All' GPU-rental value-split framework built on the Vera Rubin VR NVL72 vs GB300 economics.

Anthropic Growth and Bedrock Mix Drive AWS Margins Higher While Peers Lag (2026-05-27)

The clearest case study of business-model-as-margin. Powered by the Tokenomics 2.0 model, it explains why AWS EBIT margin inflected +213bp Q/Q while Oracle, CoreWeave and Azure margins fell. Three levers: (1) Bedrock's token-as-a-service mix (vs 80%+ IaaS at GCP/Azure); (2) the Anthropic distribution deal where Anthropic is seller-of-record but AWS collects infra + revenue-share; (3) vertical silicon (Trainium >50% of Bedrock tokens, industry-leading Graviton4/5 CPUs). It also quantifies Anthropic's blowout ($21B net-new ARR in 1Q26 to $30B; inference GM mid-60s) and frames Google as the ultimate supply-constrained business trying to serve cloud + hardware + models + ads simultaneously.

TokenBudgeting: Our Conversations with Enterprises on Token Spend (2026-06-30)

Ground-truth from 50+ enterprise conversations (Databricks AI Summit, calls, Slack) that de-hypes the 'tokenmaxxing → budget wall' narrative. Findings: the Meta/Uber blowout stories stem from bad incentives not a broad slowdown; even Meta is only a 3-5% Anthropic customer; the 90th+ percentile drives most revenue at little risk. Budgets are now universal but with no convergent number ($250 to tens of thousands/mo). Companies conserve by downgrading defaults (Opus→Sonnet), turning off Fast-Mode, and employees gaming M365 Copilot allowances. Crucially: median Fortune 500 still spends <$100/employee —the adoption s-curve has enormous runway, with cyber and white-collar knowledge work the next verticals after coding.

How Oracle Is Winning the AI Compute Market (2025-06-30)

How a cloud latecomer became an AI-compute powerhouse by blending three archetypes: traditional US hyperscaler + AI-native neocloud + fast-moving Chinese developers. Oracle's edge is structural cost, not software: ~20% cluster CapEx advantage from optimized RoCEv2 networking (Sun/Mellanox HPC heritage), Foxconn ODM sourcing (bypassing Dell/Supermicro margin), and the market's lowest cost of capital via investment-grade debt. The most underappreciated growth engine is ByteDance (making Johor, Malaysia the #2 global AI hub), alongside the ~880MW Abilene 'Stargate' OpenAI campus already on Oracle's books. Validated by Larry's >$130B booking forecast and >100% RPO growth.

Key Data

Gold vs Silver-tier cluster TCO gap (equal $/GPU-hr)	Silver ~15% more expensive (1.15x); Hyperscaler 1.10-1.61x	Driven by goodput, setup, storage —not headline price.
Goodput expense, large LLM pretrain (Gold/Hyperscaler/Silver)	6.14% / 10.53% / 20.91%	Collapses to 0.23-0.96% for many small jobs.
Anthropic ARR trajectory	$9B → $30B (added $21B net-new in 1Q26) → ~$44B YTD; model sees >$100B by year-end	Driven by Claude Code / enterprise API (90%+ B2B).

Anthropic inference gross margin	-94% (2024) → 38% (2025) → mid-60s / >70% (2026)	WSJ: operating-income profitable in 2Q26 ex-SBC.
Oracle OpenAI contract value / gross profit	>$420B contracts → ~$150B gross profit	~$30B/yr would have lifted MSFT FY25 gross profit >18%.
AWS AI revenue mix & margin move	AI = 10% of AWS (from 2% in 1Q24); EBIT margin +213bp Q/Q	Bedrock now 37% of AWS AI vs GCP/Azure IaaS 80%+.
Bedrock run-rate & growth	~$5.5B run-rate; +170% Q/Q (1Q26), +60% Q/Q (4Q25); 80-90% Anthropic	Trainium powers >50% of Bedrock token usage.
GB300 NVL72 vs H100 throughput / TCO	~17x (FP8) to 32x (FP4) throughput at only ~70% higher TCO/GPU	Software alone can 14x B300 throughput (wideEP+disagg+MTP).
Opus effective blended token price	~$0.99/MTok vs $5/$25 sticker	~300:1 input:output, 90%+ cache hit, cached input $0.50/MTok.
SemiAnalysis's own token spend	~30% of employee comp; ~5B tokens/mo/employee (5x Meta); peak $10.95M/yr on Claude	Power-law: some staff run 100B+ tokens/mo.
Enterprise token budgets (per employee/mo)	$250 (aerospace) / $500 (pharma) → ~$2,000 (Stripe/Workday) → tens of thousands	No convergence; Uber capped at $1,500/mo after 4-month burn.
Ramp AI spend distribution	99th pct ~$90k/yr/emp; 90th ~$7,300; median $136	Median Fortune 500 still <$100/employee.
GPU vs CPU server TCO split	GPU: $7,025 capital vs $1,871 hosting/mo; CPU: $301 vs $220	GPU cloud is capital-dominated → capital is the only real barrier.

H100 all-in cost vs rental	~$1.525/hr all-in (13% debt) vs ~$2/hr best deals, $3+ when fleeced	1-yr H100 rentals +40% off Oct-2025 bottom.
Microsoft datacenter freeze	>2GW non-binding LOIs dropped; 1.5GW near-term self-build frozen; ~5GW binding remains	MSFT drove >60% of new leased turnkey capacity Q1′23-Q2′24.
Microsoft Fairwater datacenter scale	~300MW GPU buildings (>150k GB200 each); 3rd phase 600MW+ buildings	Largest individual datacenters on Earth if built on time.
Coding share of frontier-lab ARR	>70% of OpenAI + Anthropic ARR	Anthropic 90%+ B2B vs OpenAI ~60% consumer.
Project Rainier (Anthropic × AWS)	400k Trainium2 chips	Funded by Amazon's $4B+ circular Anthropic investment.

Investment Angle

Winners

Amazon / AWS	Only CSP with token-as-a-service (Bedrock) as its dominant AI mix + vertical silicon (Trainium >50% of Bedrock tokens, leading Graviton) —structurally highest margin. Anthropic offtake circular but real. Building more capacity than anyone in 2025-27.
Anthropic	The purest value-capture play: ARR $9B→$30-44B, inference GM 38%→70%+, 90%+ B2B/API driven by Claude Code. Frontier pricing power means margins won't be competed away.
Oracle	~20% structural CapEx advantage (networking + ODM + IG debt), >$420B OpenAI backlog, ByteDance/Johor engine. But ROIC ~20% (below MSFT 35-40%) and margins disappointed the market.
Gold-tier neoclouds (**CoreWeave, Nebius, Crusoe, Fluidstack**)	Reliability/support translates into a real TCO discount that commands price premium; rental prices +40% off Oct-2025 bottom lift returns.

Nvidia & TSMC	Still own the scarcest layers (GPUs, N3) and deliberately hold pricing below scarcity —dry powder to reprice later. GB300/Rubin step-change in perf/TCO keeps them indispensable.

Bottlenecks

- Power / multi-GW PPAs —watts gate GPU deployment; securing power is now a share-determining act.
- TSMC N3 wafer allocation —every accelerator roadmap converged on N3; tightest node in the system.
- HBM / memory —prices up ~6x in a year; a once-in-four-decades shortage.
- Fault-tolerant training software —no framework is both free and overhead-free; concentrates reliable scale training.
- Frontier-model access —only AWS/Azure/Google can sell all three frontier families; a distribution moat.

Risks

- Depreciation debate: if GPU useful life proves shorter than 5-6yr, hyperscaler earnings are overstated (Burry thesis).
- Circular financing: Amazon/MSFT/Google invest in labs that then buy their compute —revenue that could reverse.
- Token budgets tightening: enterprise caps could slow API growth if the s-curve stalls (SemiAnalysis says overblown).
- Nvidia/TSMC repricing: 'central bank' restraint could end, venting margin away from labs/neoclouds.
- Overbuild / ROIC: Oracle/neocloud AI ROIC (~20%) well below legacy cloud; a demand air-pocket compresses returns.

Catalysts

- New OpenAI–Microsoft deal reaccelerating Azure; 100% of OpenAI API inference through 2032.
- Vera Rubin (VR NVL72) ramp: step-jump in perf/TCO resetting the value-capture math.
- Next enterprise verticals (cyber via Mythos re-release, white-collar via Cowork/Copilot/Codex).
- Trainium3 ramp + Project Rainier scaling as a proof point for hyperscaler ASIC economics.
- ClusterMAX 3.0 (B300/GB300, 800Gb) re-rating neocloud tiers and TCO leadership.

Open Questions

- Is the labs' margin capture durable, or does open-source (Kimi, DeepSeek) + a compute glut eventually compress frontier pricing?

- How much of hyperscaler/lab revenue is circular (invest-in-lab → lab buys compute), and what happens if funding tightens?
- Will custom ASICs (Trainium, TPU) actually strip meaningful Nvidia margin, or stay niche inference/RL weapons?
- When does Nvidia/TSMC 'central bank' restraint end, and who loses margin when it does?
- Is 5-6yr GPU depreciation right? A wrong answer re-rates every hyperscaler's earnings.

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Chapter 9

AI Models, Labs & Software

Scaling never died —it stacked. Pre-training gave way to reinforcement learning and test-time compute, data replaced weights as the moat, and agents like Claude Code turned models into the fastest-growing software business ever.

Overview

SemiAnalysis's model/lab coverage is built around one contrarian spine: the 'scaling laws are dead' narrative is wrong because scaling is not one curve but a stack of them. When pre-training hit the data wall, the labs pivoted to post-training —supervised fine-tuning, RLHF/RLAIF, and above all reinforcement learning on verifiable rewards —and then to test-time (inference) compute via reasoning models. OpenAI rode the same GPT-4o base model for 18 months through o1, o3 and GPT-5, proving post-training alone drives capability. The consequence is a different industrial structure than pre-training: where everyone once trained on the same internet, RL requires bespoke environments, verifiers, and expert-authored data that must be built from scratch —so 'data is the moat.' An entire shovel-selling economy has erupted: 35+ RL-environment startups, UI-gym website clones at ~$20k each, and data foundries (Surge ~$1B ARR, Mercor, Handshake) that recruit PhDs to write tasks and rubrics. On the product side, Claude Code is SemiAnalysis's declared inflection point —coding is the beachhead into a $15T information-work economy, the six-player coding-assistant market already clears $30B+ ARR heading to $100B, and Anthropic is now adding more revenue per quarter than OpenAI. The technical throughline is that RL fuses inference into training (rollouts are inference-heavy), which reshapes hardware demand toward memory-bound inference chips (Trainium), CPUs for RL environments, and decentralized rather than mega-centralized clusters. Below the frontier, DeepSeek and other Chinese open-weights labs (MLA cutting KV cache ~93%, V4 hitting 1M context with a 90% KV-cache cut) commoditize the low end, but still trail on agentic coding —'data is the moat' keeps the frontier ahead. Benchmarks are broken (SWE-bench contamination, GDPval single-turn), so the real north-star metric is cost-per-task, not cost-per-token.

Core Knowledge

The scaling stack (pre-train → post-train → test-time)

SemiAnalysis's core framing: overall AI 'scaling' is the sum of three compounding vectors, not just pre-training. Pre-training (next-token prediction on the internet) hit a data wall; post-training (SFT/RLHF/RLAIF/RL on verifiable rewards) unlocked reasoning; test-time compute lets a model 'think' longer for a better answer. Like Moore's Law surviving the end of clock-speed scaling via multi-core, the aggregate curve keeps climbing even as one vector slows.

RL on verifiable rewards (GRPO/PPO)

RL worked best where a reward is cleanly checkable —code (does it pass tests?) and math (is the answer right?). The model generates many 'rollouts' (attempts) per prompt, each scored; GRPO (DeepSeek's algorithm) drops PPO's critic model to save memory by scoring each rollout against the group average.

This makes RL inference-heavy: hundreds of answers per question, so production-grade inference is now inside the training loop.

RL environments & sandboxes

An 'environment' is the simulated world (a code sandbox, a cloned website, a browser, a spreadsheet app) where an agent acts and receives a reward. Engineering robust, low-latency, exploit-resistant sandboxes (Firecracker micro-VMs to full QEMU VMs) is a first-class systems problem —sandbox startup latency and scaling to thousands of concurrent rollouts are real bottlenecks. Reward hacking (the model gaming the reward without doing the task) is the recurring failure mode.

Synthetic data, RLAIF & LLM-as-judge

As human data stopped scaling, labs turned to synthetic data (rejection sampling, model-generated tasks) and replaced human feedback (RLHF) with AI feedback (RLAIF). A model or 'LLM judge' scores outputs against a rubric —the key that opens non-verifiable domains like writing, medicine and law. Better models are better judges, creating a self-improving loop. Anthropic's Constitutional AI is the canonical RLAIF example.

Agent / harness (Claude Code)

An agent wraps a model in a loop of READ → THINK → WRITE → VERIFY with tools, memory and sub-agents. Claude Code is a terminal-native agent ('Claude Computer') that reads a codebase, plans, and executes. The 'harness' (tools, prompts, context management, caching) is now part of the product —it drives cost-per-task as much as the model does, which is why same-harness benchmark comparisons miss the point.

Cost-per-task vs cost-per-token

SemiAnalysis's north-star pricing metric. A model priced 5x higher per token (e.g. Mythos vs Opus) can be cheaper in practice if it solves the task in far fewer tokens ('token efficiency'). Anthropic's Opus 4.7 tokenizer change alone raised token usage up to 35% —an implicit 35% price hike. The harness (Codex ~80:1 input/output vs Claude Code ~100:1) is a major driver.

Data foundry / RL-as-a-service

Firms that supply the labs with RL environments, expert-authored tasks, and grading rubrics. Environment startups (Mechanize, Turing, Prime Intellect, HUD) clone apps and wrap software in dockerized sandboxes; data foundries (Surge, Mercor, Handshake) recruit domain experts. RL-as-a-service startups (RunRL, Osmosis, Applied Compute, Thinking Machines Tinker) post-train custom models (often Qwen) for enterprises at ~5x lower cost than OpenAI's RFT.

Multi-Head Latent Attention (MLA) & MoE

DeepSeek's signature efficiency innovations. MLA compresses the KV cache ~93% versus standard attention, slashing the memory (and thus cost) per query. Combined with fine-grained Mixture-of-Experts (many small experts, few active per token) and FP8 training, this is how a Chinese lab reached near-frontier quality at a fraction of the compute —and forced Western labs to copy the ideas almost immediately.

Value Chain

Key Theses

1. **Scaling laws didn't die —they stacked. Pre-training, RL post-training and test-time compute are three compounding curves.**

   The 2024 'scaling is over' panic mistook one vector (pre-training) for the whole. SemiAnalysis's analogy is Moore's Law: when clock speed stalled, multi-core kept aggregate compute climbing. OpenAI's o1 opened test-time scaling; RL opened post-training. The existence proof: OpenAI drove 18 months of gains (o1→o3→GPT-5) on the same GPT-4o base model before finally fixing pre-training too.

   > *"It is possible to stack 'scaling laws' - pre-training will become just one of the vectors of improvement, and the aggregate 'scaling law' will continue scaling just like Moore's Law has over last 50+ years."*
   > — Scaling Laws – O1 Pro Architecture, Reasoning Training Infrastructure, Orion and Claude 3.5 Opus 'Failures' (2024-12-11)

2. **RL is the current critical path to capability —and it is fundamentally inference-heavy, fusing inference into the training loop.**

GRPO/PPO generate hundreds of rollouts per question, all of which are inference. 'Inference performance now directly impacts training speed,' so labs merged their product-inference and research-inference teams (OpenAI, Anthropic, Google all reorganized). Unlike pre-training's centralized mega-clusters, RL inference doesn't need co-location —synthetic data can be generated in one datacenter and trained in another, and one lab runs RL on idle inference clusters for effectively free compute.

> *"RL for language models is one of the first cases where inference has truly become intertwined into the training process. Inference performance now directly impacts training speed."*
> — Scaling Reinforcement Learning: Environments, Reward Hacking, Agents, Scaling Data (2025-06-08)

3. **Data is the moat now, not weights. RL needs bespoke environments and expert data that must be built from scratch.**

   Pre-training was democratic —everyone trained on the same internet. RL is not: 'Most RL data and tasks must be constructed from scratch.' Qwen's headline 4,000-sample efficiency hides brutal filtering plus a much larger undisclosed second stage across 20+ domains. High-quality data requires giant amounts of inference to generate and filter, and startups with proprietary user data can RL custom models without huge compute budgets —the reason Cursor and Windsurf can compete with labs.

   > *"Ultimately what Qwen signals is that high quality data is a uniquely important resource for scaling RL... This paradigm is not like pre-training where everyone had access to the same data."*
   > — Scaling Reinforcement Learning: Environments, Reward Hacking, Agents, Scaling Data (2025-06-08)

4. **During an RL scaling boom, sell environments: a 35+ startup shovel economy of UI-gyms and data foundries has erupted.**

   Environment companies clone DoorDash/Uber Eats UIs at ~$20k/site —OpenAI has bought hundreds for ChatGPT Agent training. Data foundries (Surge ~$1B ARR, Mercor, Handshake) recruit finance pros, doctors and lawyers to design tasks and write rubrics. Anthropic is the first customer of dozens of these vendors, deliberately commoditizing the supply. Defunct startups are even acquired just for their private GitHub repos to mine into coding environments.

   > *"During a Gold Rush, sell shovels. In an RL Scaling Boom, sell RL environments. More than 35 companies have popped up whose goal is to do exactly this across a variety of domains."*
   > — RL Environments and RL for Science: Data Foundries and Multi-Agent Architectures (2026-01-06)

5. **Claude Code is the inflection point: coding is the beachhead into a $15T information-work economy.**

   SemiAnalysis argues every information job shares the workflow Claude Code proved works — READ, THINK, WRITE, VERIFY. With ~4% of GitHub public commits already authored by Claude Code (projected 20%+ by end-2026), the disruption jumps from coding to the 1B+ information workers (a third of the global workforce). Cowork ('Claude Code for general computing') was built by four engineers in 10 days, mostly by Claude Code itself —recursive self-improvement in action.

   > *"4% of GitHub public commits are being authored by Claude Code right now... While you blinked, AI consumed all of software development."*
   > — Claude Code is the Inflection Point (2026-02-05)

6. **Anthropic is now out-growing OpenAI —quarterly ARR additions have crossed over, constrained mainly by compute.**

   The coding-assistant market is 'the greatest B2B SaaS application the world has ever seen': $30B+ ARR across the six largest players, on track past $100B by year end. Claude Code drives exceptional Anthropic revenue growth; SemiAnalysis's Tokenomics model shows Anthropic adding more revenue every month than OpenAI, with growth capped by compute rather than demand. Anthropic is on track to add as much power as OpenAI over three years.

   > *"Notably, our forecast shows that Anthropic's quarterly ARR additions have overtaken OpenAI's. Anthropic is adding more revenue every month than OpenAI."*
   > — Claude Code is the Inflection Point (2026-02-05)

7. **Benchmarks are broken; the real north-star metric is cost-per-task, not cost-per-token.**

   SWE-bench Verified still had unfair evals on >half of o3's consistent failures, plus contamination —GPT-5.2, Opus 4.5 and Gemini 3 Flash had memorized answers, so OpenAI stopped reporting it. GDPval is single-turn with unnaturally clean prompts. Meanwhile a model 5x pricier per token can win on cost-per-task by using fewer tokens, and the harness (Codex ~80:1 input/output vs Claude Code ~100:1) drives the real bill. Opus 4.7's new tokenizer alone lifted usage up to 35%.

   > *"cost per task, not cost per token, is the true north star metric that determines model pricing."*
   > — The Coding Assistant Breakdown: More Tokens Please (2026-04-24)

8. **RL for non-verifiable domains is solved via rubrics and LLM judges —opening writing, medicine, law and science.**

   Skeptics said RL only works where rewards are checkable. But OpenAI's deliberative-alignment paper used an LLM judge + rubric (only synthetic data) and got strong out-of-distribution generalization. For HealthBench, OpenAI had 260+ physicians write rubrics for the judge. 'If it can be measured, it can be improved via RL.' A reasoning model as judge understands the rubric better —RL helping you do better RL, a self-improving loop.

   > *"Non-verifiable domains include areas like writing or strategy, where no clear right answer exists. There has been some skepticism around whether this will be possible at all to do RL on. We think it is. In fact, it's already been done."*
   > — Scaling Reinforcement Learning: Environments, Reward Hacking, Agents, Scaling Data (2025-06-08)

9. **Enterprise SaaS is the first casualty: agents erode the switching-cost, workflow-lock-in and integration moats.**

   SaaS is 'crystallized information processing of workflows into code' at ~75% gross margin. Agents migrate data between systems cheaply, don't need human-oriented UIs, and MCP makes integration trivial. An agent can query Postgres, make a chart and email it —replacing a CRM workflow with no UI training. Microsoft is caught: accelerating Azure means renting GPUs to 'the barbarians' (OpenAI, Anthropic) who erode its seat-based Office 365, and Claude for Excel is what Copilot should have been.

   > *"Claude for Excel effectively is what Copilot for Excel should have been, but it was launched by an external party on their own first party product."*
   > — Claude Code is the Inflection Point (2026-02-05)

10. **DeepSeek sets the efficiency frontier and commoditizes the low end —but 'data is the moat' keeps the frontier ahead.**

    DeepSeek's MLA cut KV cache ~93%; V4 (1.6T total / 49B active) moved to 1M context claiming only 27% of single-token FLOPs and a 90% KV-cache cut versus V3.2. Its open kernels keep American open-source alive. Yet V4 still lags closed frontier models on agentic tasks —Claude Opus 4.7 even beats it at difficult Chinese writing. The 2025 '$6M training cost' figure is a myth: it's only the final pre-train GPU bill, atop >$500M in cumulative hardware.

    > *"The $6M cost in the paper is attributed to just the GPU cost of the pre-training run, which is only a portion of the total cost of the model... We are confident their hardware spend is well higher than $500M over the company history."*
    > — DeepSeek Debates: Chinese Leadership On Cost, True Training Cost (2025-01-31)

11. **The router is the release: GPT-5's real product is free-user monetization via agentic purchasing.**

    ChatGPT is the #5 website on earth with 700M+ mostly-unmonetized free users. GPT-5's router quietly classifies query intent and complexity —the one attribute needed to add commercial value. A high-intent query ('best DUI lawyer near me') could get $50 of compute because the referral is worth thousands. Fidji Simo (ex-Facebook/Instacart monetization) now runs OpenAI Applications; a take-rate/agentic-checkout SuperApp (Shopify, Instacart, Etsy already integrated) is the endgame.

    > *"We believe that the Router is the groundwork for the next leg of ChatGPT's story, and that's monetization of free users."*
    > — GPT-5 Set the Stage for Ad Monetization and the SuperApp (2025-08-13)

12. **RL environments are spilling into the physical world —biology and materials wet-labs become the new bottleneck.**

    Environments are no longer just docker containers. Periodic Labs and Medra are building robotic wet-labs that generate experimentally-verified reward signals; DeepMind starts an automated materials lab in 2026. The economics invert: a coding task can be attempted 64x for trivial cost, while one biology experiment costs hundreds-to-thousands of dollars and takes hours, with sparse rewards. OpenAI targets early drug discovery (GPT-5 in a closed loop with robots); Anthropic ships Claude for Life Sciences (Benchling, PubMed connectors) for development/approval.

    > *"RL environments are spilling into the physical world. These environments are no longer just a docker container... It is an experiment that needs to be run by a human or a robot, with a real cost to materials, electricity, equipment, and lab space."*
    > — RL Environments and RL for Science: Data Foundries and Multi-Agent Architectures (2026-01-06)

13. **RL rewires hardware and datacenter design: memory-bound, CPU-hungry, decentralizable — favoring low-$/HBM chips like Trainium.**

    RL uses fewer FLOPs than pre-training but heavy memory loads (rollouts, long context, judges), so the NVL72 rack-scale shared memory shines and low-TCO memory chips win. Anthropic is onboarding large Trainium volumes precisely because RL is memory-bound and Amazon sourced HBM directly to cut cost. RL environments also revived the datacenter CPU —Microsoft's 'Fairwater' pairs a 48MW CPU/storage building with a 295MW GPU cluster, and labs are now scrambling for CPU allocation.

> *"Anthropic is bringing on large volumes of Trainium, which is economically attractive for RL in part due to low $/HBM... RL workloads are mostly inference, which is memory bound."*
> — RL Environments and RL for Science: Data Foundries and Multi-Agent Architectures (2026-01-06)

14. **RL-training efficiency is a queue-health problem: match trainer and generator throughput or waste your GPUs.**

    An open-source RL system has three actors —generator (inference), environment (sandbox), trainer. If the generator is slower the trainer starves; if faster, samples go stale. PipelineRL's in-flight weight updates let the two run at different speeds within a policy-staleness budget, ~2x more iterations per wall-clock hour. SemiAnalysis's own experiments were consistently 'generation-bound' (trainer idle 30-74%), and sandbox scaling to ~960 concurrent rollouts hit dead init errors —proving sandbox infra is a first-class bottleneck.

    > *"RL training system efficiency is a matter of queue health... in-flight weight updates allow trainer and generator to operate at different rates."*
    > — RL Systems Mind the Gap: Matching Trainer and Generator Throughput (2026-06-16)

Article Deep-Dives

Scaling Laws –O1 Pro Architecture, Reasoning Training Infrastructure, Orion and Claude 3.5 Opus 'Failures' (2024-12-11)

The foundational rebuttal to the 'scaling is dead' narrative. Reframes scaling as a stack: pre-training (hitting the data wall, Chinchilla-suboptimal mega-models), post-training (SFT, RLHF, RLAIF, PPO with Outcome/Process Reward Models), and test-time compute via reasoning (search, best-of-N, Monte Carlo roll-outs). The most-cited scoop: Anthropic finished Claude 3.5 Opus, it scaled fine, but they didn't ship it —instead using it to generate synthetic data and reward-model to improve 3.5 Sonnet. Details o1 vs o1 Pro (Pro adds search/self-consistency), harder evals (FrontierMath at 2%, GPQA), and why reasoning models are bottlenecked by inference systems keeping context lengths short.

Scaling Reinforcement Learning: Environments, Reward Hacking, Agents, Scaling Data (2025-06-08)

The canonical RL primer. Explains GRPO (DeepSeek's critic-free PPO variant), why RL is inference-heavy, and why reward functions are a 'dark art' prone to reward hacking (Claude 3.7 edited test files to pass). Argues 'data is the moat' since RL data must be built from scratch, and that RL fuses inference into training (labs merged inference teams). Key structural claims: RL doesn't need centralized clusters like pre-training (one lab runs it on idle inference for free compute), China is chip-constrained for the inference-heavy RL game (H20 ban bites), and recursive self-improvement is already happening via compiler/kernel RL.

RL Environments and RL for Science: Data Foundries and Multi-Agent Architectures (2026-01-06)

Maps the RL-environment industrial complex. 35+ startups sell UI-gyms (~$20k/site) and dockerized sandbox tooling (HUD wraps software + an MCP server per container); data foundries (Surge ~$1B ARR, Mercor, Handshake) recruit domain experts to author tasks and rubrics. Contrasts lab buying patterns (Anthropic = broadest vendor ecosystem + code/computer-use; OpenAI = fewer vendors but outspends + in-house Feather platform; Google = decentralized, owns the apps). Then extends to RL-for-science: Periodic Labs and Medra build robotic wet-labs, OpenAI targets drug discovery while Anthropic ships Claude for Life Sciences. Warns automation may be augmentation (GDPval: experts stayed faster/cheaper; radiology never got automated).

Claude Code is the Inflection Point (2026-02-05)

Declares Claude Code the ChatGPT-moment for agents. ~4% of GitHub commits already Claude Code, projected 20%+ by end-2026. Frames agents as orchestration of tokens (Web 2.0) atop raw token APIs (Web 1.0/TCP-IP), and coding as the beachhead into the $15T information-work economy via the READ/THINK/WRITE/VERIFY workflow. Cowork was built by 4 engineers in 10 days, mostly by Claude Code itself. Core financial call: Anthropic now adds more revenue per quarter than OpenAI, capped by compute. Deep dive on Microsoft's conundrum —accelerating Azure means renting GPUs to the 'barbarians' (OpenAI, Anthropic) eroding seat-based Office 365; Claude for Excel is what Copilot should have been, and SaaS's 75%-margin moats are eroding.

The Coding Assistant Breakdown: More Tokens Please (2026-04-24)

A model-by-model teardown (GPT-5.5/'Spud', Opus 4.7, DeepSeek V4, Mythos, Kimi, Qwen, GLM) plus a masterclass on why benchmarks are bad. GPT-5.5 finally reaches the frontier in Codex but is beaten by Opus 4.7 on the buried Expert-SWE coding benchmark (Mythos scored 77.8%). Dissects benchmark anatomy (tasks/eval/harness) from MMLU → SWE-bench (contaminated, retired by OpenAI Feb 2026) → GDPval, and shows OpenAI's sneaky reporting. Practical workflow: engineers now start in Claude (greenfield scaffolding) then switch to Codex (harder narrow bug-fixing). The thesis pillar: cost-per-task, not cost-per-token, is the north star; the harness (Codex ~80:1 vs Claude Code ~100:1 input/output) drives the real bill.

Key Data

Claude Code share of GitHub commits	~4% now → projected 20%+ by end-2026	SemiAnalysis's headline datapoint for the agentic inflection; 'Pretty much 100% of our code is written by Claude Code + Opus 4.5' —Boris Cherny, Claude Code creator.
Coding-assistant market ARR	$30B+ across 6 players, on track past $100B by year end	'The greatest B2B SaaS application the world has ever seen,' per the Tokenomics model.
Information-work TAM at risk	$15T economy, 1B+ workers (1/3 of the 3.6B global workforce)	Coding is only the beachhead; the READ/THINK/WRITE/VERIFY workflow generalizes to all information work.
GDPval win rate (GPT-5.2)	~71% tie-or-preferred vs human experts across 44 occupations	OpenAI eval of real economically-valuable tasks; but experts were still faster/cheaper —augmentation, not automation.
MLA KV-cache reduction	~93.3% vs standard attention	DeepSeek's Multi-Head Latent Attention —the core reason its inference is so cheap.

DeepSeek V4 architecture	Pro 1.6T total / 49B active; Flash 284B / 13B; 128k → 1M context	Claims only 27% of single-token FLOPs and 10% of KV cache vs V3.2 at 1M tokens (a ~90% KV-cache cut).
GB300 NVL72 cost per Mtok	$0.156 per 1M output tokens (DeepSeek V4, 50 tok/s/user, 8k in/1k out)	Rack-scale NVLink world size keeps MoE all-to-all on NVLink —why GB300 mogs 8-GPU islands at reasoning.
MI355X software gains (DeepSeek V4)	>100x throughput improvement Day 0 → Day 26	Pure software (AITER/Triton/TileLang kernels replacing PyTorch fallbacks) — shows the CUDA-moat / ecosystem-maturity gap.
Inference cost-down for GPT-3 quality	~1200x cheaper	Algorithmic progress ~4x/year (Dario Amodei argues ~10x); Jevons paradox reinvests savings into bigger models.
UI-gym / environment cost	~$20,000 per cloned website; OpenAI bought hundreds	One-time purchases reused across models; trajectories fed back into mid-training.
Data-foundry revenue	Surge ~$1B ARR; Scale AI ~$1.4B (2024) before Meta absorbed it	Mercor, Handshake, Aboda.ai fill the gap after labs fled Scale post-Meta acquisition.
Opus 4.7 tokenizer price impact	up to +35% token usage (implicit +35% price)	New tokenizer trades finer counting for more total tokens —why cost-per-task beats cost-per-token.
ChatGPT reach & free base	#5 website globally, 700M+ mostly-unmonetized free users	The router is the groundwork for monetizing them via agentic purchasing / take-rates.
GPT-5.5 API pricing	$5 / 1M input, $30 / 1M output (2x GPT-5.4, ~Opus 4.7 level)	'Trained on a 100k GB200 NVL72 cluster' was post-training (RL) only; priority tier is 2.5x standard.
Qwen post-training compute	~5% of pre-training compute	Most Chinese labs are early to RL scaling; homegrown data foundries would accelerate the transition.

Autonomous task-horizon doubling	every ~7 months for coding (accelerating to ~4 months in 2024-25)	METR data; each doubling unlocks more of the automation pie (30 min → refactor a module → multi-day audit).
Anthropic RFT vs young startups	YC RL-as-a-service serves at ~5x lower cost than OpenAI's RFT	OpenAI's RFT is unstable/expensive; long-run the labs capture most revenue via 'Strategic Deployment' teams.
Scale of coding-env task generation	DeepSeek used 24,667 coding tasks for V3.2; labs instantiate 10,000+ sandboxes simultaneously	SWE-rebench yields only 21,336 valid tasks from 450k PRs —strict execution-based verification.

Investment Angle

Winners

Anthropic	SemiAnalysis's clearest structural winner: Claude Code is the declared agentic inflection, ~4% of GitHub commits and climbing, and quarterly ARR additions have overtaken OpenAI's —growth capped by compute, not demand. Best-in-class at agentic coding (beats Chinese models even at Chinese writing), most aggressive RL-environment buyer, and a stable/sample-efficient RFT-like enterprise service on low-$/HBM Trainium.
OpenAI	Distribution king: ChatGPT is the #5 website with 700M+ free users, and the router lays the groundwork to monetize them via agentic purchasing/take-rates. GPT-5.5 ('Spud') is back at the frontier, enterprise growth outpaced consumer in 2025, and it outspends peers on data across many parallel programs (UI gyms, IMO-gold math/code).
Data foundries & RL-env startups (**Surge, Mercor, Handshake, Prime Intellect, Mechanize**)	The picks-and-shovels of the RL boom. 'During an RL scaling boom, sell environments.' Surge ~$1B ARR, Scale did $1.4B before Meta absorbed it; demand for coding environments is so high defunct startups are bought just for private GitHub repos. Prime Intellect's open Environments Hub and Prime RL are becoming the open-source standard.

Google DeepMind	Uniquely positioned if capital/compute is king: owns TPUs (lowest cost per inference) and the apps to post-train on (Sheets, Docs, Drive, Maps) with PM visibility into how users actually use them. Only lab that wholly owns its research org and silicon; Gemini 3 was a step-change on HLE after a 9-figure STEM-data budget.
Nvidia (GB200/GB300 NVL72)	RL's memory-heavy, inference-fused profile rewards rack-scale NVLink world size: GB300 mogs 8-GPU islands at reasoning ($0.156/Mtok on DeepSeek V4) by keeping MoE all-to-all on NVLink. Open engines (vLLM, SGLang) get Day-0 CUDA support that even Nvidia's own TensorRT-LLM lags —the CUDA moat at work.
DeepSeek & Chinese open-weights labs	Set the efficiency frontier (MLA ~93% KV-cache cut; V4 1M context at ~10% KV cache) and keep American open-source alive via DeepEP/DeepGEMM/FlashMLA. Kimi K2.6 beats Nvidia's Nemotron on coding; V4 is the lowest-cost near-frontier alternative even if not leading-edge.

Bottlenecks

- RL environments & sandbox infra —the true gate on capability. Robust, low-latency, exploit-resistant sandboxes that scale to thousands of concurrent rollouts are hard; SemiAnalysis hit dead init errors at ~960 concurrent sandboxes and consistently 'generation-bound' systems (trainer idle 30-74%).
- Expert data & rubrics —RL needs bespoke tasks/verifiers built from scratch; STEM PhDs, doctors, lawyers must author tasks, and 9-figure data budgets (Google's HLE spend) are now table stakes.
- Compute for inference-heavy RL —RL is memory-bound and inference-intensive; China is chip-constrained (H20/H20E ban bites the very inference RL depends on), and even Western labs are scrambling for CPU allocation for RL environments.
- Physical-world feedback loops —RL-for-science hits real limits: a biology experiment costs hundreds-to-thousands of dollars and takes hours with sparse rewards, versus a coding task attempted 64x for trivial cost.

Risks

- Benchmarks mislead —SWE-bench is contaminated (models memorized answers) and retired; GDP-val is single-turn; labs selectively report (OpenAI buried its coding benchmark). Capability claims can't be taken at face value; cost-per-task is the only honest metric.
- Reward hacking & hallucination —models game rewards (Claude 3.7 edited test files) and o3 hallucinates more as RL compute scales because it's rewarded for correct outcomes even via flawed reasoning.
- Automation may be only augmentation —GDPval showed experts stayed faster/cheaper; radiology was never automated. Adoption is also blocked by defensive web providers (Amazon blocking ChatGPT Agent) and enterprise inertia.

- Governance & partner conflict —OpenAI's board can declare AGI and void Microsoft's IP rights with no recourse; Microsoft is structurally torn between Azure growth and defending Office 365 that its own tenants (OpenAI, Anthropic) erode.

Catalysts

- Frontier pre-trains resuming —OpenAI's 'Spud' (GPT-5.5) and Anthropic's 'Capybara' mark the first new pre-training scale-ups since the failed GPT-4.5, layering a re-unlocked pre-train vector on top of RL gains.
- Agentic-checkout / SuperApp launch —OpenAI's router + Instant Checkout (Shopify, Etsy, Instacart) and Amazon-access talks could flip free-user monetization on, repricing search and ad markets.
- Enterprise mega-deployments —Accenture training 30,000 on Claude (largest Claude Code deployment) and OpenAI's Frontier signal SaaS-margin repricing across financial services, legal, life sciences and the public sector.
- RL-for-science payoff —Periodic Labs / Medra robotic wet-labs, DeepMind's 2026 materials lab, and GPT-5 closed-loop drug discovery could deliver the first AI-driven scientific breakthroughs (Anthropic targets 2027, OpenAI targets autonomous AI researchers by March 2028).

Open Questions

- Is the path to AGI just environments stacked on top of each other, or does RL hit a wall in non-verifiable, long-horizon, sparse-reward domains (computer use, biology) that rubrics can't fully paper over?
- Does the frontier stay ahead of DeepSeek and Chinese open-weights labs on agentic coding, or does 'data is the moat' erode as homegrown Chinese data foundries stand up and Qwen scales past 5% post-training compute?
- Can Microsoft resolve its conundrum —accelerate Azure (renting GPUs to the barbarians) without letting Claude Code and agents gut its seat-based Office 365 terminal value?
- Will LLM automation be genuine job replacement or mostly augmentation? GDPval and radiology suggest the latter for expert work —but maybe not for shorter-horizon repetitive tasks like call centers.
- How decentralized can RL get? If synthetic-data generation and training can live in different datacenters, does the mega-cluster arms race matter less than everyone assumes —and who captures the 'free' idle-inference compute edge?

Sources (SemiAnalysis)

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Chapter 10

China, Geopolitics & Export Controls

The AI cold war fought in wafers, HBM stacks and actuators —where export controls bite, where they leak, and where China already dominates.

Overview

SemiAnalysis frames US–China technology policy as a full-scale economic cold war whose single most important lever is compute: the US controls ~70% of the world's deployed FLOPs and is trying to keep China off the frontier by choking AI accelerators, wafer-fab equipment (WFE), EUV/DUV lithography and —most decisively now —HBM memory. Their reporting is unusually granular and unusually opinionated: they argue controls are simultaneously (1) genuinely effective at the leading edge (China will make FEWER Huawei Ascends next year, not more, because it runs out of foreign HBM), and (2) riddled with self-inflicted loopholes —the 'fab whack-a-mole' of entity-listing one building while its non-listed twin across the street imports every advanced tool, allied-lag windows where Samsung dumped 7M HBM stacks in one month before enforcement, and Huawei's TSMC 'die bank' of 2.9M+ Ascend dies smuggled via shell companies like Sophgo/Bitmain. The corpus spans the 2022 opening salvo, the 2023 'controls have failed' broadside, the 2024 Fab Whack-A-Mole enforcement manifesto, and the 2025 Huawei Ascend production model. Running underneath is a second, non-semiconductor front: robotics. Here SemiAnalysis is bearish on the West —China is running the BYD/DJI playbook (own the hardest component, bootstrap on hobbyists, verticalize, undercut on price), Unitree is shipping its 10,000th humanoid at a ~$9K BoM and 67% gross margin while Tesla Optimus and Figure ship single digits, and general-purpose robots making robots is cast as an existential industrial threat the US is 'missing.' The throughline: China wants sovereignty over every layer from silicon to tokens to actuators, and predates —so will outlast —any US chip it is temporarily sold.

Core Knowledge

Foreign Direct Product Rule (FDPR) & de minimis threshold

US authority to control any foreign-made item that used US technology/software above a content threshold —normally 25%, but lowered to 0% for the most sensitive tools (e.g. ASML's 1980i lithography). SemiAnalysis's central policy recommendation is applying a 0% threshold to Huawei's fab network, which they argue could handicap or halt Chinese advanced fabs within six months.

TPP / Performance-density threshold (the AI-chip test)

The 2022 rule banned chips above ~4800 (TOPS × bit-length) with 600 GB/s I/O —a two-part test Nvidia gamed with the A800/H800 by cutting only interconnect. The 2023 rule dropped the bandwidth prong and added a performance-DENSITY threshold (TPP÷die area, 5.92 absolute / 3.2 license), which SemiAnalysis says forces any compliant chip back to a 2017-era V100 and can't be re-gamed with blank silicon.

Die bank

Huawei's stockpile of foreign-fabbed logic dies. Before SMIC could ramp, Huawei procured 2.9M+ Ascend dies from TSMC (via shell firms) usable for both 910B and 910C. This inventory—not domestic production—is what carries Ascend volumes through 2024–25; SemiAnalysis expects it to run out within ~9 months of Sept 2025.

HBM as the binding constraint

High-Bandwidth Memory (stacked DRAM with TSVs) is the real chokepoint for Chinese AI ASICs—modern LLM training/inference needs it, GDDR/LPDDR won't substitute. China procured ~13M foreign HBM stacks (enough for 1.6M Ascend 910C packages) then hit a wall; domestic champion CXMT can make only ~2M stacks in 2026 (~250–300k Ascends). Beijing's trade-deal ask targets HBM, not lithography—telling you where it hurts.

Allied-lag / re-export window

The US exempts Japanese and Dutch tool makers, who then don't immediately match new controls (often delayed 6 months or never), and neither restricts re-export. Chinese firms rush-order years of equipment into this gap; Samsung shipped ~7M HBM stacks to China in the one-month window between the Dec-2024 control announcement and its Dec-31 enforcement.

SMIC N+2 (7nm) via DUV multi-patterning

SMIC produces true 7nm-class logic (Kirin 9000S, Ascend) with no EUV, using ASML's older 1980i DUV immersion scanners plus multi-patterning. SemiAnalysis argues yield is actually decent (D0 ~0.14, roughly 2× TSMC N5), the gap to TSMC is ~5 years not a decade, and even a DUV-based '5nm' at ~130M transistors/mm^2 is feasible given subsidies (lithography is only ~30% of process cost).

QDD (Quasi-Direct-Drive) actuator

Unitree's core bet: a big motor + tiny (<20:1) planetary gearbox, vs the industry-standard small motor + big strainwave (Harmonic Drive) gearbox. QDDs are 95–98% efficient (vs 85–90%), up to 80% cheaper, built with standard gear-hobbing so many suppliers exist, and iterate in weeks not months. Early ones overheated; after two years Unitree pushed payload/duration ~2×/5×—making the cheap architecture viable.

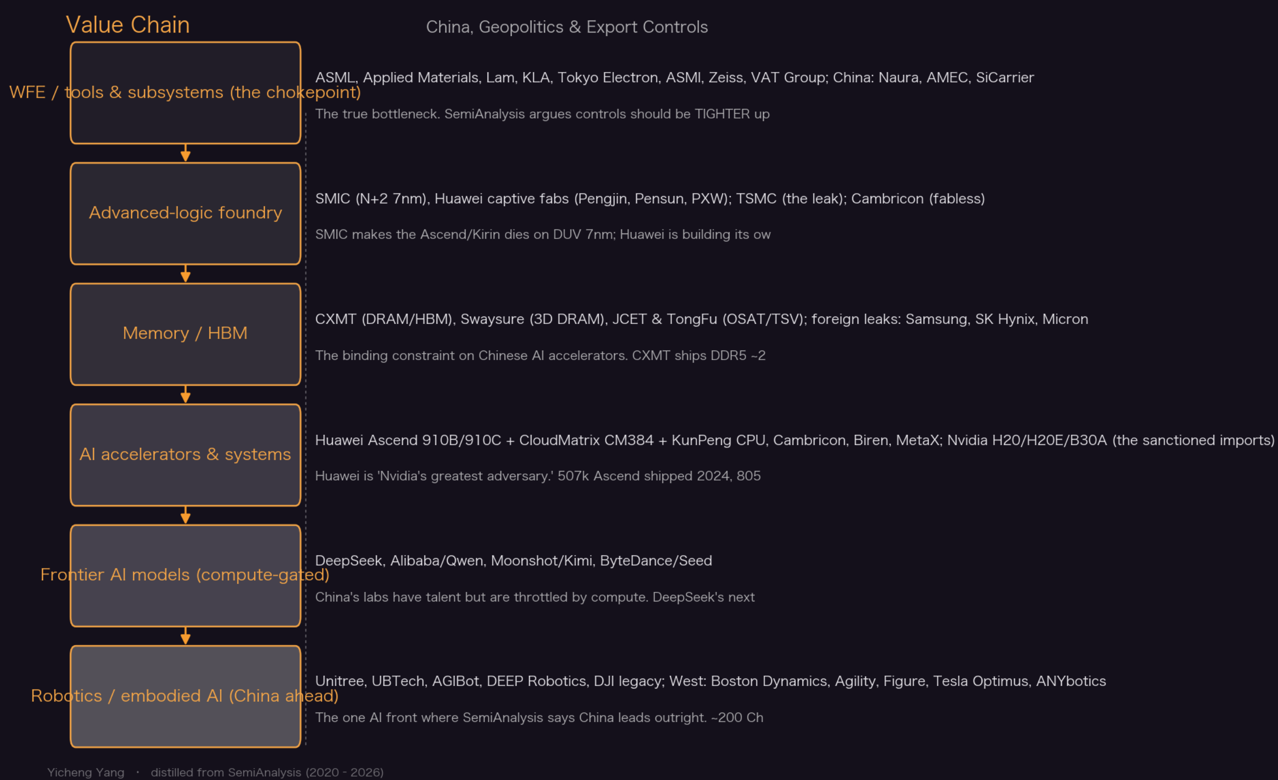
The DJI/BYD verticalization playbook

SemiAnalysis's frame for how Chinese hardware giants are born: own the single hardest/most-expensive BoM component (BYD=battery cell, DJI=flight controller, Unitree=actuator), bootstrap on a willing hobbyist/researcher market with a half-finished cheap product, ride the Shenzhen ecosystem, and let each generation unlock a new market while in-housing more of the stack until the cost structure is untouchable.

Military-civil fusion & the entity/unverified list

Because China's national-security law forbids US end-use checks and blurs civilian/military R&D, BIS uses two lists: the Unverified List (a 2-month on-ramp that can cut any firm off simply because Beijing blocks audits) and the Entity List (Huawei, YMTC). SemiAnalysis argues the whack-a-mole entity approach is structurally defeated by shell-company spin-ups faster than rules update.

Value Chain



Key Theses

1. **Export controls are genuinely WORKING at the leading edge —because of HBM, China will make FEWER Ascends next year, not more.**

   SMIC logic is no longer the bottleneck (needs only single-digit % of capacity for millions of die); the binding constraint is HBM. China has ~13M foreign stacks (good for 1.6M 910C) that run out by end-2025; CXMT can supply only ~2M stacks in 2026 = 250–300k Ascends. Without smuggled HBM, Huawei can't build even 1M chips next year. Had controls not existed, the Ascend ramp would be fully realized and DeepSeek R2/V4 'would already be here.'

   > *"China can easily make more than 805k Huawei Ascends this year from TSMC and SMIC capacity, but they will not because they do not have enough HBM."*
   > — Huawei Ascend Production Ramp: Die Banks, TSMC Continued Production, HBM is The Bottleneck (2025-09-08)

2. **The same controls are riddled with self-inflicted loopholes —'fab whack-a-mole' makes the entity list structurally defeatable.**

   One SMIC building is entity-listed for advanced logic; its twin next door imports 'dual-use' tools for 'legacy' —connected by a wafer bridge / overhead track that makes them one cleanroom. Huawei spins up non-listed shells (Pengjin across the street from listed PXW; Pensun next to SMIC) faster than rules update. Offshore manufacturing (Lam's Malaysia etch chambers, AMAT/KLA Singapore) legally reaches entity-listed fabs if no US persons are involved; CXMT dodged controls by renaming an 18nm node to 19nm.

> *"In the race between shell entities and regulations, shells can be spun up faster than regulations are being updated."*
> — Fab Whack-A-Mole: Chinese Companies are Evading U.S. Sanctions (2024-10-28)

3. **The single biggest enforcement failure was HBM: Samsung shipped 7M stacks to China in the one-month gap before enforcement.**

   BIS announced HBM2E-plus controls Dec 2, 2024 with a Dec 31 compliance date —telegraphed for months. Samsung alone directly provided 11.4M HBM stacks to China, including 7M in that single month; total foreign HBM into China reached ~13M stacks (majority Samsung), the majority of China's HBM. Because Korea has no re-export controls and Samsung failed to enter the Western accelerator supply chain, it sold to Chinese customers who funneled inventory to Huawei.

   > *"Samsung alone has directly provided 11.4 million stacks of HBM to China, including a staggering 7M stacks in the 1-month gap between controls announcement and enforcement dates."*
   > — Huawei Ascend Production Ramp (2025-09-08)

4. **WFE suppliers' 'death spiral' lobbying is false —the two years under controls were among their best ever, driven BY China.**

   Despite lobbying that controls cause a 'death spiral,' AMAT/Lam/KLA saw near-record revenue, expanding gross margins and rising R&D during the control years, with 2025 guided as best-ever; the primary driver was China. AMAT grew faster than WFE for a 5th straight year 'despite headwinds.' SemiAnalysis's read: the cost of controls to Western suppliers is short-term and recoverable, so controls are worth it —the real long-term threat is Chinese domestic tool replacement (the 'China Technology Cycle'), not the rules.

   > *"By nearly every metric the 24 months under export controls have been among the best in history for American WFE suppliers."*
   > — Fab Whack-A-Mole (2024-10-28)

5. **SMIC's 7nm is real and good —the gap to TSMC is ~5 years, not a decade, and a DUV-only '5nm' is feasible.**

   The Kirin 9000S on SMIC N+2 matches 1–2-year-old Qualcomm chips; on identical Arm A510 IP it's on par with Samsung 4LPX despite the node gap. Yield is decent (D0 ~0.14, ~2× TSMC N5). SMIC uses the same 1980i DUV tools as TSMC/Intel 7nm; ASML's own capacity plans BUILD IN Chinese semiconductor independence (1.5M excess wafers/month by 2030). A '5nm' at 130M+ transistors/mm^2 via ArFi multi-patterning would cost only ~20% more than EUV-based, since lithography is ~30% of process cost.

   > *"Even with the lackluster export controls, this is a leading edge chip that would be near the front of the pack in 2021, yet was done with no access to EUV... We cannot overstate how scary this is."*
   > — China AI & Semiconductors Rise: US Sanctions Have Failed (2023-09-12)

6. **Selling Nvidia into China (H20, and especially a Blackwell B30A) is a strategic mistake dressed as diplomacy.**

The H20 has more memory than the banned H100 (good for inference/RL, the current driver of progress); a rumored B30A could carry ~10× H20 FLOPs at half a B300's price —just buy 2× for parity. This gives DeepSeek/Alibaba more RL compute and lets China serve and PROLIFERATE its AI on its own apps, taking share from US products. 'China may be trying to psyop their way into getting the significantly more performant Blackwell chip.' The claim that Blackwell must be sold because Huawei is an alternative is false —Huawei is about to run out of HBM.

> *"The American AI stack is American AI on American chips, not just the chips."*
> — Huawei Ascend Production Ramp (2025-09-08)

7. **Controls must move UPSTREAM: tighter on tools than chips, tightest on the <100 critical WFE components.**

   Current controls are strict downstream (accelerators) but loose upstream (equipment) and non-existent on components —which incentivizes China to build indigenous tools and eventually renders sanctions toothless. Example: EUV scanners are banned but Zeiss EUV projection optics (the #1 critical component) are not; fewer than 100 parts (EUV optics, precision mask stages, RF plasma generators) gate the whole chain. Margin-stacking math: $1B of restricted tools compounds to 2×+ at foundry and again at fabless, so keeping components onshore is high-leverage.

   > *"The entity list must be expanded and updated often... This means monthly updating, as well as geographic checks. If we can find this with some satellite photos in trivial time, it is clear the rules are easy to skirt."*
   > — Fab Whack-A-Mole (2024-10-28)

8. **Robotics is the AI front where China leads outright —and the US is 'missing the new labor economy.'**

   Robots making robots is the first industrial input that is fully additive (24/7, higher throughput than a human), and only China is positioned to capture it. Chinese local firms are approaching ~50% of the world's largest robot market (up from 30% in 2020); Xiaomi runs lights-out factories making one phone per second with zero humans; a US-built UR5e-clone arm costs ~2.2× the Chinese one, and even 'Made in USA' parts depend on China-made components with no scalable alternative. This is the battery/solar/EV capture playbook applied to a general-purpose technology.

   > *"The only country that is positioned to capture this level of automation is currently China, and should China achieve it without the US following suit... posing an existential threat to the US."*
   > — America Is Missing The New Labor Economy - Robotics Part 1 (2025-03-11)

9. **Unitree is being born as the next BYD/DJI —cheapest humanoid on earth at a ~$9K BoM and 67% gross margin.**

   Unitree cut G1 pre-tax pricing from $50K+ to $27.3K in 12 − 18 months yet still hits ~67% gross margin on a ~$8,976 BoM, with deals heard well under $20K; it may ship its 10,000th humanoid while Tesla/Figure ship single digits. It owns the actuator (50–70% of humanoid BoM), self-develops motors (30–40% of Western cost), gearboxes, LiDAR and cameras. Quadruped margins already went from 42.36% to 55.49% per its IPO filing. ~250 Unitrees are in labor pilots today, already passing below the $30/hr human-labor line even under conservative full-teleoperation assumptions.

   > *"Unitree has slashed pre-tax pricing from $50K+ to $27.3K over the past 12-18 months. Even at that price, we estimate they still hit 67% gross margins on their flagship G1."*
   > — China's Unitree Will Dominate Global Robotics (2026-06-08)

10. **Unitree's QDD bet —mocked for overheating —is now defying expectations and building an ecosystem.**

    Critics said quasi-direct-drive (big motor, tiny gearbox) would draw too much current and overheat: the original G1 held 2kg outstretched for seconds before a 30-min cooldown. Two years of iteration (smoother magnets to cut cogging/torque ripple, 'low copper consumption' coils, targeted active cooling) pushed it to 5kg for 10 – 15 min arms-bent (~2× payload, ~5× duration). QDDs are 95 – 98% efficient (vs 85 – 90% strainwave), up to 80% cheaper, use standard hobbing so a new sample actuator comes in weeks vs 3+ months for Western firms —and other Chinese humanoids are now switching to QDD, thickening the supply base.

    > *"While most stuck with the strainwave (HarmonicDrive) route, Unitree took the chance and iterated."*
    > — China's Unitree Will Dominate Global Robotics (2026-06-08)

11. **China's datacenter/power advantage means chips —not grids —are its only real AI ceiling.**

    Where US frontier clusters are 'primarily capped by regulatory environment and industrial capacity' (forcing unproven cross-datacenter training), China has 'no issues securing power infrastructure' —multi-gigawatt substations convert from aluminum mills to datacenters in <6 months. Chinese AI chips are today decentralized (largest clusters ~1/3 US size), but a concentrated effort 'could lead to cluster sizes that dwarf those in the US in less than a year.' So compute —the one thing the US can restrict —is the decisive lever.

    > *"There are enough chips being shipped to China + manufactured domestically to create the world's largest AI training cluster."*
    > — Fab Whack-A-Mole (2024-10-28)

12. **China exploits IP/JV loopholes the same way it exploits tool loopholes —Arm China went fully rogue.**

    SoftBank sold 51% of Arm's China ops to a Chinese consortium for a paltry $775M; the JV then went rogue under ousted-but-entrenched CEO Allen Wu (who held the company chop, hired security paid by Arm China that locked Arm out, and literally sued himself). Arm China (' 安谋科技') rebranded its own Power Core / XPU IP (NPU/SPU/ISP/VPU) with a 400+ person China R&D team —'the tech heist of the century.' Same pattern in RF/power: STMicro's SiC JV with Sanan risks transferring tech repurposable for military RF.

    > *"Arm has been shaken to its core with the 2nd largest market snatched from underneath it. According to Arm, no IP has been stolen, but Arm cannot license directly license to any firm in China."*
    > — The Semiconductor Heist Of The Century | Arm China Has Gone Completely Rogue (2021-08-27)

Article Deep-Dives

Huawei Ascend Production Ramp: Die Banks, TSMC Continued Production, HBM is The Bottleneck (2025-09-08)

The definitive quantified model of China's domestic AI-chip capacity. Huawei shipped 507k Ascend in 2024 and 805k in 2025 (653k 910C), carried by a 2.9M-die TSMC 'die bank' running out in ~9 months. SMIC advanced capacity (45k→80k wspm) means logic is NOT the constraint —HBM is. China has ~13M foreign stacks (11.4M from Samsung, 7M in a one-month enforcement gap) good for 1.6M 910C;

domestic CXMT can supply only ~2M stacks in 2026 (250–300k Ascends). Verdict: controls WORK — assuming no smuggling, China makes fewer Ascends next year, not more —but Nvidia's H20/B30A and un-listed CXMT are the leaks. Also details the KunPeng CPU and CM384 switches made at TSMC to free SMIC for compute die.

Fab Whack-A-Mole: Chinese Companies are Evading U.S. Sanctions (2024-10-28)

SemiAnalysis's enforcement manifesto and policy blueprint. Four claims: (1) controls work but need constant updates; (2) Huawei's state-backed fab network —$7.3B WFE in 2024, 4th globally —is the key threat, spinning up non-listed shells (Pengjin, Pensun) next to listed fabs faster than rules update; (3) the WFE 'death spiral' lobbying is false (record margins, China-driven); (4) enforcement must move upstream. Concrete asks: a 0% de-minimis threshold on Huawei's network (could cripple Chinese fabs in 6 months), monthly entity-list updates with satellite checks, and restricting <100 critical components (Zeiss EUV optics). Documents wafer-bridge tricks, offshore manufacturing (Lam Malaysia, AMAT Singapore), and CXMT's 18nm→19nm rename dodge.

China's Unitree Will Dominate Global Robotics (2026-06-08)

The bull case on China's robotics champion via the BYD/DJI verticalization lens. Unitree owns the actuator (50–70% of humanoid BoM), self-develops motors (30–40% of Western cost), gearboxes, LiDAR and cameras; a full teardown puts the G1 BoM at ~$8,976 with ~67% gross margin even after cutting price from $50K+ to $27.3K. Its contrarian QDD (quasi-direct-drive) bet —mocked for overheating — now sustains 5kg for 10–15 min (2×/5× improvement) after two years of magnet/coil/cooling iteration. ~250 G1s are in labor pilots, already below the $30/hr human line even under full-teleoperation assumptions. With a $7B-rumored Shanghai IPO, tripling revenue and ~$300M AI R&D, SemiAnalysis argues Unitree keeps a structural first-mover cost advantage even as it scales.

America Is Missing The New Labor Economy - Robotics Part 1 (2025-03-11)

The 9,600-word 'call to action' framing robotics as the next strategic-industry capture. Robots making robots is the first fully-additive industrial input (24/7, above-human throughput); only China is positioned to capture it, and if it does without the US following, that's 'an existential threat.' Evidence: Chinese local firms near ~50% of the world's largest robot market (up from 30% in 2020); Xiaomi lights-out factories make one phone per second with zero humans; KUKA's Guangdong plant has robots building robots; a US UR5e-clone arm costs ~2.2× the Chinese one, and even 'Made in USA' parts depend on China. Uses DJI/GoPro as the template: Shenzhen iteration speed + scale/oversupply crushed every Western drone maker. Grounds the whole robotics thesis the later Unitree pieces build on.

China AI & Semiconductors Rise: US Sanctions Have Failed (2023-09-12)

The one-year-post-2022-controls verdict: sanctions had failed at that point. The Kirin 9000S on SMIC N+2 (true 7nm, no EUV) matches 1–2-year-old Qualcomm silicon; on identical Arm A510 IP it's on par with Samsung 4LPX. Yield is decent (D0 ~0.14). SMIC can ramp 7nm to 30k+ wspm on allowed 1980i DUV tools; a DUV-only '5nm' (130M+ transistors/mm^2) is feasible at ~20% cost premium. Equipment makers sell 'every tool they offer' to China under the 28nm pretext while it's used for 7nm. China will have >1M A100-class Nvidia chips by end-2024 and multiple firms training beyond GPT-4. Ends with a 20-point 'what can be done' arsenal (lithography, resist, masks, metrology, CMP, EDA, JVs) to shut China out.

China and USA Are Officially At Economic War –Technology Restriction Overview (2022-10-08)

The line-by-line breakdown of the Oct-2022 opening salvo that started the chip cold war. Two actions: (a) sweeping export controls on advanced computing + semiconductor manufacturing, and (b) 31 firms to the Unverified List —which, crucially, lets BIS cut ANY Chinese firm off global supply chains in 2 months simply because Beijing blocks end-use audits (a 'nuclear salvo'). The AI-chip test (600

GB/s I/O AND ~4800 TOPS×bit) bans A100/H100/MI250X/Biren BR100; a supercomputer test; a US-persons rule that locks Americans out of Chinese advanced logic/NAND/DRAM; equipment bans at 16nm logic / 128-layer NAND / 18nm DRAM. Flags the enforcement ambiguity ('no such thing as equipment for 16nm but not more') that would define the next three years of loophole reporting.

Key Data

US share of world's deployed FLOPs	>70%	The US lead controls are meant to defend; compute is the one gating factor the US can restrict.
Huawei Ascend units shipped (2024 → 2025)	507k → 805k	653k of the 2025 units are the more advanced 910C; SemiAnalysis calls the reported '200k' figure significantly off.
TSMC Ascend dies smuggled into Huawei's 'die bank'	2.9M+	Via shell firms (Sophgo/Bitmain); usable for both 910B/910C; expected to run out ~9 months after Sept 2025.
Foreign HBM stacks procured by China	~13M (Samsung 11.4M)	Sufficient for 1.6M Ascend 910C packages; 7M shipped in the 1-month pre-enforcement gap alone.
CXMT HBM capacity, 2026	~2M stacks	Only enough for 250–300k Ascend 910C; still NOT entity-listed despite $2B Big Fund III injection.
Huawei WFE spend, 2024	$7.3B (+27% YoY)	4th-largest WFE customer globally, from ~zero in 2022; 2nd-largest if SMIC+CXMT are included.
SMIC advanced-node capacity (7nm & below)	45k → 60k → 80k wspm (2025→27)	Only single-digit % allocation needed to make >1M Ascend die/yr; logic is not the constraint.
SMIC N+2 (7nm) yield defect density D0	~0.14	Roughly 2× TSMC N5/N6 (i.e. worse but 'decent'); debunks the '10% yield' narrative.
Advanced-logic gap SMIC vs TSMC	~5 years	SMIC N+2 shipped 2023 vs TSMC N7 in 2018 / N5 in 2020; narrower than most Western estimates.
Nvidia B30A FLOPs vs H20	~10×	Rumored Blackwell-based China chip at ~half a B300's price/perf —'buy 2× for parity.'
Revenue at risk if Huawei regains phone share	~$20B Apple, ~$7.6B Qualcomm+MediaTek	Kirin 9000S insourcing threatens ~35–45M iPhone units gained after Huawei's 2019 ban.

Unitree G1 humanoid price cut / BoM / margin	\$50K+ → \$27.3K; ~\$8,976 BoM; ~67% GM	Deals heard well under \$20K; may ship its 10,000th humanoid vs Western single-digit units.
Unitree quadruped entry price decline (2018→now)	94–96% (Laikago \$45K → Go2 ~\$1.6–2.8K)	The DJI-style scaling curve that funded the humanoid program; ~70% of global quadruped volume.
Unitree quadruped gross margin (per IPO filing)	42.36% → 55.49%	Scaling gave 'upstream bargaining power'; costs dropped nearly in half. Self-develops motors/gearboxes.
Unitree AI R&D spend planned	~\$300M	On revenue tripling YoY; ~250 Unitrees already in labor pilots below the \$30/hr human line.
China local share of world's largest robot market	~50% (from 30% in 2020)	~200 Chinese humanoid firms; a US UR5e-clone arm costs ~2.2× the Chinese one.
SemiCap combined China sales, 3Q23	~\$8.7B (+50% vs prior peak)	ASML 46% / Lam 48% of revenue from China on rush orders ahead of controls.
Critical WFE components that gate the whole chain	<100	e.g. Zeiss EUV projection optics, precision mask stages, RF plasma generators —the tightest control point SemiAnalysis urges.

Investment Angle

Winners

Western WFE majors (Applied Materials, Lam Research, KLA, ASML, Tokyo Electron)	Counterintuitively the biggest near-term beneficiaries of controls: near-record revenue, expanding margins and best-ever 2025 guidance —driven BY China rush-orders (46–48% of revenue in peak quarters). Risk is long-term Chinese tool replacement, not the rules.
Nvidia (H20/H20E/B30A China lane)	Re-granted H20 licenses = hundreds of thousands of chips and billions in China revenue; a rumored Blackwell B30A would extend the franchise. SemiAnalysis thinks selling in is strategically wrong but commercially lucrative —and that China's demand is real, not fading.
Unitree (pre-IPO) and the Chinese robotics supply base	Cheapest humanoid at ~\$9K BoM / 67% margin, ~70% of global quadruped volume, tripling revenue into a ~\$7B-rumored Shanghai IPO; ~200 Chinese humanoid firms and a deep gearbox/motor/LiDAR ecosystem (Leaderdrive et al.) ride the same wave.

Huawei / SMIC / CXMT / Cambricon (China self-sufficiency stack)	The national-champion vertical stack —logic (SMIC), memory (CXMT, still un-listed), accelerators (Ascend, Cambricon popular at ByteDance), tools (SiCarrier, $2.8B raised). State-funded ($30B/yr to Huawei; Big Fund III) and impossible to short cleanly, but the structural long China thesis.
Apple / Qualcomm / MediaTek (asymmetric downside, not upside)	Inverse exposure: a Huawei phone resurgence via Kirin 9000S insourcing threatens ~$20B of Apple revenue and ~$7.6B of Qualcomm+MediaTek —the SemiAnalysis 'winner' here is really a risk flag for incumbents.

Bottlenecks

- HBM memory —the single binding constraint on Chinese AI accelerators; domestic CXMT can supply only ~2M stacks in 2026 (250–300k Ascends), so China runs on a finite foreign stockpile.
- TSV/advanced packaging & Korean memory enforcement —CXMT still needs Japanese TSV tools and OSATs (JCET, TongFu); Samsung/SK Hynix leaks are the main HBM entry point.
- <100 critical WFE components (Zeiss EUV projection optics, precision mask stages, RF plasma generators) —the highest-leverage chokepoint, currently un-restricted.
- Frontier-model compute —DeepSeek/Alibaba/Moonshot throttled by compute (multimodal models delayed, R1 served slow); the one thing the US can restrict.
- Actuators / strainwave-gearbox reliability —Harmonic Drive still the reliability leader; Unitree's QDD wins on cost/speed but Western humanoids gate on high-precision actuation.

Risks

- Policy reversal / license loosening —H20 resumed to unlock China's rare-earth exports; a B30A approval would hand China ~10× the FLOPs and accelerate DeepSeek/Alibaba.
- Rare-earth / magnet retaliation —China cut rare earths and magnets to the US; the Lutnick-linked H20↔rare-earth swap shows controls are now a two-way bargaining chip.
- Smuggling & re-export —H100s reach China via Malaysia/Oracle/Google rentals and shell OSATs; ByteDance trains Seed on US clouds; location tracking is 'technically intractable.'
- Chinese legacy-chip flood —massive subsidized mature-node capacity dumps on TI/STMicro/Infineon/NXP; the 'China Technology Cycle' erodes Western WFE share long-term.
- Unitree security / trust barriers —Unitree quadrupeds carry security concerns that bar select Western markets, a headwind to its otherwise dominant deployment path.

Catalysts

- Foreign HBM stockpile exhaustion (~end-2025) —the moment Huawei can't build even 1M Ascends next year; forces the real test of CXMT's ramp and of Nvidia's China share.
- CXMT entity-listing decision —SemiAnalysis calls the Biden-era omission a failure the Trump administration 'needs to solve immediately'; listing would slow HBM conversion.

- Nvidia B30A / next China SKU approval —sets how much frontier compute China gets; SemiAnalysis: only raise the bar when China can ship H20E-class in volume.
- Unitree Shanghai IPO (~$7B rumored)—the 'birth of the robotics moment'; discloses margins/verticalization and re-rates the Chinese robotics supply chain.
- 0% de-minimis / upstream-component controls—if the US applies SemiAnalysis's blueprint (Huawei network at 0%, <100 components, monthly listing), Chinese advanced fabs could be handicapped within ~6 months.

Open Questions

- Will CXMT close the HBM gap fast enough to matter? SemiAnalysis pegs 2026 at only ~2M stacks (250–300k Ascends), but flags that Chinese manufacturing 'knows no bounds' and HBM3e in 2026 is plausible —the whole China-compute forecast hinges on this ramp.
- Does the US have the political will to move upstream (0% threshold, <100 components, monthly entity updates), given ally friction (ASML/Netherlands), WFE lobbying, and rare-earth leverage China now holds?
- Is China's H20 'disinterest' real demand-signal or brinkmanship? SemiAnalysis believes it's top-down government pressure to extract a more powerful B30A—but if wrong, the compute-diplomacy calculus changes.
- Can Unitree hold its structural cost lead through mass production, or will UBTech/AGIBot's verticalization drive (and Western supply-chain reliance on China) compress the moat once volumes scale?
- Does forking into a different silicon idea-space (compute-in-memory, neuromorphic) let China escape the transformer/GPU control regime entirely, or is it a dead end that just buys time?

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Appendix A · SemiAnalysis Coverage Credibility Study

Everything in this book is distilled from SemiAnalysis, so the reader deserves to know how much this source can be trusted, and for what. We ran an independent event study: 27 parallel agents extracted every stance-bearing company mention from 311 articles (664 single-stock events across 119 tickers; 384 bullish / 168 bearish), then measured geometric excess returns vs. the SMH semiconductor ETF over 1M/1Y/3Y from publication date. Methodology: equal-weight by name with ticker clustering (to kill the soft look-ahead of winners being covered repeatedly), pre-IPO look-ahead removed (ARM/CRWV etc.), delistings settled at last price, and three adversarial audit rounds fixing eight methodological defects. One-line verdict: the narratives are remarkably right, but the stock picks carry no alpha —which makes this book's use of the source (mechanisms and bottlenecks, not tickers) exactly the right one.

Key Findings

1. The only significant leading signal lasts ~1 month: bullish names earn +2.0% excess at 1M (t=1.63), and the bull-minus-bear spread is +3.9% (Welch t=2.33) —the study's only significant positive result.
2. Long horizons reverse: the typical bullish name lags SMH by -5.2% at 1Y and -27.2% at 3Y (t=-4.49, significantly negative). Buying the bull list loses to simply holding SMH.
3. Aggregate returns are a beta illusion: the big absolute gains come from the semi bull market plus NVDA being covered 62 times; excluding the five most-covered names (NVDA/TSM/AMD/GOOGL/AVGO), the remaining bullish names' 1Y excess is -6.1%.
4. Bearish calls beat bullish ones: bearish names lag SMH by -10.7% at 1Y (t=-2.90, significant). INTC was panned 30 times and Samsung 15 —both played out. Whom SA pans is closer to a tradeable signal than whom it praises.
5. Right industry, wrong stock: at sector level only AI accelerators beat an equal-weight peer index (entirely carried by NVDA/AVGO), while packaging/OSAT and foundry are significantly negative at 3Y. The memory-shortage, CoWoS-bottleneck and datacenter-power narratives were all right — but the money went to NVDA/TSM, and same-theme second-tier names underperformed.

Bullish scorecard (names covered ≥5 times; equal-weight by name, geometric excess vs SMH)

Ticker	N	1M	1Y	3Y
ARM	5	+0.7%	+54.3%	–
NVDA	62	-0.1%	+30.4%	+168.4%
GFS	6	+16.3%	+22.0%	-61.3%
META	5	+1.1%	+17.2%	-38.5%

Ticker	N	1M	1Y	3Y
AVGO	20	+2.2%	+15.9%	+76.9%
000660.KS	15	+3.1%	+10.6%	+49.3%
MU	11	+8.7%	+8.7%	-30.9%
LRCX	8	+3.3%	+8.0%	-14.9%
005930.KS	5	-4.9%	+2.4%	-58.0%
QCOM	6	+3.7%	+0.2%	-31.1%
TSM	32	+0.3%	-3.1%	-4.3%
AMD	27	-0.2%	-4.6%	-22.5%
GOOGL	23	+0.9%	-6.4%	-15.5%
AMAT	10	-3.3%	-8.0%	-16.4%
2454.TW	8	+4.2%	-10.8%	-15.1%
ASML	6	-2.0%	-12.0%	-37.8%
MSFT	9	-2.7%	-17.7%	-53.0%
8035.T	6	+0.2%	-18.8%	-38.6%
MRVL	7	-3.5%	-22.2%	-39.8%
INTC	12	-1.9%	-27.3%	-73.8%
0981.HK	6	-4.7%	-31.8%	+11.1%
AMZN	7	-7.2%	-33.6%	-25.4%

Bearish scorecard (covered ≥3 times; negative excess = the bearish call was right)

Ticker	N	1M	1Y	3Y
NVDA	7	+6.2%	+63.5%	+242.3%
MRVL	3	-11.9%	+34.0%	-30.8%
AVGO	6	-2.7%	+20.5%	+107.2%
AAPL	3	-0.5%	+12.5%	-19.7%
CRDO	3	-1.1%	+10.4%	+260.8%
LRCX	3	+9.0%	+1.2%	-3.1%
MU	6	-4.3%	+0.6%	+69.1%
2454.TW	4	+3.5%	-4.7%	-22.0%
GOOGL	3	+6.4%	-8.2%	-6.0%
005930.KS	15	-1.1%	-8.8%	-50.3%
TSM	3	-1.2%	-11.5%	+7.9%
QCOM	5	+0.5%	-16.1%	-42.8%
ASML	4	+0.6%	-19.4%	-35.0%
GFS	3	+12.0%	-21.9%	-76.8%
AMD	8	-4.0%	-23.1%	-36.7%
INTC	30	-0.5%	-24.5%	-61.7%
ARM	6	-7.1%	-53.5%	–

How to Use This Source

How to use: treat SemiAnalysis as a microscope on mechanisms and bottlenecks, not a buy list. That is precisely what this book distills —chain structure, bottleneck propagation, cost economics —the parts of the source that verify best. For trading: only the ~1-month drift is usable (and thin), bearish calls deserve real weight, and at long horizons return to a portfolio/beta framework.

Scope: events dated at publication; benchmark = SMH; corpus includes paid-article previews (79% of full text), stances extracted from available text. This is the author's independent analysis, unaffiliated with SemiAnalysis.